Economic Booms and the Structural Transformation of American Agriculture

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Abstract

This paper presents evidence that competition for inputs, particularly land, leads to a process of regional structural transformation when non-agricultural sectors grow. Using data from 1978 to 2012 on counties in the United States from the Agricultural Census, it finds that regions with higher growth in the number of people employed in non-agricultural industries have less growth in agricultural production, as well as cropland, and experience a shift towards crops rather than animal products and towards using more machinery. A shift-share Bartik instrument strategy confirms that this is caused by the non-agricultural sector.

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1 Introduction

A major theory in economics holds that developing countries typically undergo a structural transformation in the process of development, whereby they shift from agriculture to manufacturing and then on to services (Gollin and Kaboski, 2023). In particular, this is often said to be driven by increased agricultural productivity freeing up labor for manufacturing (Emerick, 2018; Asher et al., 2023). However, another possible part of the

story is one where non-agricultural sectors grow, outcompeting agriculture. This paper tests this explanation within the United States, using county-level data to examine the question of whether regional economic booms lead to less agricultural production from 1978-2012.

For an idea of how this might play out, take the Santa Clara Valley. In the 21st century, it is part of Silicon Valley, an urbanized center of high-tech innovation. But in the middle of the 20th century, it was known for an entirely different industry: its orchards, which produced a large portion of the country's prunes and apricots (Todd, 2022). However, as San José expanded into a major commercial hub, the price farmers could get for selling their land increased drastically, as did the property taxes, spurring the former orchards to be converted into housing developments and offices. Nowadays, few orchards remain in the region once known as the "Valley of Heart's Delight" and "Garden of the World" (Tsu, 2013).

In this paper, I try to answer the question of whether economic booms crowd out agricultural production, based on county-level agriculture and industry data for the entire United States, testing whether what happened in the Santa Clara Valley is a nationwide story. In particular, I focus on the tension between two models of structural transformation: one in which land is not used in the production of non-agricultural goods and another in which both sectors use land in production. While both predict the same overall effect on production, where agricultural production declines, the first predicts that changes come through competition for labor and the second predicts that changes come through competition for labor and land. The latter more closely matches media narratives that emphasize the role of competition for land in localized agricultural declines (Bowler, 2025). Additional potential mechanisms, such as environmental effects from pollution or countervailing local demand effects, add more complications to this picture. Furthermore, I examine how these changes led to additional adjustments within the agricultural sector, shifting other characteristics of agricultural production like crop choice and ownership structure.

I find evidence from descriptive results that an increase in the employed population, my principal proxy for an economic boom, is associated with a decrease in the value of agricultural production in a county, as well as the number of acres used to produce crops. This is robust to many alternative specifications, including changing the time period of analysis to some extent. Meanwhile, there is unclear evidence of an effect on various measures of labor. Furthermore, I find evidence from a shift-share instrument based on Bartik (1991) that this is a causal relationship, where an economic boom leads to decreased cropland. I also establish that this leads to a shift towards crops rather than animal products and increased usage of machinery. Although I cannot rule out other mechanisms, I also find some suggestive evidence that this is driven by competition for land. This fits with the theoretical model in which land is used for production in both sectors, although other results demonstrate the need for a more complex theory of structural transformation.

2 Theory and Background

This paper belongs to the literature about structural transformation, the process through which economically developing countries typically go from a principally agricultural economy to one based on manufacturing and thenceforth one based on services (Gollin and Kaboski, 2023). Previous work on closed economies has highlighted two main mechanisms for structural transformation: the Baumol effect, in which economic production shifts to sectors with slower productivity growth, and decreased demand for agricultural products as a share of total spending when income rises (Emerick, 2018; Ngai and Pissarides, 2007; Baumol, 1967). An additional potential explanation is differences in capital intensity (Dennis and İşcan, 2009).

However, both the Baumol effect and the income effect primarily operate in a closed economy; this is the default in models of structural transformation, with some exceptions based on Matsuyama (2009) (Bustos et al., 2016). For the Baumol effect, this is nearly a necessity; otherwise an increase in one sector's productivity would likely lead to more production, due to comparative advantage. Income effects also rely on people mostly consuming food that is grown within their own region, which assumes limited tradability of food either because of transport costs or trade barriers such as tariffs.

My setting in this paper is an open economy: the counties of the United States. The Commerce Clause and its subsequent interpretation through Supreme Court cases like Gibbons v. Ogden has ensured that the United States has few internal trade barriers other than transport costs, and so agricultural products are widely traded and consumed in places other than where they were grown. I hypothesize that this open economy shows a pattern similar to structural transformation, where increased economic performance in other sectors, as measured through the share employed, leads to a decrease in agricultural production.

The Baumol effect cannot explain this pattern, because in an open economy it is counterweighted by comparative advantage. I also reject income effects as a plausible explanation, because my setting is at a time when agriculture is already a small share of the US economy, but yet the United States is a major food producer. As such, other mechanisms must be necessary to explain this pattern which resembles structural transformation, other than the ones typically used to account for structural transformation. Bustos et al. (2016) lays out a model in an open economy in which structural transformation instead comes from technical change affecting different factors and industries (Matsuyama, 1992). I follow that work by focusing on explanations involving competition for inputs being used to their comparative advantage: land, labor, water, and the like, which are useful to both the agricultural and non-agricultural sector but are limited in their availability. I also hypothesize possible mechanisms that could lead to the opposite finding, particularly cross-sectoral agglomeration effects.

Many other papers frame American regional transformation as a process of structural transformation, but typically concentrate on earlier stages of the process (Caselli and Coleman II, 2001). For instance, Boone and Wilse-Samson (2023) analyzes changes in farm labor in response to the Great Depression as a form of structural change. Furthermore, Fiszbein (2022) argues that regions that initially have a more diverse array of crops grown experience structural transformation during the Industrial Revolution more quickly because they had more opportunities for cross-sectoral spillovers. Eckert and Peters (2022) also develops a model of regional structural change and applies it to the United States between 1880 and 1920.

2.1 Theoretical Model

Assume a small open economy (i.e. a U.S. county) composed of two sectors: the agricultural and non-agricultural sectors, much like those in Asher et al. (2023) and Bustos et al. (2016). The agricultural sector produces a tradable good A. The non-agricultural sector produces a tradable good N. Because both goods are tradable, prices p_A and p_N are exogenous. To start, I assume that N is produced using only labor, and A is produced using both labor and land in a Cobb-Douglas production function. a represents the sectoral total factor productivity, while θ represents the labor elasticity of output. Additionally, I assume constant returns to scale. This translates to the following equations:

$A = a_A * L_A^{\theta} * T_A^{1-\theta}$ $N = a_N * L_N$

This means that the following equation provides the marginal product of agricultural labor, which is equivalent to the wage in equilibrium:

$MPL_A = a_A * T_A^{1-\theta} * L_A^{\theta-1} * \theta$

I can then solve for agricultural production as a function of the wage.

$L_A = T_A * \frac{a_A * \theta}{w}^{\frac{1}{1-\theta}}$

$A = a_A * \frac{a_A * \theta}{w}^{\frac{\theta}{1-\theta}} * T$

$\frac{dA}{dw} = a_A * T * \frac{a_A * \theta}{w}^{\frac{2\theta-1}{1-\theta}} * a_A * \theta * w^{-2} * -1 * \frac{\theta}{1-\theta}$

On the other side, the marginal product of non-agricultural labor is as follows, which must also equal the wage in equilibrium:

$MPL_N = a_N$

Because the non-agricultural good is only produced using labor, a positive shock to its total factor productivity is equivalent to a positive shock to the amount of labor used to produce it; if demand is held constant, that is the only possible source of a positive shock to the amount of labor. A positive shock to total factor productivity also leads to an increased

wage. Because all quantities in the derivative of agricultural production with respect to wage are positive other than the -1, it must be negative, meaning that an increase in wage leads to a decrease in agricultural production. All land is used in agriculture, so this means that there must be a decrease in the quantity of labor used for agricultural production.

However, there is a strong theoretical case for assuming, contra to Asher et al. (2023), that the non-agricultural sector uses land also, and that land can move between sectors.[1] This may be in the form of housing or recreational uses, but also for land directly involved in the production of non-agricultural goods such as the plot on which a factory or shop sits. I model this by instead assuming that N is also produced in a Cobb-Douglas production function with constant returns to scale where γ represents the labor elasticity of output. Thus, it is highly akin to the Heckscher-Ohlin model.

All the equations given above for the agricultural sector still hold true for this context, but this requires new derivations for the non-agricultural sector, as provided below.

$$MPL_N = a_N * T_N^{1-\gamma} * L_N^{\gamma-1} * \gamma$$

To figure out the wage, I then must set $MPL_N = MPL_A$ because of the assumption that the wages in the two sectors must be the same in equilibrium. Therefore:

$$a_N * T_N^{1-\gamma} * L_N^{\gamma-1} * \gamma = a_A * T_A^{1-\theta} * L_A^{\theta-1} * \theta$$

Furthermore, there is now a rental price r of land, since it can be used for things other than agriculture. This is equivalent to the marginal product of land, which must now be equalized in both sectors.

$$MPT_A = a_A * L_A^{\theta} * (1-\theta) * T_A^{-\theta}$$

$$MPT_N = a_N * L_N^{\gamma} * (1-\gamma) * T_N^{-\gamma}$$

This then lets me generate marginal rates of technical substitution for each sector, which by the equimarginal principle must be set equal to the ratio of input costs. So:

$$MRTS_A = \frac{\theta}{1-\theta} * \frac{T_A}{L_A} = \frac{w}{r}$$

$$MRTS_N = \frac{\gamma}{1-\gamma} * \frac{T_N}{L_N} = \frac{w}{r}$$

This provides the following equations for the cost of producing one unit of A or N:

$$c_A = \frac{1}{a_A} * \frac{w}{\theta}^{\theta} * \frac{r}{1-\theta}^{1-\theta}$$

$$c_N = \frac{1}{a_N} * \frac{w}{\gamma}^{\gamma} * \frac{r}{1-\gamma}^{1-\gamma}$$

[1] If land cannot move between sectors, it is by and large identical to assuming that land is only in the agricultural sector.

I can set the price equal to the marginal cost. Because Cobb-Douglas has constant returns to scale, this is equivalent to setting the price equal to the unit cost. This produces the following equations, using the hat algebra notation of Jones (1965), where a hat denotes the proportional change of a variable as in dx/x:

$$\widehat{p_A} = -\widehat{a_A} + \theta * \widehat{w} + (1-\theta) * \widehat{r}$$

$$\widehat{p_N} = -\widehat{a_N} + \gamma * \widehat{w} + (1-\gamma) * \widehat{r}$$

Holding prices constant, since they are determined exogenously by other regions, this gives the following results:

$$\widehat{a_A} = \theta * \widehat{w} + (1-\theta) * \widehat{r}$$

$$\widehat{a_N} = \gamma * \widehat{w} + (1-\gamma) * \widehat{r}$$

I set a_A to be 0, which in this context is equivalent to assuming that a change in a_N, the total factor productivity of the non-agricultural sector, does not directly adjust the total factor productivity of the agricultural sector. This allows me to solve the system of equations assuming that the wage is endogenous, which is equivalent to stating that there is no labor migration. Thus:

$$\widehat{w} = \widehat{a_N} * \frac{\theta-1}{\theta-\gamma}$$

$$\widehat{r} = \widehat{a_N} * \frac{\theta}{\theta-\gamma}$$

As such, the response to an increase in the total factor productivity of the non-agricultural sector affects both wages and rent, but it depends on which sector has a greater labor elasticity of output. If $\theta > \gamma$, then it leads to an increase in rent and a decrease in wages, but if $\theta < \gamma$, it leads to a decrease in rent and an increase in wages instead.

If land T and labor L are both immobile across regions and there is no land or labor outside the sectors of the model, this is equivalent to the model in Jones (1965), with modifications to allow for shocks to a_N. Thus I define $x_{i,j}$ to represent the amount of factor i used to produce one unit of commodity j (corresponding to a in Jones's paper), so that $x_{L,A} = \frac{L_A}{A}$ and $\lambda_{i,j}$ to represent the share of factor i that is used in the production of commodity j, such as $\lambda_{L,A} = \frac{L_A}{L}$. Thus:

$$x_{L,A}A + x_{L,N}N = L$$

$$x_{T,A}A + x_{T,N}N = T$$

Differentiating this with $\widehat{L}$ and $\widehat{T}$ equal to 0 gives the following equations:

$$\lambda_{L,A}\widehat{A} + \lambda_{L,N}\widehat{N} = -\lambda_{L,A}\widehat{x_{L,A}} - \lambda_{L,N}\widehat{x_{L,N}}$$

$$\lambda_{T,A}\widehat{A} + \lambda_{T,N}\widehat{N} = -\lambda_{T,A}\widehat{x_{T,A}} - \lambda_{T,N}\widehat{x_{T,N}}$$

Substituting in that $x_{i,j}$ is equal to the factor used in that sector divided by its output formula, alongside the formulas from the marginal rates of technical substitution, produces the following set of equations when $\widehat{a_A}$ is set to be 0:

$$\widehat{x_{LA}} = (\theta - 1) * (\widehat{w} - \widehat{r})$$

$$\widehat{x_{LN}} = (\gamma - 1) * (\widehat{w} - \widehat{r}) - \widehat{a_N}$$

$$\widehat{x_{TA}} = \theta * (\widehat{w} - \widehat{r})$$

$$\widehat{x_{TN}} = \gamma * (\widehat{w} - \widehat{r}) - \widehat{a_N}$$

I can then insert that back into the differentiated land and labor constraints to get the following equations:

$$\lambda_{L,A}\widehat{A} + \lambda_{L,N}\widehat{N} = -\lambda_{L,A}(\theta - 1)(\tfrac{-\widehat{a_N}}{\theta - \gamma}) - \lambda_{L,N}((\gamma - 1)(\tfrac{-\widehat{a_N}}{\theta - \gamma}) - \widehat{a_N})$$

$$\lambda_{T,A}\widehat{A} + \lambda_{T,N}\widehat{N} = -\lambda_{T,A}\theta(\tfrac{-\widehat{a_N}}{\theta - \gamma}) - \lambda_{T,N}(\gamma(\tfrac{-\widehat{a_N}}{\theta - \gamma}) - \widehat{a_N})$$

Solving them as a system of equations generates the following results:

$$\frac{\widehat{A}}{\widehat{a_N}} = \frac{\theta(\lambda_{T,N} - \lambda_{L,N}) - \lambda_{T,N}}{(\theta - \gamma)(\lambda_{L,A} - \lambda_{T,A})}$$

$$\frac{\widehat{N}}{\widehat{a_N}} = 1 + \frac{\gamma(\lambda_{L,A} - \lambda_{T,A}) + \lambda_{T,A}}{(\theta - \gamma)(\lambda_{L,A} - \lambda_{T,A})}$$

I can interpret this as meaning that a positive shock to total factor productivity in the non-agricultural sector leads to a decrease in total agricultural production and an increase in total non-agricultural production, because all $\lambda_{i,j}$ are greater than zero and the denominator of both must be positive, as when $\lambda_{L,A} > \lambda_{T,A}$, that means that agriculture is the proportionally more labor-intensive sector and thus $\theta > \gamma$, as well as vice versa.

Additionally, I can calculate the following:

$$\widehat{L_A} = \widehat{x_{L,A}} + \widehat{A}$$

$$\widehat{L_N} = \widehat{x_{L,N}} + \widehat{N}$$

$$\widehat{T_A} = \widehat{x_{T,A}} + \widehat{A}$$

$$\widehat{T_N} = \widehat{x_{T,N}} + \widehat{N}$$

By dividing both sides by $\widehat{a_N}$, I can thus identify the signs of a shock to total factor productivity in the non-agricultural sector on labor and land usage in both sectors.

$$\frac{\widehat{L_A}}{\widehat{a_N}} = \frac{-\lambda_{L,N}}{(\theta - \gamma)(\lambda_{L,A} - \lambda_{T,A})} < 0$$

$$\frac{\widehat{L_N}}{\widehat{a_N}} = \frac{\lambda_{L,A}}{(\theta - \gamma)(\lambda_{L,A} - \lambda_{T,A})} > 0$$

$\frac{\widehat{T_A}}{\widehat{a_N}} = \frac{-\lambda_{T,N}}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} < 0$

$\frac{\widehat{T_N}}{\widehat{a_N}} = \frac{\lambda_{T,A}}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} > 0$

This means that the agricultural sector then experiences a decrease in both land and labor, and so therefore there must also be an increase in both the land and labor allocated to the non-agricultural sector.

Take the wage bill W and the rent bill R to be the wage or rent times the amount of labor or land demanded for a given industry. Because the equations are in Cobb-Douglas form, the payments for each factor are constant shares, and thence $\frac{\widehat{W_A}}{\widehat{a_N}} = \frac{\widehat{A}}{\widehat{a_N}}$, and so on and so forth for the rest. Thus the wage and rent bills will increase for the non-agricultural sector, and decrease for the agricultural sector.

However, there are some conditions here. This result only applies if θ is not equal to γ, and furthermore relies on the county producing both types of goods and being in a small open economy. Additionally, the hat algebra is a first-order approximation and so is more accurate for smaller changes as opposed to larger ones. Constant returns to scale, as mandated by the Cobb-Douglas nature of the production functions, may also not necessarily hold.

A third model is one in which there is imperfect labor mobility. Take L_U to represent the total labor supply in the United States, and w_U to represent the wage in the entire United states, which may differ from w. μ is a parameter restricting mobility; if there is no labor mobility allowed, $\mu = 0$, while perfect labor mobility is equivalent to $\mu = \infty$. η is a constant representing the share of the total population that is in the county in the scenario where $w = w_U$. Thus:

$L = L_U * \eta * (\frac{w}{w_U})^{\mu}$

As such:

$\widehat{L} = \widehat{L_U} + \widehat{\eta} + \mu(\widehat{w} - \widehat{w_U})$

Assuming that L_U and w_U are constant due to it being a small open economy, that is equivalent to:

$\widehat{L} = \widehat{\eta} + \mu * \widehat{w}$

In practice, I assume that there are no shocks to county attractiveness, and so set $\widehat{\eta}$ to be equal to zero. Since w and r are not defined with respect to the labor pool, I can substitute in the derivation of $\widehat{w}$ above and have it still apply, to get that:

$\frac{\widehat{L}}{\widehat{a_N}} = \frac{\widehat{\eta}}{\widehat{a_N}} + \mu(\frac{\theta-1}{\theta-\gamma})$

So while land remains the same, since it is still immobile,

$$\lambda_{L,A}\widehat{A} + \lambda_{L,N}\widehat{N} = \widehat{L} - \lambda_{L,A}\widehat{x_{L,A}} - \lambda_{L,N}\widehat{x_{L,N}}$$

This means that:

$$\frac{\widehat{A}}{\widehat{a_N}} = \frac{\theta(\lambda_{T,N}-\lambda_{L,N})-\lambda_{T,N}}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} + \frac{\lambda_{T,N}*\widehat{L}}{\widehat{a_N}(\lambda_{L,A}-\lambda_{T,A})}$$

$$\frac{\widehat{N}}{\widehat{a_N}} = 1 + \frac{\gamma(\lambda_{L,A}-\lambda_{T,A})+\lambda_{T,A}}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} - \frac{\lambda_{T,A}*\widehat{L}}{\widehat{a_N}(\lambda_{L,A}-\lambda_{T,A})}$$

I can then use this to get the effects of a shock to TFP in the non-agricultural sector on each sector's factor usage:

$$\frac{\widehat{L_A}}{\widehat{a_N}} = \frac{-\lambda_{L,N}+\lambda_{T,N}*\mu(\theta-1)}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} < 0$$

$$\frac{\widehat{L_N}}{\widehat{a_N}} = \frac{\lambda_{L,A}-\lambda_{T,A}*\mu(\theta-1)}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} > 0$$

$$\frac{\widehat{T_A}}{\widehat{a_N}} = \frac{\lambda_{T,N}(-1+\mu(\theta-1))}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} < 0$$

$$\frac{\widehat{T_N}}{\widehat{a_N}} = \frac{\lambda_{T,A}(1-\mu(\theta-1))}{(\theta-\gamma)(\lambda_{L,A}-\lambda_{T,A})} > 0$$

Consequently, adding in imperfect labor mobility does not affect the overall conclusions of the model in terms of sign, although the precise values of the effects are somewhat different depending on the value of μ. While perfect labor mobility may generate different predictions, there is clearly not perfect labor mobility between counties because of the costs of moving and making new social ties in a different community.

A corollary of this is that, in a world with only total factor productivity shocks, an increase in the labor observed in the non-agricultural sector must be due to a positive TFP shock for the non-agricultural sector. This is necessary to connect the theory, which deals with an unobserved parameter of TFP, to the empirics, which can only measure labor shocks because there is not sufficient data on TFP for the time period and geography of interest.

If I were instead to impose a shock to η and keep w and r constant, this would model an amenity shock for mobile labor instead of a TFP shock to the non-agricultural sector. In this case, I get the following results:

$$\frac{\widehat{L_A}}{\widehat{\eta}} = \frac{\widehat{T_A}}{\widehat{\eta}} = \frac{\lambda_{T,N}}{\lambda_{L,A}-\lambda_{T,A}}$$

$$\frac{\widehat{L_N}}{\widehat{\eta}} = \frac{\widehat{T_N}}{\widehat{\eta}} = \frac{-\lambda_{T,A}}{\lambda_{L,A}-\lambda_{T,A}}$$

In this model, the change in each factor must go in the same direction within a sector, but what direction that is depends on the values of $\lambda_{L,A}$ and $\lambda_{T,A}$. As such, shocks to amenities do not produce the same theoretical predictions as shocks to total factor productivity in the non-agricultural sector.

However, there are many complications of the setting that this model does not account for. Not all goods are tradable, and so they may have local prices that differ from national prices. It is also reasonable to model multiple types of workers, with some, like undocumented immigrants, being more likely to work in agriculture than others; these types are not evenly distributed regionally. Lastly, it does not take into consideration other forms of linkage between the agricultural and non-agricultural sectors, like one sector producing intermediate goods for the other's final goods.

2.2 Mechanisms

Land competition. A literature has examined how competition between agricultural and non-agricultural land uses impacts the price of agricultural land. The core theory of this literature is that the price of farmland is determined by a combination of its potential for agriculture and its potential for other uses, principally residential housing (Hanson et al., 2018; Huang et al., 2006; Chicoine, 1981; Livanis et al., 2006; Shi et al., 1997). If there is an increase in demand for residential housing, this raises the price of farmland. Hanson et al. (2018) finds that, in Illinois, the impact of distance to cities on farmland prices has been decreasing over time. Johnson et al. (2023) finds that the COVID pandemic has led the expansion of the urban fringe into agricultural land to be stronger in smaller metropolitan areas compared to larger ones. Plantinga et al. (2002) decomposes agricultural land values into components due to development potential and agricultural use, finding significant regional heterogeneity in the relative importance of the two. Outside the United States, Bren d'Amour et al. (2017) predicts a loss of cropland, particularly highly productive cropland, in the developing world because of continuing urbanization, and suggests that this may lead to intensified production on current cropland and the development of new cropland.

This has also attracted attention as a driver of decreased farm numbers from journalists and government officials (Bowler, 2025; Cline, 2024; Nickerson et al., 2012; Delta Stewardship Council, 2026). A common version of this narrative blames "urban sprawl" for hurting agricultural production, typically emphasizing land used for housing workers who have moved into growing industries. Concerns about this are a partial motivation for the development of urban growth boundary policies, such as "greenbelts" and "rural buffers", that keep a perimeter of land outside a city restricted to agriculture, nature conversation, and similar uses (Handy, 2025). In an empirical test of these ideas, Islam et al. (2024) confirms that agricultural land is being converted to non-agricultural uses in the Midwest, and implies that this is due to growing urban areas.

In the theoretical model, this is equivalent to allowing mobility of land across sectors but not across regions. Consequently, this mechanism is not in play in a scenario where there is either no land mobility across sectors or no usage of land for non-agricultural activities.

Labor competition. Farmers and non-agricultural employers must compete for laborers, as indicated in the theoretical model with there being a fixed regional pool of labor. However, this conclusion may be changed if there is total labor mobility, since in that case wage is exogenous.

Prior literature has established the importance of labor market conditions for agricultural production, primarily with a focus on migration. For instance, Lin and Peri (2025) shows that Japanese internment led to decreased farm values and fewer productivity improvements by removing a population of skilled agricultural workers. In Hornbeck and Naidu (2014), a flood led to out-migration and thus increased capital intensity of agricultural production. Clemens et al. (2018) similarly found that the end of the bracero program led to an increase in mechanized harvesting, while crops that could not benefit from mechanized harvesting experienced a drop in production. Ibrahim (2025) is the most similar to this paper, as it examines what happens to agriculture locally when a mine appears as an alternative source of employment, finding that agricultural production decreases due specifically to the labor mechanism rather than pollution.

Water. Similarly, farmers and non-farmers compete for water, which is a constraining resource in certain parts of the country (Hornbeck and Keskin, 2014). While I only presented the theoretical model for land and labor, the results would hold if labor and water are instead treated as the factors of competition.

Government. Local governments can choose to prioritize different sectors of the economy. If non-farm sectors are doing better, this may lead to a transfer of local government support towards them away from farms. It also may lead to increased property tax bills, as Todd (2022) attests to for the Santa Clara Valley.

Pollution. Non-agricultural land uses can cause pollution, which may harm the productivity of agricultural land and lead to a shift away from agriculture. Fitzgerald et al. (2020) finds energy to be competing with agriculture in North Dakota, leading to dynamic adjustments. Hoy et al. (2018) finds potential consolidation of farms in areas affected by shale gas drilling, but otherwise no effects. Weber and Hitaj (2015) attributes a small increase in farm values in parts of Pennsylvania and Texas to shale gas drilling, suggesting that it is due to ownership of mineral rights. The theoretical model does not account for this potential mechanism.

Local demand. An increased local market may lead to increased demand. This should be particularly true for goods that benefit from short transport chains (Volpe et al., 2013). This is not included in the theoretical model, which ignores trade barriers and trade costs.

Agglomeration. Cross-sectoral agglomeration effects may also lead to increased agricultural production (Jacobs, 1969). These effects would be stronger if the growing sectors in a region are ones that have knowledge that transfer well to agriculture. For instance,

if there is a regional concentration of the GPS industry due to growth in another sector, that could in theory lead to increased adoption of GPS technology for agricultural purposes in that region, raising productivity. In the theoretical section, this would correspond to shocks to a_N also affecting a_A in the same direction.

These mechanisms lead to an investigation of how the structural transformation of agriculture during this time period includes other types of changes within agriculture as well as sectoral reallocation.

This second investigation is informed by the body of economic research examining the impacts of changes to agricultural production conditions. Moscona and Sastry (2023) provides evidence that crops being more affected by higher temperatures in the United States causes there to be increased innovation affecting these crops in the United States. Moscona (2025) finds similar results for crops that were more affected by the Dust Bowl, which also had major impacts on migration, agricultural land values, and the intergenerational likelihood of remaining in farming (Gutmann et al., 2016; Hornbeck, 2012; Liu, 2025; Sichko, 2024, 2025). Boll weevil infestation decreased the prevalence of fixed-rent tenancy, lowered farm wages, decreased cotton production, led to lower land values, and affected population patterns (Ager et al., 2017; Lange et al., 2009). The 1874 Rocky Mountain locust plague led to both out-migration and a decline in total farm acres (Gillen et al., 2025).

3 Data

3.1 Sources

United States Census of Agriculture. The Census of Agriculture is a comprehensive survey of American farms, forests, and ranches, that reports its results at the level of the county. In past decades, it has been conducted approximately every five years. I use the digitized version of the data from Haines et al. (2014), covering the 1978, 1982, 1987, 1992, 1997, 2002, 2007, and 2012 censuses. Data from previous years is available, but has not been used because it differs significantly in what crops it reports. Data from 2017 and 2022 is also available but not used because of a lack of County Business Patterns data.

County Business Patterns. The County Business Patterns are a yearly count of employment and other economic variables in each county of the United States, compiled from administrative data by the U.S. Census Bureau. I use a version of the data from Eckert et al. (2020), which constructs a consistent panel of counties with reported values based on 2012 NAICS industry codes. It omits most employment in direct agricultural production, as well as most kinds of government employment and some employment in households (nannies, housekeepers, cleaners, etc.). The patterns of suppression used for the data change significantly after 2016, motivating the restriction of this project to go only until 2012.

GIS Data. Maps of counties in the United States are based on 1970 borders, as reported in the 2000 Tiger/Line+ version in Schroeder et al. (2025).

Climate Data. Data on precipitation comes from NOAA's NClimDiv dataset (Vose et al., 2014). Data on drought comes from the Palmer Drought Severity Index (Kassinger, 2020). These both cover the entire United States for the entire time period of interest at the county-by-month level, which I aggregate to a yearly basis.

Supplementary Data Sources. I also get census data from IPUMS, primarily to use in calculating unemployment rates (Ruggles et al., 2025) and as an alternate measure of the labor force. I primarily use county-level tables here. To analyze county-level population and employment information on a yearly basis, I also supplement the County Business Patterns data with intercensal county-level population data, divided by age, as compiled by the National Cancer Institute's SEER program (National Cancer Institute, 2026). This covers the time period 1969-2024. Furthermore, I use the Bureau of Labor Statistics's Local Area Unemployment Statistics, calculated at the county level from 1990 to the present, in order to gain additional coverage of unemployment rates (Bureau of Labor Statistics, 2026).

3.2 Variables

Table 1 shows the many variables available to measure agricultural conditions in the United States over time, most of which come from the Census of Agriculture. The main variable of interest is the estimated value of agricultural production in a county; the advantage is that it directly measures output, rather than inputs. However, it also has some disadvantages; most importantly, prices for agricultural commodities are not the same across all regions of the United States, and may in particular be increased in regions that produce little of their own agriculture due to greater transport prices. Additionally, it may be vulnerable to misreporting, and is variable because it reflects international commodity markets and weather conditions.

I also measure the inputs of different factors of production. Most important are cropland and all farmland, both of which indicate land usage — I focus on cropland because farmland used for animals may also be used for other purposes like recreation or nowadays solar panel farms.

I also examine various measures of the agricultural labor force, but there are significant concerns with relying on any one data source for this. The County Business Patterns data omits NAICS codes 111 and 112, which correspond to direct agricultural production, leaving in only agricultural support services (U.S. Census Bureau, 2025). The United States Census and ACS report agricultural employment combined with mining in some years and are more infrequent than the County Business patterns data (Schroeder et al., 2025). The Agricultural Census lists different statistics about the agricultural workforce in different years. As such, I have the wage bill for all years and the number of full-time operators and full-time hired laborers for 1992 on, but no measure of the number of hours worked

by contracted laborers. An additional concern is that reporting on the agricultural labor force might not be accurate, however, because a large share of the American agricultural workforce is undocumented (Charlton, 2024).

Table 1: Agricultural Variables

Variable Name	**Description**
Cropland	Total acres of cropland (including tree crops besides maple trees, sod, and greenhouses) in a county.
Farmland	Total acres of farmland (cropland, pasture, and grazing land) in a county.
Irrigated Land	Total acres of irrigated land in a county (generally but not necessarily cropland).
Non-irrigated Land	Constructed as cropland - irrigated land, leading it to occasionally be negative.
Farms	Number of farms in a county, where a farm is defined as anywhere that normally would sell >$1000 in agricultural products.
Farm Acre Value	Value of land and buildings per acre, on average for farms in a county (not inflation adjusted). This measures demand for land.
Production Value	Market value of agricultural products (crops and animal products) sold in a county, total.
Crop Production Value	Market value of crops (including nursery products, etc.) sold in a county, total.
Production Value per Acre	Production Value / Farmland. This is a measure of land productivity.
Full-Time Operators and Hired Labor	All operators whose main occupation is listed as farming in the agricultural census, plus all hired workers working over 150 days of the year. Note that this omits contracted labor and part-time labor.
Hired Workers	All hired workers, whether full-time or part-time. Measured in workers, not hours.
Value of Hired and Contracted Labor	The amount paid by farms to all hired labor and contracted labor in a county, equivalent to a wage bill. Excludes operators.
Agricultural Workers, CBP	All workers in NAICS sector 1 (farming, hunting, etc.) except for NAICS 111 and 112, which comprise the core of agriculture.

Property Tax Paid	The amount of property tax total paid by farms in a county.
Production Value per Worker	Production Value / Number of Agricultural workers. This is a measure of labor productivity.
Cropland to Farmland Ratio	Cropland / Farmland. This measures whether agriculture is used for animals or plants.
Crop to Farm Production Value Ratio	Crop Production Value / Production Value. This also measures whether agriculture is used for animals or plants.
Machinery to Production Value Ratio	Value of machinery and equipment / Production Value. This is a measure of the usage of capital in agricultural production.
Fertilizer to Production Value Ratio	Value of fertilizer / Production Value. This also measures another kind of capital usage.
Net Return	The amount that a farm has in net income (subtracting off all the production costs from their revenue), summed over an entire county.
Profit Margin	Net return / Production Value. This measures how much profit people could earn off of an investment in agricultural production.
Net Losses	The number of farms in a county with net losses.
Net Loss Percentage	Net Losses / Farms. The percentage of farms in a county with net losses.

3.3 Construction

The subscript c represents a county, the subscript i represents an industry, the subscript p represents a crop, and the subscript t represents a year.

3.3.1 County Business Patterns

The data-cleaning process begins by dropping from the County Business Patterns data all agricultural industries, including ones with strong indirect links (Eckert et al., 2020). This means the removal not just of farms but also of inputs like fertilizer and farm equipment manufacturing and outputs that are middlemen rather than direct-to-consumer sales; grocery stores are included, but grocery wholesalers are not, for instance, since I assume that grocery stores are located near consumers but grocery wholesalers may be located near producers. I also remove mining, because a body of literature has found mixed effects of mining on agriculture due to both being land-intensive industries, as well as pollution from mining (Hoy et al., 2018; Weber and Hitaj, 2015; Aragón and Rud, 2013). A full list of removed NAICS codes are available in Table 70.

I then calculate how much of each county's employment, ignoring the excluded industries, is represented by a given industry. This data is used to generate a file listing employment numbers by industry by county (henceforth $\eta_{c,i,t}$) for each year between 1978 and 2012 in which there is an agricultural census.

$\eta_{i,t} = \sum_c \eta_{c,i,t}$, $\eta_{c,t} = \sum_i \eta_{c,i,t}$

$\psi_{c,i,t} = \frac{\eta_{c,i,t}}{\eta_{c,t}}$; this represents the share of a county's population employed in any eligible industry that is employed in that specific industry.

The growth $\gamma_{i,t=\alpha,t=\beta} = \frac{\eta_{i,t=\beta} - \eta_{i,t=\alpha}}{\eta_{i,t=\alpha}}$ represents the proportional change between a later census year beta and a basis census year alpha (typically 1978) in a given industry. Similarly, $\xi_{c,t=\beta,t=\alpha} = \frac{\eta_{c,t=\beta} - \eta c,t=\alpha}{\eta_{c,t=\alpha}}$ represents the change in a county's employment between a later census year and a basis census year.

3.3.2 Census of Agriculture

$\chi_{c,p,t}$ represents the number of acres of a given agricultural variable (crop or county-level statistic) in a given county in a given year. I combine some crops that are listed separately but are practically identical, such as sorghum grown for hay and sorghum grown for silage, but do not combine others, such as leaving dry and green lima beans as separate categories because they are grown in different areas of the country. I exclude sub-varieties of crops as well as ones that are inconsistently reported between years and classifications of unspecified crops. Inherently, this involves judgement calls. For some crops that are grown only in very small numbers in a given state, they are reported as having 0 acres grown in a given county, even when the true number is greater than zero, for reasons of privacy. I treat these just the same as any other zero cell, but this introduces some error into the dataset.

For all parts of this paper, only data from the 48 continental states are included, meaning that the District of Columbia, Alaska, and Hawai'i are excluded. I also exclude counties that are not listed in both the Census of Agriculture and the County Business Patterns panel; for the most part, this means excluding Virginia's independent cities, as well as a handful of counties that experienced border changes over the time period. 43 counties present in the County Business Patterns are excluded, as are 4 counties present in the Census of Agriculture, leaving a total of 3065 counties in the data, although some other ones are missing for various variables.

Take $\chi_{c,p,t}$ to represent a variable of interest at the county-year level.

$\chi_{p,t} = \sum_c \chi_{c,p,t}$

$\delta_{p,t=\alpha} = 100 * (\frac{\chi_{p,t=\alpha} - \chi_{p,t=1978}}{\chi_{p,t=1978}})$ represents the percent growth of a crop from 1978, when this paper begins, to the census year in question.

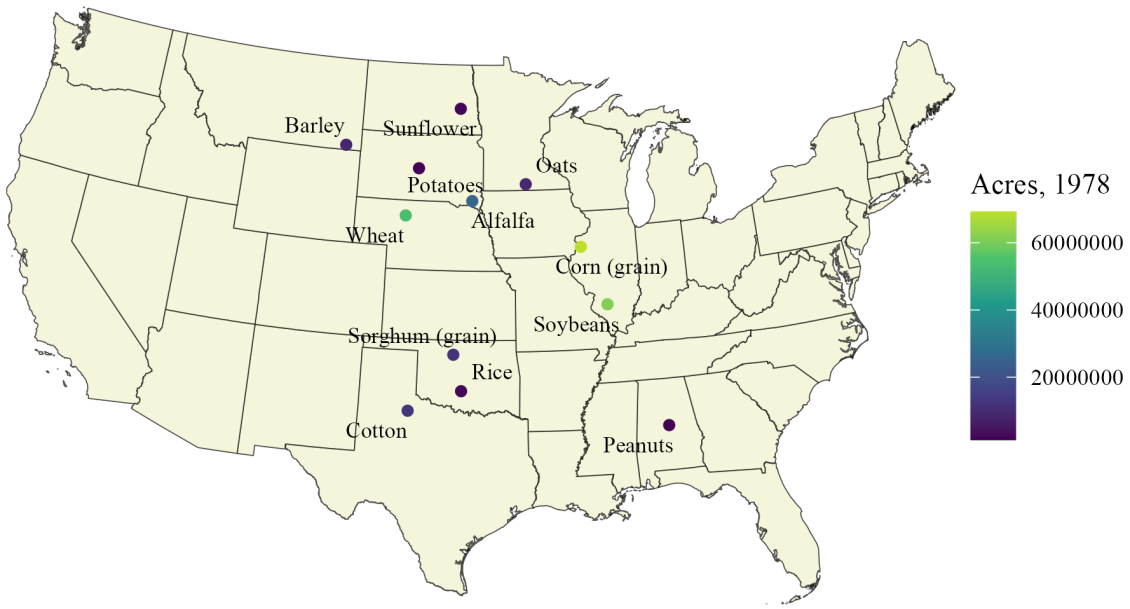
I winsorize most variables by replacing values below the 5th percentile with the 5th percentile and those above the 95th percentile with the 95th percentile, to reduce the impact of outliers.

3.4 Crop Locations

There is significant heterogeneity in terms of how much crops are concentrated in specific regions, as well as in total amounts of crop production. Figure 3 shows that apricots are highly concentrated in California's Central Valley in 1978, while wheat production is much more spread out, with the most wheat being produced in the Great Plains but a wheat industry also existing in most of the United States. I also calculate the centroid of each crop's production by weighting county-level centroids by acres of each crop in 1978, as shown for widely grown crops in Figure 1, with full results in Figure 4. This shows that crops with more total acres tend to have centroids somewhat more towards the middle of the United States, although numerically the crops are spread out all over the United States excepting the Northeast. Note that crops are not evenly distributed around the centroid, so some centroids are not near the county that grows the most acres of the crop; for instance, rice is mostly grown in California and a region around the Mississippi River, rather than in the rice centroid.

Figure 5 shows that growth in the employment variable from 1978 to 2012, after winsorization, is concentrated in metro areas, particularly ones in the "Sun Belt" region of the South and Southwest. By contrast, employment growth is low in the Midwest generally. Figure 6 shows that production value growth is highest in the Great Plains, California, and pockets of the South, while being lowest in Appalachia and the Northeast. Figure 7 shows that cropland growth is similarly highest in the Great Plains generally, but lowest in parts of the Deep South typically.

Figure 1: Centroids of Crop Production in the United States



Notes: Centroids are calculated by weighting the centroids of counties by the acres of each crop produced in the county in 1978.

4 Main Results

4.1 Regression Framework

Another way to examine the correlations is by performing basic regressions with some additional control variables, which should not be interpreted as causal. The main regression specification is as follows, where the carets denote winsorized variables and Y represents the dependent variable of interest, such as cropland and its growth:

$$\widehat{Y_{c,t=\alpha,t=\beta}} = \theta_s + \beta_1 * \widehat{EmployedPopGrowth}_{c,t=\alpha,t=\beta} + \beta_2 * \widehat{Y_{c,t=\alpha}} + \beta_3 * \widehat{EmployedPop}_{c,t=\alpha} + \beta_4 * \widehat{Y_{c,t=\alpha}}^2 + \beta_5 * \widehat{EmployedPop}_{c,t=\alpha}^2 + \epsilon_{c,t=\alpha,t=\beta}$$

$t = \alpha$ represents a starting or basis year, while $t = \beta$ represents an ending year. The main specification is the one that begins in 1978 and ends in 2012, to cover the longest time frame, but I report all possible specifications to examine temporal heterogeneity.

While this cannot be interpreted as causal, it rules out certain threats to causality by controlling for starting shares of variables and state fixed effects. The purpose of winsorizing all variables is to reduce the impact of outliers. Motivation for this comes from a suspicion that outliers are likely to represent reporting errors rather than true outliers, either because of mis-entering numbers somewhere in the process or agricultural census staff focused on one region doing a bad job. For the same reason, I weight using the log of the basis year employed population as a measure of variability. In robustness checks, I also use percentiles as an alternate strategy for reducing the impact of outliers, but report the winsorized values as the main specification because they are easier to interpret as a percentage point change.

All standard errors are Conley standard errors with a cutoff of 100 km unless otherwise specified (Conley, 1999).

I also run a simpler regression specification as follows, unweighted:

$$Y_{c,t=\alpha,t=\beta} = \beta_0 + \beta_1 * EmployedPopGrowth_{c,t=\alpha,t=\beta} + \epsilon_{c,t=\alpha,t=\beta}$$

The choice of basis year may matter because it affects whether the farm crisis of the 1980s shows up in the data (Barnett, 2000; Calomiris et al., 1986). In the 1970s, low interest rates and rising land prices led farmers to purchase more land, often taking out debt to do so. In the 1980s, however, farmers often had trouble paying these debts because of a combination of policies designed to fight inflation and low agricultural prices, leading to many bankruptcies as well as the collapse of agricultural banks and the consolidation of farmland. Glenna (2003) dissents from much of the other literature on this topic in instead emphasizing that the period of the farm crisis did not see anomalously large drops in farm numbers or certain other indicators of crisis, but instead represents a changing in the relationship between government and agriculture.

As a precursor to the actual regressions, I regress the winsorized growth variable used herein, of employed population, on winsorized growth in the total population of working age (20-64), coming from the intercensal population estimates, as well as the winsorized growth in the unemployment rate, coming from the Bureau of Labor Statistics (Bureau of Labor Statistics, 2026; National Cancer Institute, 2026). I include state fixed effects, but no other variables, and use the same Conley standard errors and weights as I use in the main regressions. Due to a lack of unemployment data before 1990, this only covers the years 1992, 1997, 2002, 2007, and 2012. Table 6 shows the results of this. There are typically significant negative coefficients on growth in the unemployment rate, indicating that the employed population expands when the unemployment rate goes down, and significant positive coefficients on growth in the total working age population, which indicates that the employed population expands when there is more net migration. This suggests that the correct theoretical model is one that allows imperfect labor mobility, but that the growth in the employed population is capturing both in-migration and changes in the unemployment rate.

4.2 Production Value

Table 2 shows the main regression specification as applied to agricultural production value. From 1978 to 2012, the model predicts that a county that grows ten percentage points more in employment will have 2.7 percentage points less agricultural production value growth. 11/28 combinations of years have a significant negative coefficient for the effect of employment growth on production value, with the others having insignificant and mostly negative effects. While this cannot be interpreted as causal — for instance, a decrease in production value could be causing increased employment growth instead — it is suggestive evidence for the hypothesis put forth in this paper. These negative effects are specifically concentrated when the ending year is 2007 or 2012, which indicates stronger effects towards the end of the time period.

This effect is robust to using percentiles without winsorization instead, as shown in Table 7.[2] The simpler regression specification with no controls, no winsorization, and no weighting produces generally similar results that are presented in Table 8, although 1978-1987 and 1987-1992 now have significant and positive coefficients. Interacting it with starting shares, as in Table 9, also does not qualitatively change the results.

It is not entirely robust, however, to restricting to just growth in manufacturing (industries starting with NAICS code 3), as shown in Table 10, which only has a significant negative coefficient in 6 of the 28 year combinations, although that includes the 1978 to 2012 time period. This may be due to manufacturing being only a minor component of the industry composition in some counties during this time period.

I also examine regional heterogeneities in the coefficients on production value. Ta-

[2]This version uses raw starting shares rather than winsorized starting shares as control variables.

Table 2: Winsorized Change in Agricultural Production Value, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.022	0.031	0.011	-0.017	-0.033	-0.110**	-0.268***
	(0.027)	(0.026)	(0.027)	(0.028)	(0.027)	(0.034)	(0.049)
N	3047	3028	3051	3051	3036	2994	3007
R2	0.191	0.161	0.128	0.146	0.132	0.142	0.236
Starting 1982							
Employment Growth		0.025	0.018	-0.016	-0.029	-0.093**	-0.232***
		(0.027)	(0.024)	(0.025)	(0.025)	(0.030)	(0.045)
N		3031	3057	3057	3040	3000	3012
R2		0.203	0.151	0.165	0.137	0.139	0.231
Starting 1987							
Employment Growth			0.002	-0.052+	-0.069*	-0.160***	-0.324***
			(0.028)	(0.030)	(0.029)	(0.035)	(0.060)
N			3033	3033	3021	2980	2991
R2			0.133	0.176	0.119	0.155	0.257
Starting 1992							
Employment Growth				0.012	-0.007	-0.112**	-0.292***
				(0.031)	(0.035)	(0.037)	(0.063)
N				3045	3033	2990	3002
R2				0.153	0.107	0.140	0.264
Starting 1997							
Employment Growth					0.060+	-0.056	-0.229***
					(0.036)	(0.036)	(0.069)
N					3038	2997	3010
R2					0.097	0.165	0.286
Starting 2002							
Employment Growth						-0.069	-0.264**
						(0.050)	(0.086)
N						2994	3005
R2						0.231	0.304
Starting 2007							
Employment Growth							0.050
							(0.069)
N							2977
R2							0.192

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

ble 11 shows that the Northeast follows a significantly different pattern, frequently with positive coefficients, while the Midwest, South, and West all have negative coefficients when examined over longer time frames. The pattern being observed in multiple regions is evidence against it being driven by a factor limited to a specific region, such as the increase in ethanol production in the Midwest towards the end of my time period (Le and Galvez-Soriano, 2026).

The overall story, in which economic booms are associated with decreases in agricultural production, is highly consistent with the theoretical model, both in the version with land mobility and in the simpler version without it.

4.3 Agricultural Factors of Production

4.3.1 Land

I begin with the analysis of the association between changes in the employed population and changes in farmland, as shown in Table 12. Here, there is a consistent negative association, significant in 21 of the 28 possible combinations of starting and ending years. Table 3 shows similar results, with significant negative coefficients in 14 of the 28 combinations. Ten percentage points more employment growth in a county is associated with 0.29 percentage points lower growth in cropland. This fits with the predictions of the theoretical model in which land can move between sectors, but is not concordant with the model in which land is always in the agricultural sector. Consequently, it shows the importance of taking into account land mobility in models of structural transformation.

By contrast, irrigated land has nearly exclusively insignificant effects in Table 13 which point in a variety of directions, although from 1978 to 2012 ten percentage points more growth in the employed population is associated with 2.47 percentage points less growth in the amount of irrigated land. Table 14 shows significant and negative effects for non-irrigated acres of cropland by themselves.[3]

The effects on cropland are also robust to the simpler regression specification in Table 15. However, they are again not robust to using manufacturing growth only, as shown in Table 16.

Table 17 shows regional heterogeneity in the effects on cropland, as calculated by estimating the model separately for each region as defined by the Census Bureau. The specification corresponds to that of Table 3, with the typical 1978 starting year. There is typically a negative association between normalized changes in cropland growth and employment across all regions, particularly over long time periods, but this is most likely to be statistically significant in the Midwest.

[3]A small number of observations per year have a negative amount of non-irrigated land because I calculate it as total cropland minus irrigated land. These have been kept in the data; they may represent misreporting or irrigated land that is not currently in use for crops.

Table 18 shows that the negative effects are consistent even when interacted with starting shares of cropland and employed population, which are not significant.

4.3.2 Labor

As noted above, there are a variety of ways to measure agricultural labor, which each have different strengths and disadvantages. From 1992 onward, the Agricultural Census lists the number of hired laborers working 150 or more days, which I add together with the number of operators mainly employed in farming to get a rough measure of full-time employment. Table 19 shows the results from this, which have a positive and significant coefficient in 6 of the 10 year combinations. From 1992 to 2012, ten percentage points more employment growth in a county is associated with 1.24 percentage points more full-time farm operators and hired laborers.

I also examine the wage bill, including contracted labor, in Table 20. This also has a positive effect in 18 of the 28 year combinations. I furthermore examine the growth in the number of people employed in agriculture based on County Business Patterns data in Table 21, although this excludes people directly involved in crop production in NAICS codes 111 and 112. Here, effects are mostly insignificant.

As robustness checks, I also find that counties with more employment growth have consistently higher growth rates for full-time operators, as shown in Table 22, as well as hired laborers by themselves, either full or part time, in Table 23. I additionally check the Census and ACS results for the employed population in agriculture and related industries, and also find a positive if less consistent effect in Table 24, with 8 year combinations having a positive and significant coefficient but one having a negative and significant coefficient.

While this is not causal, this indicates that there are additional dynamics in the labor market that are not captured by a simple theoretical model, since both versions predicted that the amount of labor in agriculture would decrease. For instance, this may be explained by there being highly separated agricultural and non-agricultural labor markets, labor mobility effects, or non-Cobb-Douglas production technology.

4.4 Crop-by-County Results

Here, I use a variation of the model where the dependent variable is the number of acres listed as being dedicated to a specific crop in the Census of Agriculture. Figure 2 demonstrates that the effects are largely driven by widely grown crops, principally legumes and grains. Crops that have less total acreage typically, such as vegetables and fruits, often have insignificant effects, and occasionally even have significant and positive effects (without correcting for multiple hypothesis testing). This pattern is by and large consistent no matter the choice of basis or ending years, although unsurprisingly it is clearer when assessed over a longer period of time. Table 25 shows acres for widely grown crops over time. Corn, soybeans, and wheat consistently have significant negative effects, where soybeans in

Table 3: Winsorized Change in Total Cropland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.035**	-0.016+	0.011	-0.005	-0.013*	-0.029***	-0.029***
	(0.011)	(0.009)	(0.009)	(0.007)	(0.006)	(0.005)	(0.005)
N	2939	2935	2973	2973	2967	2965	2965
R2	0.116	0.352	0.339	0.351	0.401	0.424	0.531
Starting 1982							
Employment Growth		-0.014	0.005	-0.004	-0.017*	-0.035***	-0.035***
		(0.011)	(0.011)	(0.009)	(0.007)	(0.006)	(0.006)
N		2963	3001	3001	2993	2992	2987
R2		0.295	0.299	0.317	0.395	0.408	0.511
Starting 1987							
Employment Growth			0.017	0.006	-0.015+	-0.040***	-0.043***
			(0.012)	(0.011)	(0.009)	(0.008)	(0.008)
N			3003	3003	2994	2995	2994
R2			0.168	0.197	0.286	0.303	0.425
Starting 1992							
Employment Growth				-0.006	-0.018	-0.061***	-0.060***
				(0.013)	(0.012)	(0.012)	(0.013)
N				2975	2968	2968	2966
R2				0.111	0.206	0.285	0.433
Starting 1997							
Employment Growth					-0.028+	-0.066***	-0.063**
					(0.016)	(0.017)	(0.020)
N					2985	2984	2980
R2					0.190	0.303	0.440
Starting 2002							
Employment Growth						-0.090***	-0.046+
						(0.022)	(0.025)
N						3042	3036
R2						0.229	0.382
Starting 2007							
Employment Growth							0.014
							(0.028)
N							3038
R2							0.248

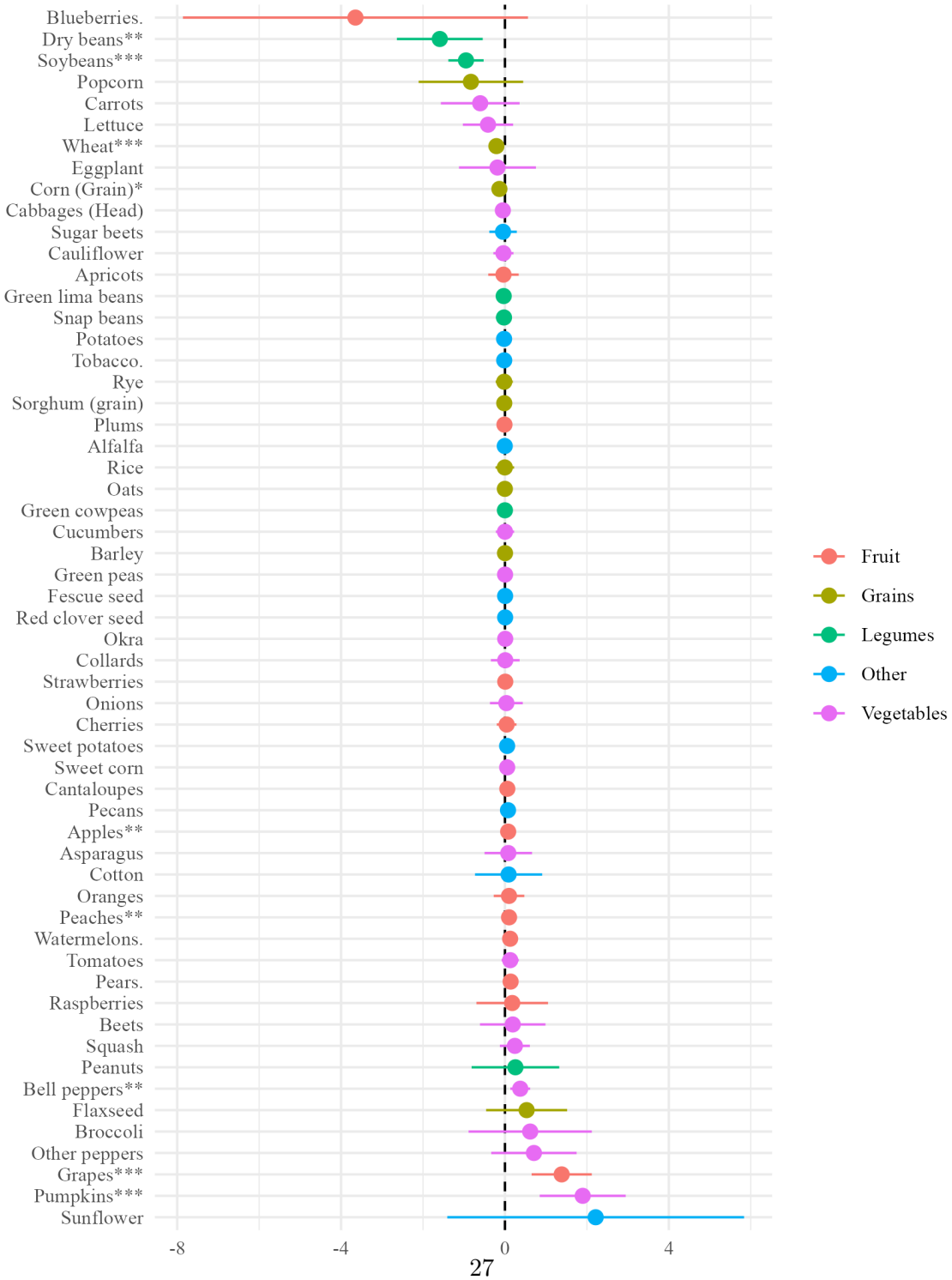
+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

particular are negatively affected, while there are insignificant effects on alfalfa and cotton in most year combinations.

Table 26 shows the results when acres are aggregated by type of crop instead. Here, there are consistent negative effects for grains, legumes, and other (which includes cotton, potatoes, and tobacco), consistently insignificant effects for vegetables, and a significant and positive effect for fruit in the longest time period only. Table 27 aggregates the results by production quintile instead (not estimating the first quintile because many of those crops are not grown at all in some years), also finding that significant effects are concentrated in the quintile of crops that are most widely grown.

There are various plausible mechanisms that could lead to this pattern. One consideration is that, for other crops, local demand effects may be reducing the change that would otherwise occur in employment; farmers' markets typically sell mostly fruits and vegetables rather than grains. There are also many more counties that grow them, increasing the power of the regression design used herein to detect effects. Furthermore, large farms in the United States disproportionately focus on these crops and may be more responsive to changes in input competition, or may face lower moving costs.

Figure 2: Effect of Change in Employed Pop. on Change in Land per Crop, 1978-2012



5 Causal Results

5.1 Causality

The question that this paper asks is not just whether economic booms, as proxied for by increases in the labor force, are correlated with decreases in crop production, as shown above, but whether economic booms are causative of decreases in crop production. As such, there are two questions that I want to answer: Could crop production alternately be causing the economic booms? Or is there some third factor that causes both of them?

Much research on structural transformation supports the idea of reverse causality; the Baumol effect and income effect mechanisms are both driven by changes in agriculture spilling over to other industries. However, this literature typically examines contexts where agriculture is a major sector of the economy, such as earlier American history or the developing world (Eckert and Peters, 2022; Dennis and İşcan, 2009). While agriculture is still undoubtedly a major industry in the United States towards the end of the 20th century, it is a smaller sector in comparison to what other researchers have examined. For instance, Emerick (2018) finds that an agricultural boom leads to a boom in employment in other industries, using an instrument of short-term rainfall. Emerick hypothesizes that this is due to the increased demand generated by the increased agricultural output. In the theoretical model, this would be equivalent to a shock to a_A affecting the non-agricultural sector, although the conclusions those papers reach rely on substantially different models that often focus on a closed economy.

There are a limited number of additional factors that could cause changes in both general economic conditions and local crop production. Natural disasters could easily reduce the amount of cropland in an area by lowering the land's natural productivity. However, literature on the employment effects of natural disasters often finds negative results, indicating that estimates that include areas affected by natural disasters may underestimate the effects of economic booms on cropland area (Boustan et al., 2020; Barattieri et al., 2023; Kukeli, 2025; Anttila-Hughes and Hsiang, 2013).

A decrease in the immigrant population of an area (whether due to a shock to immigration enforcement or attitudes) may also lead to a decrease in crop production (Clemens et al., 2018). However, fewer people almost certainly means fewer employed people, so it is not a major threat to causality. In the inverse, an increase in immigration to a specific area could result in more people being employed and a greater population, the main measure of an economic boom used herein, but would not lead to a decrease in crop production through channels other than ones related to an economic boom. Immigration may also respond to an economic boom rather than cause one.

Another scenario in which there might be a non-causal correlation between economic booms and decreased cropland would be that one region of the country happens to have

both a booming economy and some other factor that leads crop production to be lowered. For instance, if people are heavily moving to the Southwest because of cheap housing, but the Southwest also happens to be in a drought, this might induce a correlation. This is inevitably a much harder scenario to control for, but it also motivates the use of spatially clustered standard errors with a 100 kilometer cutoff, following Conley (1999), as well as the state fixed effects. Similarly, some type of infrastructure investment could generate both economic booms and affect agricultural production, but most infrastructure that benefits other sectors, such as better road and rail networks for shipping goods, should also benefit agriculture.

5.2 Shift-Share Instruments

One way to handle many of the concerns about causality in response to the descriptive results is a shift-share instrument, which is commonly used to study regional economic conditions (Borusyak et al., 2024, 2022; Adão et al., 2019; Broxterman and Larson, 2020). For a shift-share instrument to be interpreted as causal, changes in the instrumental variable must only affect changes in the dependent outcome variable through changes in the variable that it is an instrument for (Goldsmith-Pinkham et al., 2020).

The classic shift-share design for regional growth is the instrument from Bartik (1991), which is highly similar to that of Blanchard and Katz (1992). This instrument's shares are the pre-existing amounts of industries in a county, which it uses to predict regional growth by combining them with each industry's nationwide growth or the leave-one-out version thereof (Goldsmith-Pinkham et al., 2020; Borusyak et al., 2024). An advantage of this instrument is that it can remove concerns from reverse causality, which are relevant here, as well as removing amenity shocks (Breuer, 2022). In practice, this mostly reflects changes in tradable sectors such as automobile manufacturing, because non-tradable sectors tend to be more evenly dispersed.

A major framework for instrumental variables splits them into ones based on exogenous shares and ones based on exogenous shifts (Borusyak et al., 2024, 2022). I follow Goldsmith-Pinkham et al. (2020) and assume that the shares are exogenous: what industries a county starts with can only affect changes in its agricultural production through economic growth. This assumption is violated if we assume that certain industries (that are growing either more or less than normal) have industry-specific colocation benefits with agriculture that only appear after a certain time period, similar to the logic of Fiszbein (2022). This might also arise through high spatial correlation of industries; for instance, if new and highly productive crop varieties are developed that rely on rainy climates, but industries that rely on dry climates are disproportionately growing.

A slight variation on the Bartik (1991) instrument works well for the context of this project. I generate a matrix of county-by-industry employment combinations, grouping NAICS codes at the 4-digit level, for each year in which the agricultural census was taken

between 1978 and 2007. I also generate the growth rates of each industry for each combination of years from 1978 to 2012. I then multiply these two matrices to get a predicted county-level growth rate based on the national-level growth rate.

Mathematically, the first is equivalent to a matrix A, where rows represent industries and columns represent counties, and the value in each cell is $\phi_{c,i,t=\alpha}$. The second is a matrix B, consisting of a single column, where each row represents an industry and each cell is $\gamma_{i,t=\alpha,t=\beta} = \frac{\chi_{i,t=\beta} - \chi_{i,t=\alpha}}{\chi_{i,t=\alpha}}$. I winsorize the shifts in matrix B and then calculate $A^T * B$, where each cell is $\widehat{\xi_{c,t=\alpha,t=\beta}}$, approximating $\xi_{c,t=\alpha,t=\beta}$. Many papers also add in a control for the share of the industries in the non-shift-share specification for each county that are measured in the shift-share, but that is unnecessary here since I use the same industries to measure growth rates in both.

Percentile employment growth, as shown in Figure 5, is strongest in major metro areas and weakest in the Great Plains, followed up by New England. In comparison, the shift-share estimate shown in Figure 8 estimates less economic growth in the South, and more in the Great Plains and Mountain West comparatively, but the maximum values observed in the shift-share are noticeably lower.

5.3 Empirical Model

I use two-stage least squares to estimate the causal effect of a change in the total employed population of a county on a change in the amount of cropland in that county, or similar dependent variables. As in the OLS model, variables are consistently winsorized to remove outliers. The first stage is as follows:

$$\widehat{\xi_{c,t=\alpha,t=\beta}} = \theta_s + \beta_0 * \widehat{\overline{\xi_{c,t=\alpha,t=\beta}}} + \epsilon_{c,t=\alpha,t=\beta}$$

$$\widehat{Y_{c,t=\alpha,t=\beta}} = \theta_s + \beta_1 * \widehat{\xi_{c,t=\alpha,t=\beta}}' + \epsilon_{c,t=\alpha,t=\beta}$$

This is equivalent to getting a percentile estimate of county-level growth in the employed population from a starting year to an ending year due to county-level growth in the industries that are in that county in the starting year, and then regressing percentile change in cropland (or similar variables) on that variable. I include state fixed effects but not other controls because the portion of county-level growth represented by national-level growth trends should be purely exogenous from any county-level factors, like starting shares of cropland and employment.

I report the F-statistic of the first-stage as a test of whether an instrument is weak or not, as well as the coefficient β_0 from the first stage. A common heuristic is that the F-statistic should be > 10, but Lee et al. (2022) suggests a higher value of 104.7 as the threshold. I continue to use Conley standard errors. Adão et al. (2019) suggests an alternate method of calculating standard errors for shift-shares that takes into account different types of places having similar characteristics, but that method is not designed

to deal with transformations of the shift-share variable such as winsorization, and is also strongly affected by shifts being nearly collinear, which exists in my data as a result of the imputation processes used by Eckert et al. (2020).

Table 28 shows the first stage results (with the sample of counties with recorded production values). One finding of note here is that the F-stat is above 10 in all year combinations, and above 104.7 in most year combinations, particularly when examined over longer time periods.

A caveat is that the Bartik shift-share instrument only captures the growth in the number of people employed in a county that is driven by nationwide trends, and this might not correspond to the overall impacts of that growth (Goldsmith-Pinkham et al., 2020). For instance, this may be due to regional heterogeneity, where the instrument better reflects growth in some regions that have a different relationship than the overall relationship. A partial solution to this is to examine the Rotemberg weights, which show the misspecification elasticities of individual industries as if they were their own instrumental variable (Rotemberg, 1983).[4] Table 29 suggests that this is driven by primarily services as well as some manufacturing sectors such as footwear. Some Rotemberg weights are negative.

One concern here is that these sectors may be ones in many cases that respond to employed population growth rather than lead to it, not fitting with the theory of the Bartik instrument. For instance, a shock to a_A, agricultural total factor productivity, could lead to in-migration and so more people employed in these sectors to serve the new workers in the agricultural industry. For that reason, I test robustness to restricting only to manufacturing (NAICS codes beginning with 3) as a proxy for sectors that are traded, but do not use them as the main specification because it involves calculating them off less data. I also check robustness by aggregating to the 2-digit NAICS code level, in case of concerns about the imputation processes used to get the industry-level data (Eckert et al., 2020).

5.4 Results

Table 4 show the results for the main 2SLS specification for agricultural production value. 18/28 coefficients are negative and significant, typically mirroring the years in which the coefficients are negative and significant for the non-shift-share specification. However, in many cases, both the standard errors and the raw coefficients are larger. For instance, from 1978 to 2012 a percentage point of employment growth translates into 0.268

[4]Because of the structure of the bartik.weight package in R, I have to drop a few missing counties and industries. Rotemberg weights are not designed to handle transformed variables, so I report the Rotemberg weights with winsorized shifts (except in the NAICS 2-digit code specification, where there are not enough to winsorize) but without winsorizing the final shift-share. This should cause additional caution in interpreting these results.

percentage points of agricultural production value growth in the OLS, but 1.15 percentage points in the shift-share. One reason for this is that 2SLS is the local average treatment effect for employment growth affected by the instrument, which may differ from growth in general as measured through OLS's average treatment effect. The results are very similar to what I get when I aggregate NAICS codes to the 2-digit level instead of the 4-digit level, as shown in Table 30. If I use the 4-digit level but restrict to manufacturing (NAICS codes beginning with 3), I also get similar results to the main shift-share, albeit with slightly lower magnitudes, for production value in Table 31.

Table 32 shows the results for the main 2SLS specification for cropland. 18/28 coefficients are negative and significant, with the others all being insignificant and mostly being negative. Typically, the coefficients are more likely to be significant when measured over a longer span of time. By and large, this confirms the overall story of the descriptive results. Table 33 shows that the results are broadly similar when farmland instead of cropland is treated as the variable of interest.

For labor measures, the shift-share also has significant positive coefficients in 6 of 10 year combinations when the number of full-time operators and hired laborers is the variable of interest, as shown in Table 34, although 2007-2012 is significant and negative. However, the wage bill in Table 35 shifts to being significantly negative in 6 of the 28 year combinations, all of them ones ending in 2012, while also being significantly positive in two others. The census results in Table 36 also switch to pointing mostly in the negative direction, typically significantly, but also are significantly positive in a single specification and marginally positive in others. All in all, this paints an unclear picture of what the true effects of growth in the employed population on agricultural labor are.

Table 4: Winsorized Change in Agricultural Production Value, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.215+	-0.229**	-0.173	-0.241*	-0.317**	-0.602***	-1.150***
	(0.114)	(0.087)	(0.110)	(0.117)	(0.109)	(0.148)	(0.188)
N	3047	3028	3051	3051	3036	2994	3007
R2	0.899	0.897	0.890	0.893	0.893	0.891	0.901
Starting 1982							
Employment Growth		-0.100	-0.111	-0.284*	-0.218*	-0.490***	-0.936***
		(0.121)	(0.220)	(0.117)	(0.103)	(0.138)	(0.179)
N		3031	3057	3057	3040	3000	3012
R2		0.902	0.894	0.896	0.894	0.891	0.899
Starting 1987							
Employment Growth			0.092	-0.246	-0.123	-0.628***	-1.554***
			(0.350)	(0.221)	(0.127)	(0.157)	(0.231)
N			3033	3033	3021	2980	2991
R2			0.892	0.899	0.892	0.892	0.904
Starting 1992							
Employment Growth				-0.003	-0.168+	-0.727***	-1.640***
				(0.166)	(0.102)	(0.145)	(0.224)
N				3045	3033	2990	3002
R2				0.899	0.894	0.895	0.909
Starting 1997							
Employment Growth					-0.248*	-0.808***	-1.936***
					(0.121)	(0.140)	(0.242)
N					3038	2997	3010
R2					0.893	0.901	0.914
Starting 2002							
Employment Growth						-1.038**	-2.034***
						(0.332)	(0.555)
N						2994	3005
R2						0.907	0.914
Starting 2007							
Employment Growth							-0.445
							(0.413)
N							2977
R2							0.904

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

6 Mechanisms

This paper began by setting out a list of mechanisms by which economic booms could causally affect agricultural production. Of these, five correctly predicted the observed negative relationship between the two: competition for land, competition for water, competition for labor, local government support, and pollution. The next task is to disentangle the relative importance of these five mechanisms, and also to examine any possible countervailing effects.

One method for doing this is based on the observation that, where there is a preëxisting scarcity of a factor of production (land, water, labor), a change in its availability is likely to have a stronger effect on agricultural production. However, this relies on the assumption that there are diminishing marginal returns to increasing quantities of each factor, as found in the Cobb-Douglas functions of the theoretical model. I consider that to be a reasonable assumption, and so I divide the regressions into quintiles based on various measures of scarcity. I use the non-shift-share specification because the results have been shown to be broadly similar, and it appears to give somewhat more precision.

Some of the previous results are suggestive of certain mechanisms, although they are far from confirmed. In particular, there is an unclear and possibly positive effect on labor, which indicates that it is probably not the main driver of the results. By contrast, economic booms lead to a decrease in cropland.

As a more general test of mechanisms I examine the impacts of unemployment rate growth and working-age population growth as alternative independent variables, the same variables that I decomposed employment growth into in Table 6. Since people who are unemployed still typically use some land and water (for housing) and still vote (addressing local government support), stronger effects on working-age population growth indicates that those mechanisms are more likely to be responsible. By contrast, pollution and heat island effects mostly come from labor, so unemployed people are unlikely to contribute to those.

For total production value, Table 37 shows that growth in total working-age population has broadly similar effects as growth in the employed population. From 1992 to 2012, a county with ten percentage points more growth in that population measure is associated with 4.56 percentage points less growth in total production value. However, the coefficients in Table 38 are consistently statistically insignificant instead. Cropland tells a similar story for total working-age population, as shown in Table 39. But for growth in the unemployment rate, there is a negative and significant coefficient from 1992 to 2012 instead in Table 40, which is contrary to the theoretical predictions, since growth in the unemployment rate should lead to less employed population and thus more growth in cropland.

This is highly suggestive of competition for land, or possibly water, being a main mechanism by which more employed people in a county translates into decreased agricultural production. However, it is not fully confirmative, so I examine the mechanisms further.

6.1 Land

A reasonable measure of scarcity in land is the number of people per unit land. For the sake of using consistent data sources, I represent this as the employed population in the starting year from the County Business Patterns data per square meter of the county in 1970. The idea here is that if there are more people per unit land, adding additional people places more demands on the land than it would were those people added instead to a less populous county.

Because of the possibility of nonlinearities, I use a specification where I divide the counties into quintiles based on the employed population per unit area in 1978, thereby keeping them consistent across all years. Table 41 show the results for production value. Table 42 show the results for cropland. Higher quintiles indicate more people per unit area. For both variables, there are effects of mixed sign and frequently insignificant in the two quintiles with the smallest employed population per area, but more consistently negative and sometimes significant effects in the three quintiles with more employed population per area. Furthermore, these effects become larger in magnitude in the higher quintiles, particularly the fourth quintile.

Additionally, this story likely indicates that employment growth should cause an increase in the value of an acre of farmland, because there is increased demand for each unit of land, which has constant supply.[5] Table 43 shows that this holds true for 24/28 year combinations in the non-shift-share regression framework, particularly those with earlier starting years, although 3 also have a significant and negative coefficient. In the shift-share results of Table 44, by contrast, there are significant and negative coefficients on employment growth in 12 of the 28 year combinations, predominantly ones with later starting years. In the end, there are unclear results when it comes to the value of an acre of farmland.

These results further confirm the idea that the land competition mechanism is an important factor in explaining why economic booms cause decreased agricultural production.

6.2 Water

There are two different ways to think about where water is scarce. On one hand, what might matter is temporary water availability, so places that have a drought during the

[5] Alternately, if the farmland that costs more per acre gets disproportionately converted to other uses, it could cause a decrease in the value of an acre of farmland.

time frame studied will be the most responsive. However, this might also reflect droughts leading to more turnover among farms due to lower profits. On the other hand, I can also examine the typical amount of precipitation received in an area. However, this ignores that those regions have likely adapted to having lower average amounts of precipitation.

If there is a positive coefficient on the interaction term in the drought model, that means that, for every additional 1 unit on the Palmer Drought Severity Index, indicating it being in less drought, the coefficient on employment growth increases in raw value. This fits with the idea that drought might matter, because more drought leads to a greater negative effect of employment growth. Table 45 reports those specifications for cropland and production value, including only the 1978 row for reasons of space. There are no significant coefficients on drought in either variable.

I also interact average precipitation in the pre-1978 period and economic growth when looking at production value and cropland in Table 46. Here, there are mostly insignificant effects but occasional negative signs on the interaction term, while employment growth by itself often receives an insignificant coefficient. The negative interaction terms should be interpreted as there being a stronger negative effect of employment growth in areas that receive more precipitation, contra to the hypothesis that more precipitation leads to weaker effects. The changes to employment growth indicate high collinearity between these variables, since adding precipitation as a control does not substantially change the results. However, an important caution here is that precipitation numbers are highly regionally correlated.

When the two results are taken as a whole, it is highly unclear whether competition for water is one of the pathways through which economic booms lead to decreased agricultural production.

6.3 Labor

Compared to land and water, it is more difficult to measure a consistent component of labor scarcity over time. Papers that have analyzed the importance of the labor supply to agricultural productivity have often, as a result, focused on shocks caused by immigration policy, but in the context of this paper immigration may also have other effects (Lin and Peri, 2025; Clemens et al., 2018). One method is to use the share of the labor force that is in agriculture. The idea here is that there are two types of people, ones who prefer to work in agriculture and ones who prefer to work in non-agriculture, and they are evenly distributed. Thus, an expansion of job opportunities in non-agricultural sectors has more effect in highly agricultural areas, where more people who would prefer to work in non-agriculture are working in agriculture. However, this may not be true, because of migration and sector-specific skill investments.

Like with the measures of labor by themselves, one difficulty is that the County Business Patterns purposefully do not count direct agricultural laborers, so I instead use

the share in agriculture-related industries, those whose NAICS codes begin with 1. The assumption here is that this is highly correlated with the number of people directly working in agriculture. Alternative data sources would either not be at the county level or would not be accessible in every year.

Table 47 shows the results of an interaction of the share of the labor force in agriculture on cropland and total production value. Interaction effects on total production value are not significant, but there is a positive interaction when acres of cropland are used as the independent variable instead. This does not fit with the initial hypothesis, which would predict a negative coefficient on the interaction term. Instead, I can interpret these results as saying that places with a higher share of employment in agriculture-related industries have less negative effects of growth in the employed population.

Furthermore, the results on land competition can also be interpreted as an indication that labor competition is not the variable responsible. If more people are in an area, and hence there is more competition for land, that also means that there is a greater potential labor supply. As such, even if the percentage of people who could be willing to work an agricultural job is the same, the raw number of people in an area who could be employed on farms increases in that case. This is strengthened by rural counties typically specializing in agriculture and other natural resource industries in the American context. Figure 9 demonstrates that both the quintile measures of labor and land scarcity are anti-correlated, following this logic.

6.4 Pollution

Another channel through which an economic boom could decrease agricultural land is through pollution. Todd (2022) gives an anecdote of an apricot farmer complaining about toxic waste from Silicon Valley being put on his farm. Consequently, I can measure which industries pollute and which ones do not using the most updated version of the EPA's classification of industries by NAICS code for the Toxics Release Inventory, although this covers hazardous chemical waste more than air pollution, which might also matter (Environmental Protection Agency, 2026).[6] I can then include both types of growth separately as independent variables.

Table 48 shows that, when production value is the dependent variable, the results are noisy but generally point towards effects being driven by both kinds of growth. From 1978 to 2012, the longest time span, both have significant and negative coefficients of a similar magnitude. Looking at cropland instead, as in Table 49, paints a similar picture, although from 1978 to 2012 there is only a significant and negative coefficient on growth in less polluting industries.

[6]If a 4-digit NAICS code is split between industries that are classified and ones that aren't, I classify it as entirely being on the list.

6.5 Local Government

Todd (2022) highlights some of the ways in which local government was able to either help or hurt the orchards of the Santa Clara Valley as they experienced disruptions due to the rise of Silicon Valley. These include adjusting property taxes to be lower for agricultural businesses and changing zoning laws to favor agricultural enterprises (Division of Land Resource Protection, n.d.). If other sectors boom, local government would likely become less responsive to these agricultural concerns because of the need to focus on those sectors. However, this might also be enhanced by other components of an economic boom decreasing the amount of cropland, which then decreases the importance of agriculture to local politicians.

Since this narrative relies heavily on the impact of property taxes, a possible measure is whether economic booms lead to farms paying more property tax total. Table 50 reports more property tax being paid in 4 of the 15 year combinations in areas with more employment growth according to the OLS specification, but the shift-share design in Table 51 instead finds significant negative coefficients, making it unclear what the true effects of property tax are. As such, there is no particularly strong evidence that property tax is a major mechanism by which employment booms lead to decreased agricultural production, but I also cannot rule it out. Furthermore, it is not clear if any effect on property tax is a consequence of other things happening to agriculture in a county, such as them decreasing production, or is a cause of the changes that this paper documents in agriculture.

6.6 Heat Islands

Similarly, heat island effects from increased activities in other sectors are possible but unlikely to have such a high magnitude. Moscona and Sastry (2023) examines the impact of increased extreme heat on crops, given its significant impact on agricultural production, but relies on variation driven by shifts in climate patterns. Papers more directly on heat islands find mixed effects; Waffle et al. (2017) suggests that in some colder climates they may help agriculture, while Kabano et al. (2021) shows that they lead to a decreased growing season in tropical areas. Areas with heat islands also likely have more employed people per unit area, meaning that it is extremely difficult to disentangle them from land competition effects.

6.7 Local Demand

I can also examine whether more people working in non-agriculture increases local demand for agricultural products, although this would depend either on some of the growth in non-agricultural employment coming from migration effects or alternately on increased incomes overall in the county. The closest measure of local demand is the amount of agricultural products that are sold directly to consumers, such as in farmers' markets, although there are other ways for crops to be sold locally that are not incorporated into

this measure. This data is only available after 1992.

From 1992 to 2012, a county being 10 percentage points higher in employment growth is associated with it having 12.45 percentage points more direct-to-consumer sales, according to the OLS regressions in Table 52. However, it is only significant and positive in 4 of the 10 year combinations, while the shift-share version in Table 53 has no significant coefficients and typically negative values instead.This provides slight evidence of there being small local demand effects that are insufficient to cancel out the overall negative effects on agricultural production from increased employment. A very small proportion of total American agriculture is sold directly to consumers, representing .32% of all sales in 2012.

I am not able to measure agglomeration effects, but can conclude that they are not the dominant mechanism because they predict an opposite sign from my findings.

7 Within-Agriculture Adjustments

The previous sections have demonstrated that economic booms, as measured in employment growth, causally lead to decreased agricultural production at the county level. Furthermore, this is true even for certain specific crops. This section extends the analysis to consider adjustments between different types of agricultural production, both between different crops and between crops and animals.

7.1 Crops vs. Animal Products

One way to examine adjustments within agriculture is to look at how much of agricultural inputs and outputs are in crops compared to those in animal products. I first test the non-shift-share model on the ratio of crop product value to all agricultural products' value, as shown in Table 54. This shows that in 17 out of the 28 year combinations there is a positive and significant coefficient on employment growth, indicating that more employment growth in an area leads to a shift away from animal products towards crop products. From 1978 to 2012, this means that a 10 percentage point increase in employment growth in a county is associated with a .68 percentage point increase in growth in the crop to total production ratio. This could come from farmers switching from animal products to crops, animal farmers exiting, or crop farmers entering. The results of the shift-share in Table 55 are slightly noisier, with 10 out of the 28 year combinations having positive and significant coefficients and 2, 2002-2012 and 2007-2012, having negative and significant coefficients.

I also test the ratio of cropland to farmland in Table 56 and for the shift-share in Table 57. In the first, it has a significant negative coefficient in 8 out of the 28 year combinations and a positive coefficient in 1. In the second, the coefficient is positive and significant from 1978-2007 and 2002-2007, and otherwise insignificant. As such, this

presents evidence that this is not true (or at least is insufficiently large to show up in the data) at the land level, however, and instead represents a pattern of shifts where more of the value comes from crops but more of the land is actually in farms.

7.2 Technology

Agricultural production can differ in its technological content. I examine the value of machinery and equipment and the value of fertilizer, and focus on the ratio of its value to total agricultural product value in that county.

The ratio of machinery and equipment value to agricultural product value is the dependent variable in Table 58. In this model, the coefficient on normalized employment growth is positive and significant in 23/28 year combinations, indicating that an economic boom leads to a shift in production towards using more equipment. The results from the shift-share model in Table 59 are similar, albeit with the positive coefficients concentrated in more recent ending years. This may reflect agglomeration effects, where engineering and mechanical knowledge is transferred from other sectors, or alternately a shift in the optimal ratio of inputs due to an increase in the price of land.

The fertilizer ratio, however, shows a more complicated pattern; there are positive and significant coefficients in 3/28 year combinations in Table 60, all of which have 2012 as the ending year, as well as 1 negative and significant coefficient from 1978 to 1982. However, in the shift-share model in Table 61, 10/28 year combinations have a positive and significant coefficient.

This provides some evidence that economic booms lead to adjustments in the agricultural production function to use more machinery and possibly fertilizer. That may be because land has become scarcer. Thus, I can interpret this is confirmatory evidence that factors of production shift in importance as a result of an economic boom.

7.3 Nationwide Crop Results

An important additional question is whether this exclusively leads to crop movement, or also leads to a change in the total amount of a given crop produced. I investigate this question by calculating each crop's exposure to areas of employment growth and regressing that on the growth in the production of that crop.

I do that by generating a matrix A of all values of $\frac{\chi_{c,p,t=\alpha}}{\chi_{p,t=\alpha}}$, where rows correspond to counties and columns correspond to crops. I then also generate a matrix B of all values of $\xi_{c,t=\alpha,t=\beta}$, i.e. a column of growth rates between the starting and ending year for each county, winsorized. I further generate C as the transpose of A times B, where each cell is a measure of the growth in the employed population the area that produces a crop would experience were there no movement in where a crop is produced. I do a similar process with raw employment numbers instead of growth rates also to calculate starting employment

numbers for the area that produces a crop.

I use the following model to regress winsorized growth in the nationwide number of acres devoted to a specific crop on a winsorized version of the employment growth variable. Additionally, I weight by the log of the crop-weighted employment numbers in the starting year.

$$\widehat{Y_{p,t=\alpha,t=\beta}} = \beta_0 + \beta_1 * \widehat{C_{p,t=\alpha,t=\beta}} + \beta_2 * \widehat{Y_{p,t=\alpha}} + \beta_3 * \widehat{Y^2_{p,t=\alpha}} + \beta_4 * \widehat{EmployedPop}_{p,t=\alpha} + \beta_5 * \widehat{EmployedPop}^{\,2}_{p,t=\alpha} + \epsilon_{p,t=\alpha,t=\beta}$$

Table 5 shows the results from this. Most years have an insignificant effect, with various signs, but two are significant and negative while 2007-2012 is significant and positive. The longest time period, 1978-2012, produces an insignificant negative result, where 10 percentage points more employment growth that a crop is exposed to translates into 3.67 percentage points less nationwide growth in the cropland devoted to that crop. As such, results are inconclusive regarding whether employment growth in areas that grow a crop leads to a decrease in total production of that crop, or whether instead production is compensated for in other areas. In particular, because there are between 115 and 139 crops observed in any given combination of years, there are a low number of observations.

7.4 Profit

I also examine the financial situation of farms in counties that are experiencing economic booms. However, most variables in this section are missing from the 1978 and 1982 agricultural censuses, so my specification only begins in 1987. Table 62 shows a significant and negative coefficient on growth in the employed population when total net income (profit) of all farms in a county is taken as the dependent variable in 11 out of the 15 year combinations, similar to the shift-share results in Table 63. There is also a negative and significant coefficient on profit margin in Table 64, which I take to be the net agricultural income in a county divided by the total agricultural production value. The shift-share version, Table 65, has similar if somewhat noisier results.

Another measure of profitability is the percentage of farms that record net losses out of all total farms.[7] In the non-shift-share specification, 12 out of 15 coefficients are significant and positive, as shown in Table 66. However, that switches to significant coefficients that are both positive and negative in different years, with most being insignificant, when I examine the shift-share specification in Table 67.

I take this to be suggestive evidence that farms have lower profit margins in areas with more growth in the employed population, and this may be a mechanism that leads to lowered agricultural production if these farms then scale back operations or go out of business. However, there are other compatible narratives, such as a decrease in the number

[7]Note that this means that farms that do not fill out this question are treated as not having net losses.

Table 5: Winsorized Change in Total Cropland Nationwide by Employment Growth

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.146	-1.163	0.381	-0.111	-0.703*	-0.261	-0.367
	(0.884)	(0.877)	(0.539)	(0.372)	(0.311)	(0.225)	(0.289)
N	115	115	115	115	115	115	115
R2	0.098	0.217	0.094	0.103	0.124	0.096	0.054
Starting 1982							
Employment Growth		-0.762	-0.422	-0.103	-0.402	0.115	-0.308
		(1.069)	(1.234)	(0.473)	(0.448)	(0.309)	(0.510)
N		118	118	118	118	118	118
R2		0.120	0.069	0.126	0.085	0.087	0.052
Starting 1987							
Employment Growth			-0.903	-0.071	-0.945*	-0.185	-0.651
			(0.705)	(0.418)	(0.464)	(0.389)	(0.583)
N			133	133	133	133	133
R2			0.076	0.085	0.057	0.038	0.020
Starting 1992							
Employment Growth				-0.243	-0.820	-0.550	-0.594
				(0.493)	(0.555)	(0.467)	(0.570)
N				137	137	137	137
R2				0.059	0.036	0.016	0.019
Starting 1997							
Employment Growth					-0.464	-0.220	0.187
					(1.125)	(0.638)	(0.961)
N					139	139	139
R2					0.042	0.024	0.008
Starting 2002							
Employment Growth						1.229	0.446
						(0.762)	(0.918)
N						136	136
R2						0.029	0.023
Starting 2007							
Employment Growth							2.387*
							(1.011)
N							136
R2							0.109

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

of farms aiming to make a profit in favor of farms whose owners have other jobs and so are more motivated by non-pecuniary benefits.

7.5 Ownership Type

The Agricultural Census lists farms and acres of farmland by the kind of ownership type. There are five classifications used: individual or family proprietorship, partnership, family-held corporation, non-family-held corporation, and other, which includes structures such as trusts. Corporations provide more liability protection for farmers than sole proprietorships or partnerships, which are essentially equivalent to having multiple proprietors (Kim and Mills, 2023). Typically, corporations represent larger farms, but political rhetoric often focuses on family farms and holds them up as the American agrarian ideal (Vogeler, 1981). Figure 10 shows that most acres are in individual or family proprietorship consistently over time, although the share is decreasing.

I examine the number of acres held in each ownership type as a variable of interest, using both the non-shift-share methods in Table 68 and the shift-share methods in Table 69. Effects are generally negative across ownership structures, with individual or family sole proprietorship having the most consistently significant negative effect in the shift-share, but with negative effects on partnerships and other structures in the OLS. The positive coefficient on non-family corporation acre growth from 1978 to 2012 is .253, larger in magnitude than the coefficients on many other types of structures, but it is not significant, because few counties record having such structures.

8 Conclusion

The central question of this paper is whether economic booms lead to decreased agricultural product value in a region, and, if so, whether that operates through competition for land, competition for labor, or both. My answer to the first question is yes, but with some caveats. This relationship between more growth in the employed population and less growth in production value is strong and consistent in a less plausibly causal specification. When I switch to a shift-share instrument based on Bartik (1991), I get broadly similar results. I show that there is also a decrease in farmland and cropland during economic booms, but find more inconclusive evidence about the agricultural labor force. This also holds true for the acres dedicated to some specific large crops, but not to less widely-grown crops. Thereby, I conclude that changes in land use are an important aspect of structural transformation, contrary to previous models used in the literature in which land cannot move out of the agricultural sector.

I also investigated what mechanisms are driving this effect. I found little to no evidence of it being driven by pollution, competition for water, or competition for labor, but cannot rule out those mechanisms. I found some evidence for it being driven by competition for land, with the effects being strongest in places with a higher density of

people per unit of land. These results are as a whole inconclusive, but may support the view in some of the prior literature that competition for land is a driver of changing agricultural patterns.

Additionally, I found evidence of shifts in production within the agricultural sector. Places experiencing an economic boom also see an increase in the share of agricultural product value that comes from crops as opposed to animal products. They also switch to using more machinery, and possibly other technological inputs. However, an examination as to whether there are nationwide shifts in what crops get grown as a result of this produced inconclusive results.

This concords strongly with narratives advanced by journalists and policymakers about how land competition between agricultural and non-agricultural uses, particularly in suburban regions of the United States, is hurting local agriculture (Bowler, 2025). However, this paper goes beyond those narratives in showing that this process happens at the scale of the entire country, and causes not just decreases in agricultural land but also adjustments within types of agriculture.

Furthermore, the results of this paper can inform our understanding of structural transformation to a certain extent. Firstly, it adds to the analysis of regional transformation in open economies using more general structural transformation mechanisms, such as competition for inputs that are used to their comparative advantage. Secondly, it highlights the role of the non-agricultural sector in driving these changes. Thirdly, it confirms the importance of land use in developing models of structural transformation, since a theoretical model with land used in both sectors produces results that much more closely match the empirics than one in which land can only be used in a single sector.

There are still many open questions as a result of this research. For one, it is not clear what effects this has on total agricultural production in the United States; declines in certain places could be compensated for by increases in other places. It is also not clear whether similar patterns occur in other countries, but there are reasons to think that the findings might be very different in some other geographies; developing countries often have a larger share of their population in agriculture and so effects led by the agricultural sector are likely more important, although there is still a role for other sectors to play (Foster and Rosenzweig, 2004; Emerick, 2018; Ibrahim, 2025). Many also do not possess the kinds of residential zoning codes found in the United States. The Agricultural Census also does not permit me to distinguish between farms on different kinds of land within a county, making it difficult to examine reallocation within counties, which might also be an important aspect of this phenomenon. I also do not break down the non-agricultural sector into manufacturing and services, which each have their own patterns of production.

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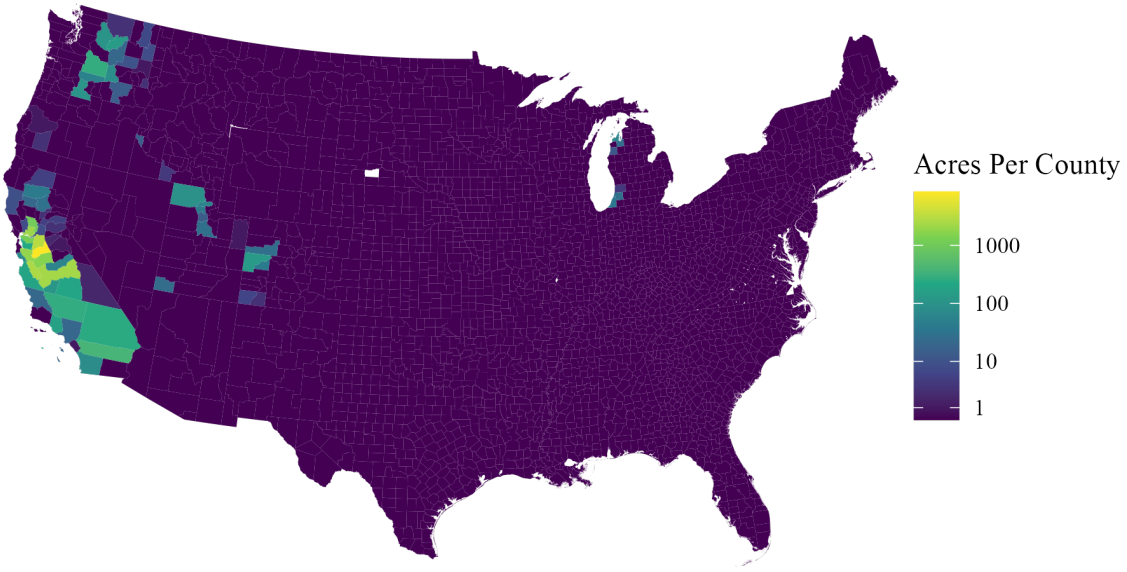
_ , Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, Alex Hayes, Lionel Henry, Jim Hester, Max Kuhn, Thomas Lin Pedersen, Evan Miller, Stephan Milton Bache, Kirill Müller, Jeroen Ooms, David Robinson, Dana Paige Seidel, Vitalie Spinu, Kohske Takahashi, Davis Vaughan, Claus Wilke, Kara Woo, and Hiroaki Yutani, "Welcome to the tidyverse," *Journal of Open Source Software*, 2019, *4* (43), 1686.

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9 Appendix A: Additional Figures and Tables

Figure 3: Acres Planted With Specified Crops, 1978

(a) Apricots



(b) Wheat

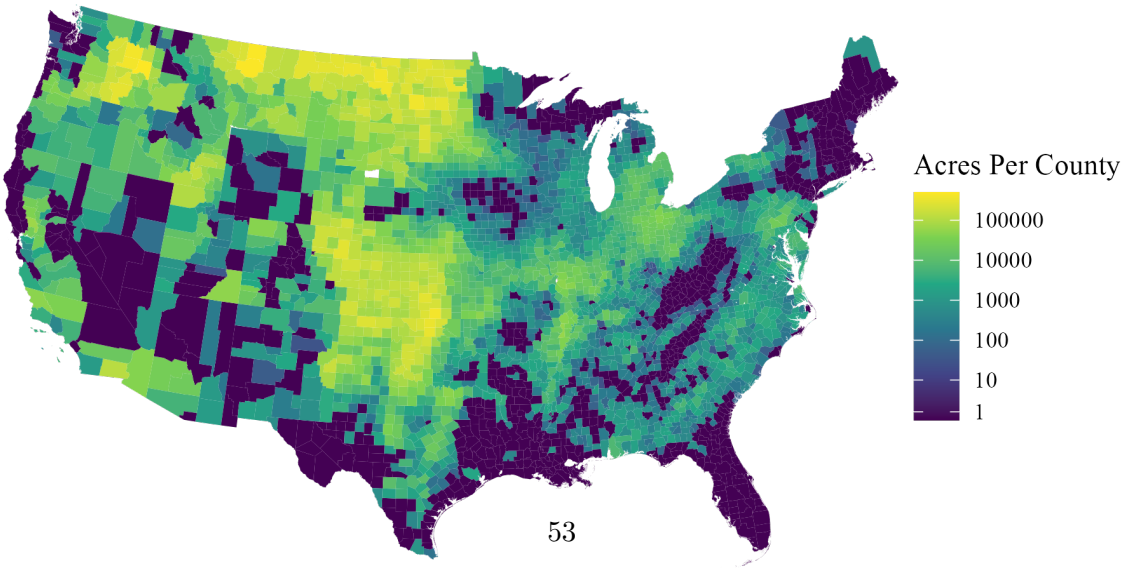


Figure 4: Centroids of Crop Production in the United States

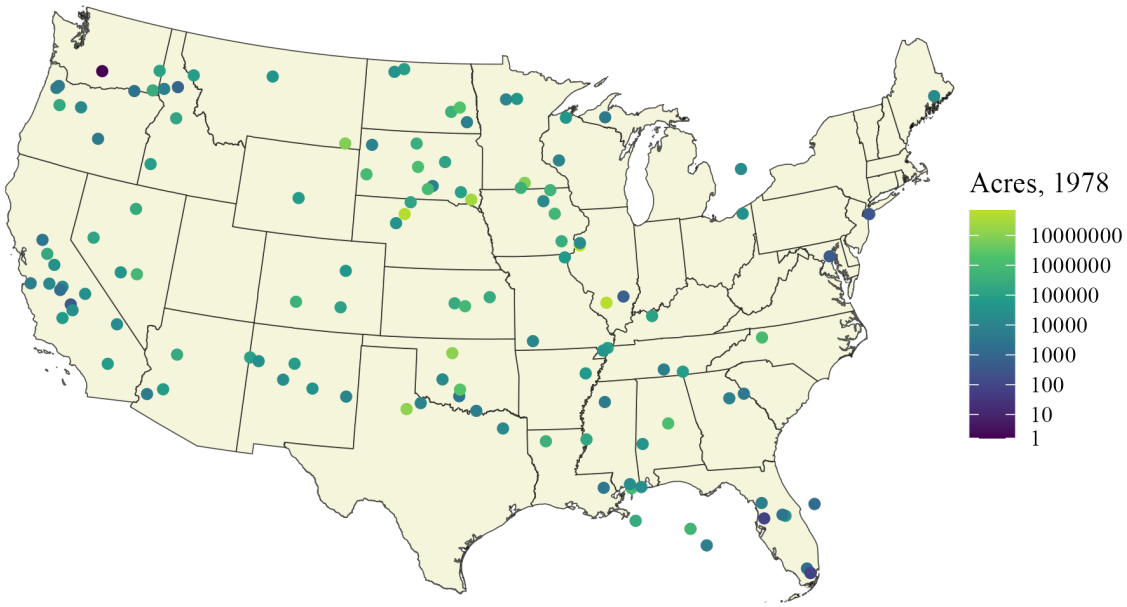


Figure 5: Winsorized Employment Growth, 1978 to 2012

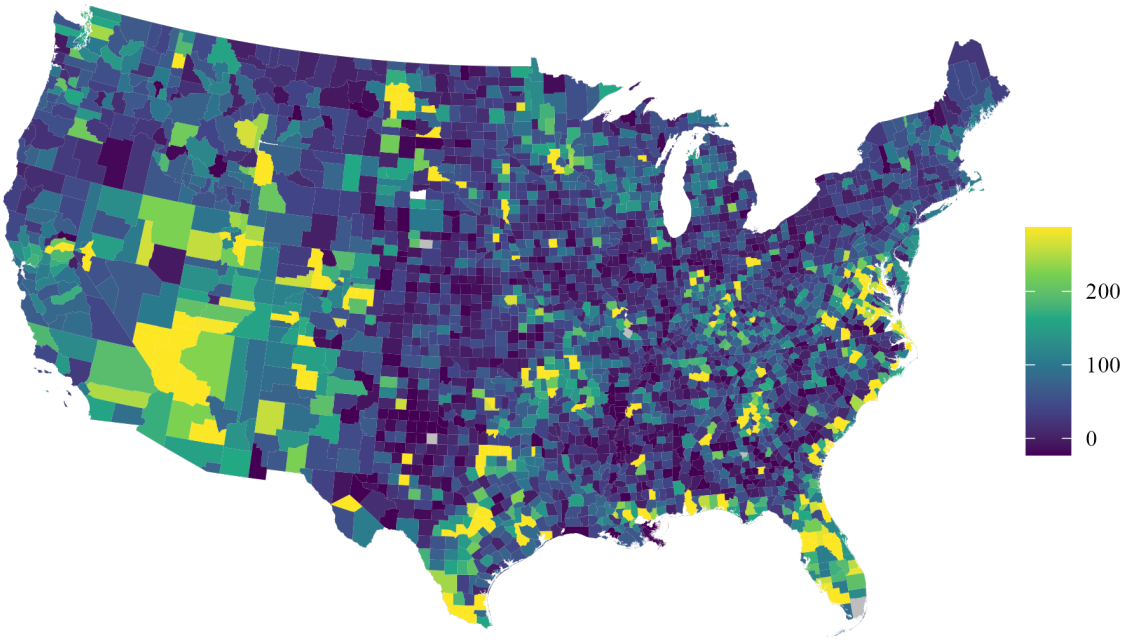


Figure 6: Winsorized Production Value Growth, 1978 to 2012

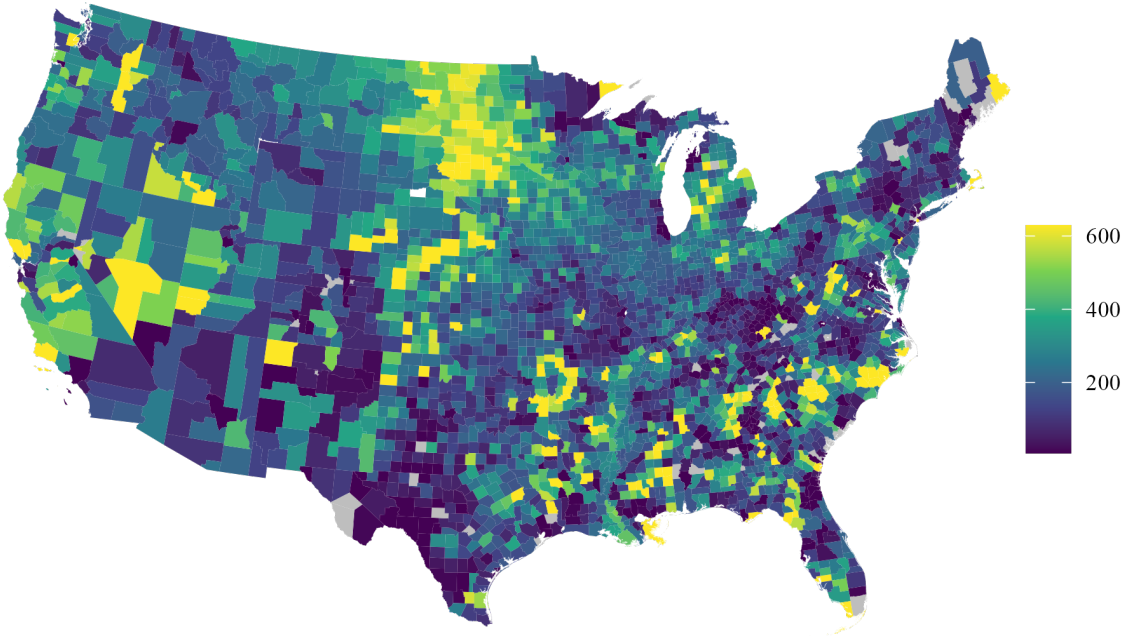


Figure 7: Winsorized Cropland Growth, 1978 to 2012

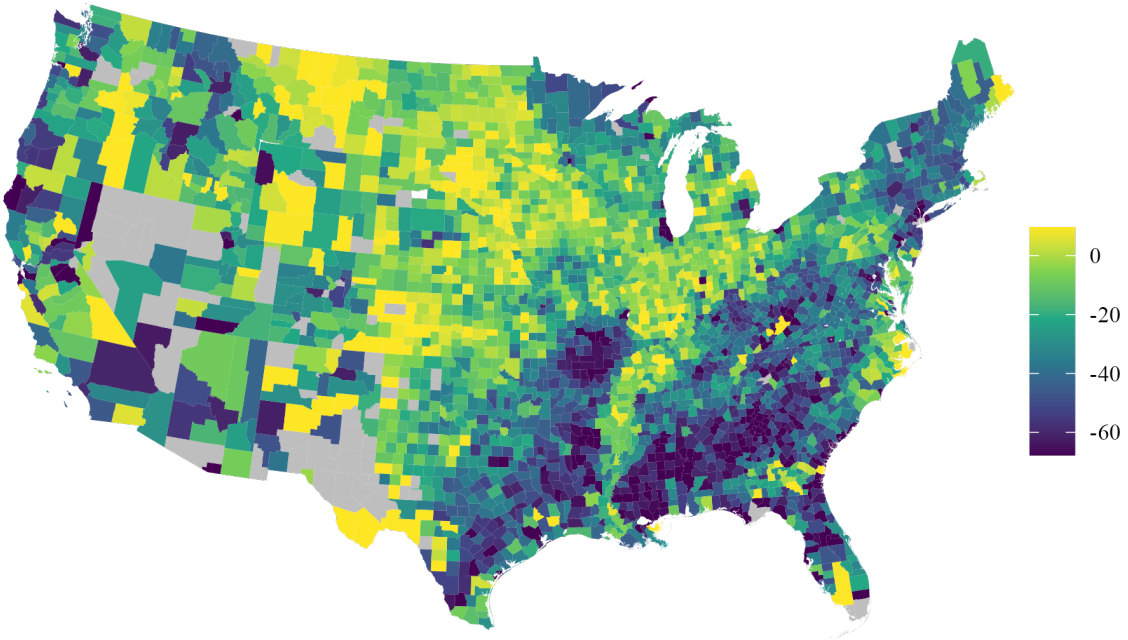


Table 6: Winsorized Change in Employed Population Growth by Unemployment and Population Growth

	1997	2002	2007	2012
Starting 1992				
Unemployment Rate Growth	-0.198***	-0.073*	-0.100*	-0.070*
	(0.031)	(0.032)	(0.042)	(0.031)
Population Growth	0.863***	1.129***	1.254***	1.079***
	(0.070)	(0.062)	(0.061)	(0.046)
N	1111	1110	1110	1110
R2	0.246	0.372	0.489	0.495
Starting 1997				
Unemployment Rate Growth		-0.004	-0.027	-0.015
		(0.011)	(0.023)	(0.016)
Population Growth		1.120***	1.186***	0.964***
		(0.059)	(0.057)	(0.047)
N		1110	1110	1110
R2		0.286	0.428	0.446
Starting 2002				
Unemployment Rate Growth			-0.173***	-0.106***
			(0.032)	(0.025)
Population Growth			1.035***	0.789***
			(0.066)	(0.052)
N			1108	1108
R2			0.261	0.312
Starting 2007				
Unemployment Rate Growth				-0.078***
				(0.015)
Population Growth				0.652***
				(0.072)
N				1110
R2				0.263

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Percentile Change in Total Production Value, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Percentile Employment Growth	-0.025	0.025	0.009	-0.027	-0.005	-0.066**	-0.119***
	(0.020)	(0.023)	(0.022)	(0.021)	(0.023)	(0.022)	(0.024)
N	3047	3028	3051	3051	3036	2994	3007
R2	0.204	0.182	0.147	0.154	0.138	0.174	0.242
Starting 1982							
Percentile Employment Growth		0.024	0.007	-0.021	-0.008	-0.069**	-0.111***
		(0.025)	(0.022)	(0.023)	(0.023)	(0.022)	(0.023)
N		3031	3057	3057	3040	3000	3012
R2		0.209	0.158	0.154	0.131	0.158	0.229
Starting 1987							
Percentile Employment Growth			-0.004	-0.051*	-0.035	-0.104***	-0.143***
			(0.021)	(0.021)	(0.022)	(0.021)	(0.023)
N			3033	3033	3021	2980	2991
R2			0.143	0.166	0.112	0.171	0.261
Starting 1992							
Percentile Employment Growth				0.005	0.001	-0.072***	-0.114***
				(0.019)	(0.024)	(0.021)	(0.020)
N				3045	3033	2990	3002
R2				0.151	0.109	0.156	0.257
Starting 1997							
Percentile Employment Growth					0.035	-0.031	-0.087***
					(0.021)	(0.019)	(0.021)
N					3038	2997	3010
R2					0.106	0.171	0.278
Starting 2002							
Percentile Employment Growth						-0.035+	-0.076***
						(0.019)	(0.020)
N						2994	3005
R2						0.243	0.306
Starting 2007							
Percentile Employment Growth							0.012
							(0.021)
N							2977
R2							0.200

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Change in Total Production Value, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Raw Employment Growth	-0.002	0.086**	0.032	-0.018	-0.002	-0.043*	-0.109***
	(0.004)	(0.032)	(0.028)	(0.020)	(0.016)	(0.022)	(0.028)
N	3047	3028	3051	3051	3036	2994	3007
R2	0.000	0.006	0.001	0.000	0.000	0.002	0.006
Starting 1982							
Raw Employment Growth		-0.024	0.052	-0.018	0.024	-0.020	-0.097**
		(0.032)	(0.052)	(0.017)	(0.035)	(0.043)	(0.035)
N		3031	3057	3057	3040	3000	3012
R2		0.000	0.002	0.000	0.000	0.000	0.004
Starting 1987							
Raw Employment Growth			0.082*	-0.035*	0.003	-0.063	-0.142**
			(0.037)	(0.017)	(0.019)	(0.039)	(0.045)
N			3033	3033	3021	2980	2991
R2			0.011	0.001	0.000	0.002	0.004
Starting 1992							
Raw Employment Growth				-0.015	-0.007	-0.112**	-0.175***
				(0.013)	(0.021)	(0.035)	(0.049)
N				3045	3033	2990	3002
R2				0.000	0.000	0.004	0.006
Starting 1997							
Raw Employment Growth					0.109	-0.165*	-0.167*
					(0.070)	(0.069)	(0.072)
N					3038	2997	3010
R2					0.003	0.004	0.003
Starting 2002							
Raw Employment Growth						-0.231+	-0.156*
						(0.124)	(0.078)
N						2994	3005
R2						0.005	0.002
Starting 2007							
Raw Employment Growth							-0.021
							(0.038)
N							2977
R2							0.000

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: Winsorized Change in Agricultural Production Value, County Level, 1978 to 2012

	(1)	(2)	(3)	(4)
Employment Growth	-0.26754***	-0.23470**	-0.22327***	-0.22085**
	(0.04924)	(0.07421)	(0.05411)	(0.07776)
Growth x 1978 Production Value		-0.00000		-0.00000
		(0.00000)		(0.00000)
Growth x 1978 Employed Pop.			-0.00000*	-0.00000
			(0.00000)	(0.00000)
Growth x 1978 Production Value x 1978 Employed Pop.				-0.00000
				(0.00000)
N	3007	3007	3007	3007
R2	0.236	0.236	0.237	0.238
Controls	✓	✓	✓	✓
State FE	✓	✓	✓	✓

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Figure 8: Winsorized Shift-Share of Employment Growth, 1978 to 2012

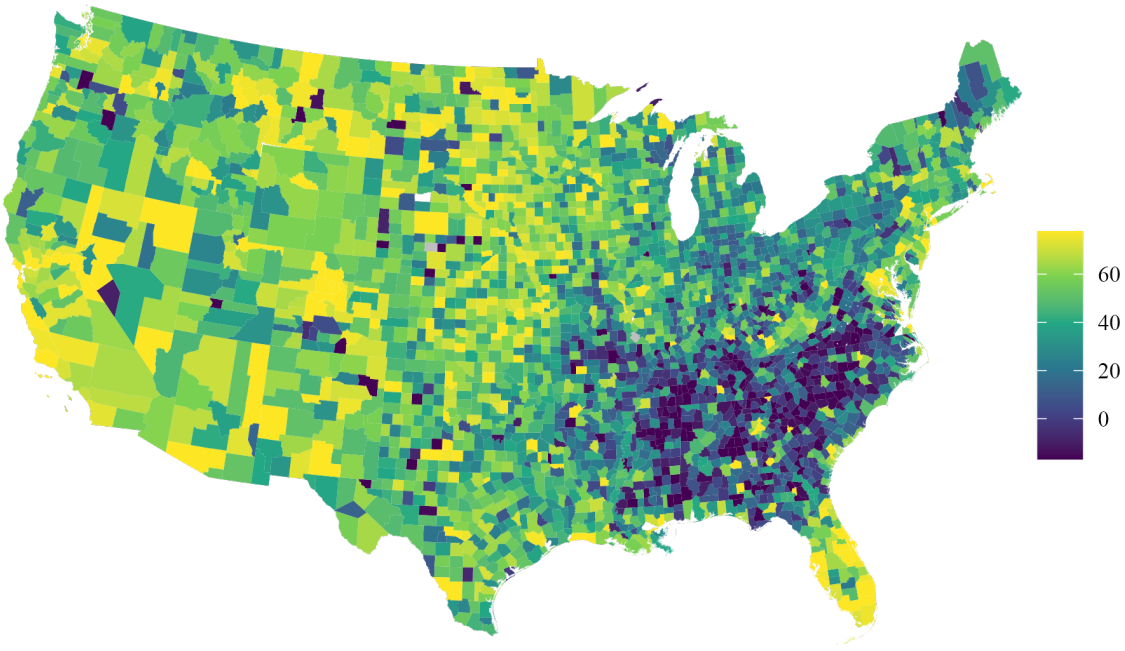


Table 10: Winsorized Change in Total Production Value, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Manufacturing Employment Growth	-0.002	-0.011	-0.016	-0.013	-0.027+	-0.030	-0.068*
	(0.009)	(0.009)	(0.010)	(0.013)	(0.016)	(0.022)	(0.034)
N	2919	2899	2947	2947	2888	2849	2847
R2	0.194	0.161	0.128	0.147	0.135	0.133	0.211
Starting 1982							
Manufacturing Employment Growth		-0.011	-0.017*	-0.017+	-0.017	-0.023	-0.045
		(0.009)	(0.008)	(0.010)	(0.014)	(0.016)	(0.028)
N		2880	2917	2916	2862	2824	2828
R2		0.203	0.153	0.168	0.141	0.133	0.209
Starting 1987							
Manufacturing Employment Growth			-0.006	0.004	0.002	-0.017	-0.041
			(0.007)	(0.009)	(0.013)	(0.016)	(0.027)
N			2899	2900	2854	2818	2815
R2			0.135	0.174	0.120	0.145	0.234
Starting 1992							
Manufacturing Employment Growth				-0.003	0.011	-0.006	-0.051+
				(0.008)	(0.014)	(0.016)	(0.029)
N				3028	2937	2897	2890
R2				0.152	0.108	0.134	0.250
Starting 1997							
Manufacturing Employment Growth					0.006	-0.005	-0.041+
					(0.012)	(0.014)	(0.023)
N					2943	2904	2899
R2					0.100	0.161	0.277
Starting 2002							
Manufacturing Employment Growth						0.005	-0.031
						(0.014)	(0.026)
N						2835	2837
R2						0.229	0.297
Starting 2007							
Manufacturing Employment Growth							-0.020
							(0.022)
N							2814
R2							0.185

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 11: Winsorized Change in Total Production Value, County Level, Regions, Beginning in 1978

	1982	1987	1992	1997	2002	2007	2012
Midwest							
Employment Growth	-0.017	0.039	0.022	-0.020	-0.046	-0.129**	-0.245***
	(0.042)	(0.036)	(0.035)	(0.038)	(0.030)	(0.040)	(0.057)
N	1052	1049	1052	1052	1048	1048	1050
R2	0.288	0.248	0.211	0.135	0.186	0.268	0.431
Northeast							
Employment Growth	0.147	0.065	0.081	0.302+	0.351**	0.541**	0.492+
	(0.100)	(0.110)	(0.123)	(0.175)	(0.111)	(0.176)	(0.279)
N	212	206	212	212	210	205	205
R2	0.208	0.204	0.175	0.196	0.220	0.185	0.145
South							
Employment Growth	-0.000	0.033	-0.005	-0.040	-0.057	-0.148**	-0.313***
	(0.039)	(0.039)	(0.042)	(0.038)	(0.039)	(0.050)	(0.070)
N	1378	1369	1380	1380	1373	1343	1351
R2	0.092	0.071	0.061	0.120	0.056	0.094	0.135
West							
Employment Growth	-0.169*	-0.010	0.018	0.008	-0.008	-0.045	-0.258**
	(0.069)	(0.068)	(0.062)	(0.068)	(0.061)	(0.064)	(0.090)
N	405	404	407	407	405	398	401
R2	0.225	0.103	0.172	0.235	0.247	0.262	0.268

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 12: Winsorized Change in Total Farmland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.018*	-0.012+	-0.005	-0.017**	-0.019***	-0.031***	-0.029***
	(0.009)	(0.007)	(0.006)	(0.005)	(0.005)	(0.004)	(0.005)
N	3035	3028	3042	3042	3013	3029	3023
R2	0.103	0.319	0.415	0.353	0.283	0.333	0.328
Starting 1982							
Employment Growth		-0.026***	-0.010+	-0.021***	-0.024***	-0.031***	-0.032***
		(0.008)	(0.006)	(0.006)	(0.005)	(0.005)	(0.006)
N		3033	3048	3048	3017	3036	3030
R2		0.341	0.423	0.340	0.279	0.326	0.325
Starting 1987							
Employment Growth			-0.005	-0.028***	-0.032***	-0.044***	-0.045***
			(0.009)	(0.008)	(0.007)	(0.007)	(0.007)
N			3042	3042	3012	3028	3022
R2			0.229	0.185	0.184	0.257	0.253
Starting 1992							
Employment Growth				-0.019+	-0.024*	-0.047***	-0.049***
				(0.011)	(0.012)	(0.009)	(0.010)
N				3045	3015	3034	3025
R2				0.105	0.165	0.219	0.206
Starting 1997							
Employment Growth					-0.011	-0.056***	-0.055***
					(0.014)	(0.012)	(0.014)
N					3013	3033	3023
R2					0.146	0.199	0.185
Starting 2002							
Employment Growth						-0.063***	-0.036*
						(0.015)	(0.017)
N						3018	3010
R2						0.140	0.140
Starting 2007							
Employment Growth							0.003
							(0.023)
N							3034
R2							0.088

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 13: Winsorized Change in Total Irrigated Land, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.232+	-0.165	-0.153	0.024	-0.126	-0.100	-0.247*
	(0.124)	(0.144)	(0.096)	(0.085)	(0.103)	(0.096)	(0.105)
N	2273	2333	2463	2463	2364	2373	2354
R2	0.121	0.247	0.179	0.199	0.230	0.226	0.240
Starting 1982							
Employment Growth		0.158	-0.113	0.107	0.044	-0.014	-0.074
		(0.169)	(0.102)	(0.085)	(0.097)	(0.104)	(0.100)
N		2382	2496	2496	2401	2409	2396
R2		0.212	0.127	0.157	0.208	0.185	0.186
Starting 1987							
Employment Growth			-0.135	0.050	-0.039	-0.065	-0.107
			(0.086)	(0.069)	(0.101)	(0.081)	(0.084)
N			2681	2681	2558	2573	2565
R2			0.061	0.070	0.125	0.123	0.142
Starting 1992							
Employment Growth				0.274*	0.060	-0.024	-0.144
				(0.124)	(0.148)	(0.094)	(0.110)
N				2717	2588	2593	2592
R2				0.092	0.157	0.138	0.123
Starting 1997							
Employment Growth					0.160	0.023	-0.112
					(0.145)	(0.104)	(0.111)
N					2584	2585	2583
R2					0.134	0.108	0.128
Starting 2002							
Employment Growth						0.031	-0.119
						(0.126)	(0.129)
N						2658	2640
R2						0.099	0.134
Starting 2007							
Employment Growth							0.062
							(0.123)
N							2667
R2							0.085

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 14: Winsorized Change in Total Non-Irrigated Cropland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.039**	-0.011	0.014	-0.003	-0.008	-0.026***	-0.023***
	(0.014)	(0.012)	(0.011)	(0.009)	(0.008)	(0.006)	(0.006)
N	2939	2935	2973	2973	2967	2965	2965
R2	0.101	0.316	0.296	0.330	0.369	0.402	0.473
Starting 1982							
Employment Growth		-0.038**	-0.003	-0.003	-0.009	-0.031***	-0.028***
		(0.014)	(0.013)	(0.011)	(0.009)	(0.007)	(0.007)
N		2962	3000	3000	2992	2992	2986
R2		0.279	0.257	0.317	0.369	0.399	0.463
Starting 1987							
Employment Growth			0.014	0.016	0.004	-0.036***	-0.032***
			(0.015)	(0.014)	(0.011)	(0.009)	(0.009)
N			3002	3002	2994	2994	2993
R2			0.155	0.253	0.306	0.344	0.415
Starting 1992							
Employment Growth				0.011	0.017	-0.047***	-0.047**
				(0.019)	(0.018)	(0.013)	(0.015)
N				3047	3038	3038	3035
R2				0.176	0.190	0.316	0.433
Starting 1997							
Employment Growth					0.013	-0.050*	-0.032
					(0.024)	(0.020)	(0.021)
N					3037	3037	3033
R2					0.140	0.277	0.409
Starting 2002							
Employment Growth						-0.095***	-0.020
						(0.025)	(0.026)
N						3040	3034
R2						0.229	0.370
Starting 2007							
Employment Growth							0.040
							(0.034)
N							3038
R2							0.227

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 15: Change in Total Cropland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Raw Employment Growth	-0.005***	-0.022+	-0.012	-0.024*	-0.008	-0.026**	-0.020**
	(0.001)	(0.013)	(0.011)	(0.009)	(0.007)	(0.009)	(0.006)
N	2939	2935	2973	2973	2967	2965	2965
R2	0.002	0.003	0.001	0.002	0.001	0.005	0.014
Starting 1982							
Raw Employment Growth		-0.039*	-0.014	-0.028*	-0.007	-0.021	-0.028***
		(0.017)	(0.016)	(0.011)	(0.013)	(0.015)	(0.008)
N		2963	3001	3001	2993	2992	2987
R2		0.006	0.001	0.002	0.000	0.000	0.002
Starting 1987							
Raw Employment Growth			0.033	-0.015	0.016	-0.018	-0.032*
			(0.026)	(0.016)	(0.020)	(0.026)	(0.015)
N			3003	3003	2994	2995	2994
R2			0.001	0.000	0.000	0.000	0.000
Starting 1992							
Raw Employment Growth				-0.010	0.017	-0.042*	-0.040**
				(0.008)	(0.015)	(0.019)	(0.014)
N				2975	2968	2968	2966
R2				0.000	0.000	0.001	0.004
Starting 1997							
Raw Employment Growth					0.036	-0.058+	-0.037+
					(0.032)	(0.034)	(0.020)
N					2985	2984	2980
R2					0.001	0.002	0.002
Starting 2002							
Raw Employment Growth						-0.188***	-0.047*
						(0.056)	(0.024)
N						3042	3036
R2						0.003	0.002
Starting 2007							
Raw Employment Growth							0.006
							(0.021)
N							3038
R2							0.000

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 16: Winsorized Change in Total Cropland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Manufacturing Employment Growth	-0.001	-0.008*	0.002	0.002	-0.001	0.000	-0.001
	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
N	2835	2825	2884	2883	2833	2834	2822
R2	0.114	0.348	0.341	0.355	0.400	0.411	0.522
Starting 1982							
Manufacturing Employment Growth		-0.007*	-0.003	0.000	-0.005	-0.003	-0.008+
		(0.004)	(0.004)	(0.003)	(0.004)	(0.004)	(0.004)
N		2825	2869	2868	2825	2824	2814
R2		0.300	0.309	0.326	0.396	0.398	0.510
Starting 1987							
Manufacturing Employment Growth			0.001	0.003	-0.001	0.000	-0.000
			(0.003)	(0.003)	(0.004)	(0.003)	(0.004)
N			2879	2880	2839	2840	2826
R2			0.174	0.203	0.283	0.292	0.425
Starting 1992							
Manufacturing Employment Growth				-0.004	0.001	0.001	-0.010
				(0.004)	(0.005)	(0.005)	(0.006)
N				2961	2883	2880	2864
R2				0.112	0.203	0.278	0.433
Starting 1997							
Manufacturing Employment Growth					0.002	0.010*	-0.005
					(0.006)	(0.005)	(0.007)
N					2904	2897	2879
R2					0.192	0.302	0.440
Starting 2002							
Manufacturing Employment Growth						0.011	-0.012
						(0.007)	(0.008)
N						2878	2866
R2						0.229	0.388
Starting 2007							
Manufacturing Employment Growth							-0.002
							(0.009)
N							2870
R2							0.254

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 17: Winsorized Change in Total Cropland, County Level, Regions, Beginning in 1978

	1982	1987	1992	1997	2002	2007	2012
Midwest							
Employment Growth	-0.028*	-0.021+	-0.019*	-0.030***	-0.021**	-0.030***	-0.018*
	(0.013)	(0.012)	(0.008)	(0.009)	(0.008)	(0.007)	(0.008)
N	1041	1041	1046	1046	1046	1046	1046
R2	0.185	0.364	0.406	0.354	0.320	0.454	0.562
Northeast							
Employment Growth	-0.006	0.002	-0.013	0.009	-0.021	-0.042*	-0.007
	(0.012)	(0.022)	(0.029)	(0.024)	(0.019)	(0.021)	(0.029)
N	206	205	207	207	205	204	204
R2	0.093	0.258	0.382	0.356	0.368	0.356	0.416
South							
Employment Growth	-0.050**	-0.019	0.015	-0.002	-0.011	-0.034***	-0.032***
	(0.017)	(0.014)	(0.013)	(0.011)	(0.009)	(0.007)	(0.007)
N	1351	1345	1360	1360	1358	1355	1356
R2	0.085	0.246	0.282	0.328	0.379	0.276	0.294
West							
Employment Growth	0.022	0.014	0.061*	0.028	-0.004	-0.005	-0.035*
	(0.033)	(0.026)	(0.027)	(0.021)	(0.021)	(0.013)	(0.014)
N	341	344	360	360	358	360	359
R2	0.099	0.207	0.166	0.210	0.234	0.362	0.385

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 9: Correlations of Land and Labor Scarcity Measures

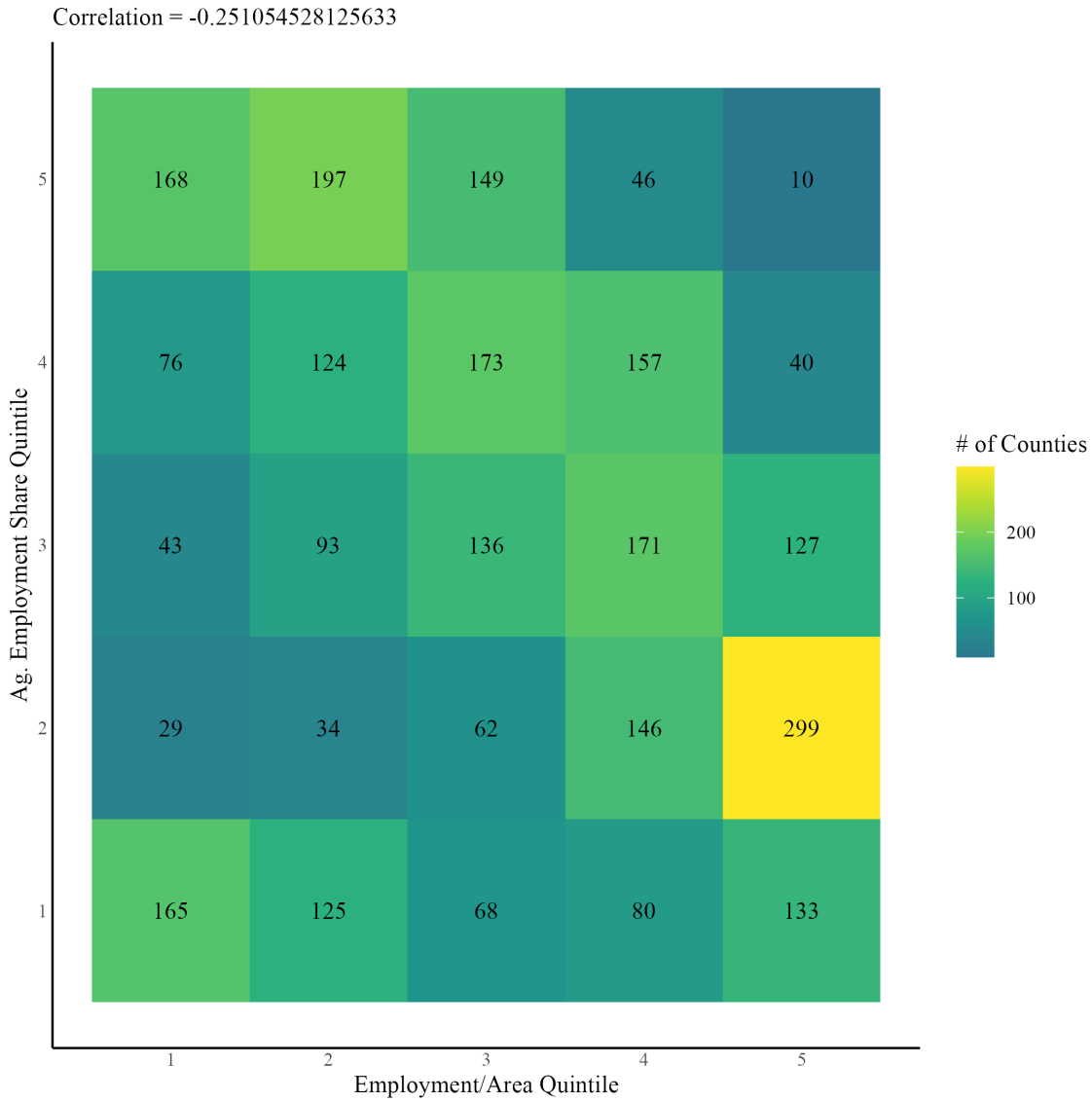


Table 18: Winsorized Change in Total Cropland, County Level, 1978 to 2012

	(1)	(2)	(3)	(4)
Employment Growth	-0.02884***	-0.02039**	-0.02247***	-0.01596+
	(0.00528)	(0.00773)	(0.00602)	(0.00913)
Growth x 1978 Cropland		-0.00000		-0.00000
		(0.00000)		(0.00000)
Growth x 1978 Employed Pop.			-0.00000	-0.00000
			(0.00000)	(0.00000)
Growth x 1978 Cropland x 1978 Employed Pop.				-0.00000
				(0.00000)
N	2965	2965	2965	2965
R2	0.531	0.532	0.532	0.535
Controls	✓	✓	✓	✓
State FE	✓	✓	✓	✓

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 10: Acres of Farmland by Type of Ownership by Year

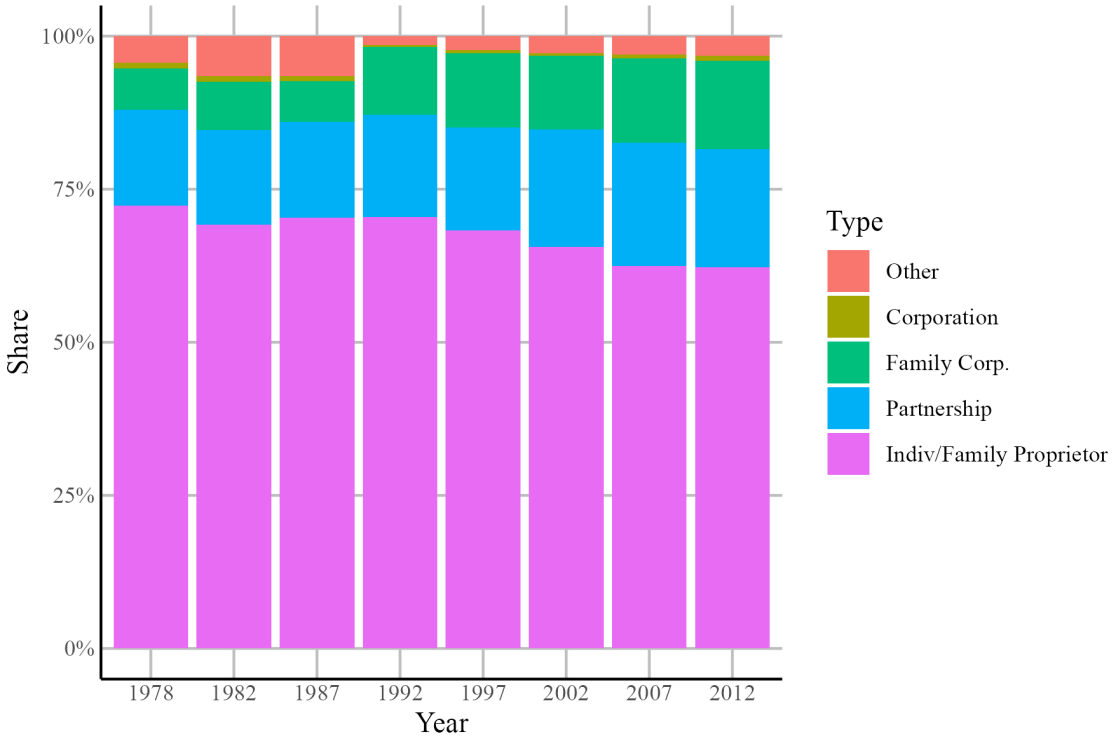


Table 19: Winsorized Change in Number of Full-Time Operators and Hired Laborers, County Level

	1997	2002	2007	2012
Starting 1992				
Employment Growth	-0.028	0.175***	0.128***	0.124***
	(0.044)	(0.034)	(0.025)	(0.027)
N	3061	2975	2802	2835
R2	0.392	0.463	0.528	0.548
Starting 1997				
Employment Growth		0.260***	0.118***	0.110***
		(0.046)	(0.025)	(0.031)
N		2976	2803	2835
R2		0.284	0.302	0.306
Starting 2002				
Employment Growth			0.021	0.018
			(0.022)	(0.025)
N			2765	2792
R2			0.196	0.229
Starting 2007				
Employment Growth				-0.004
				(0.035)
N				2679
R2				0.122

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 20: Winsorized Change in Value of Hired and Contracted Labor, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.011	0.142*	0.108*	0.055	0.212***	0.244***	0.213***
	(0.046)	(0.056)	(0.043)	(0.035)	(0.047)	(0.042)	(0.054)
N	2769	2749	2952	2952	2805	2770	2825
R2	0.122	0.090	0.087	0.100	0.124	0.161	0.165
Starting 1982							
Employment Growth		0.126*	0.063	0.024	0.139**	0.177***	0.127*
		(0.057)	(0.040)	(0.035)	(0.044)	(0.044)	(0.052)
N		2651	2829	2829	2696	2667	2719
R2		0.124	0.082	0.095	0.111	0.131	0.170
Starting 1987							
Employment Growth			0.046	0.023	0.132**	0.154***	0.116*
			(0.045)	(0.038)	(0.049)	(0.043)	(0.056)
N			2811	2811	2677	2644	2702
R2			0.092	0.082	0.099	0.141	0.153
Starting 1992							
Employment Growth				-0.002	0.178**	0.168***	0.120+
				(0.040)	(0.062)	(0.050)	(0.067)
N				3032	2859	2819	2879
R2				0.062	0.102	0.141	0.189
Starting 1997							
Employment Growth					0.400***	0.258***	0.092
					(0.084)	(0.071)	(0.089)
N					2854	2821	2877
R2					0.121	0.155	0.210
Starting 2002							
Employment Growth						0.123*	0.113
						(0.049)	(0.083)
N						2709	2756
R2						0.105	0.180
Starting 2007							
Employment Growth							0.182*
							(0.086)
N							2738
R2							0.164

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 21: Winsorized Change in Workers in CBP Agriculture-Related Industries, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.086	0.105	0.189*	0.111+	0.036	0.015	0.066
	(0.163)	(0.105)	(0.081)	(0.066)	(0.056)	(0.042)	(0.041)
N	2405	2407	2447	2458	2291	2281	2259
R2	0.114	0.122	0.166	0.157	0.170	0.147	0.170
Starting 1982							
Employment Growth		0.235	0.200*	0.125+	0.053	0.063	0.106*
		(0.160)	(0.097)	(0.068)	(0.066)	(0.051)	(0.049)
N		2413	2462	2457	2302	2292	2265
R2		0.104	0.157	0.140	0.158	0.138	0.162
Starting 1987							
Employment Growth			0.006	-0.036	-0.103	-0.040	-0.093
			(0.156)	(0.110)	(0.091)	(0.086)	(0.082)
N			2502	2494	2327	2297	2296
R2			0.179	0.165	0.189	0.174	0.183
Starting 1992							
Employment Growth				-0.411	-0.938	-0.154	0.298
				(1.763)	(1.511)	(1.040)	(0.936)
N				2893	2640	2610	2594
R2				0.111	0.134	0.121	0.133
Starting 1997							
Employment Growth					-7.822+	-3.515*	4.154
					(4.011)	(1.689)	(5.436)
N					2634	2599	2582
R2					0.132	0.125	0.132
Starting 2002							
Employment Growth						-0.067	0.007
						(0.132)	(0.120)
N						2234	2203
R2						0.058	0.091
Starting 2007							
Employment Growth			74				0.130
							(0.201)
N							2123
R2							0.077

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 22: Winsorized Change in Number of Full-Time Operators, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.013	0.063***	0.064***	0.056***	0.121***	0.079***	0.079***
	(0.022)	(0.014)	(0.013)	(0.009)	(0.016)	(0.011)	(0.013)
N	3058	3058	3058	3058	3056	3056	3056
R2	0.308	0.460	0.494	0.541	0.511	0.550	0.545
Starting 1982							
Employment Growth		-0.006	0.038***	0.032***	0.106***	0.072***	0.072***
		(0.011)	(0.010)	(0.008)	(0.016)	(0.011)	(0.013)
N		3061	3061	3061	3059	3059	3059
R2		0.394	0.438	0.501	0.474	0.524	0.513
Starting 1987							
Employment Growth			0.027*	0.015+	0.132***	0.077***	0.072***
			(0.011)	(0.009)	(0.019)	(0.014)	(0.016)
N			3059	3059	3057	3057	3057
R2			0.313	0.398	0.455	0.494	0.469
Starting 1992							
Employment Growth				0.022*	0.157***	0.086***	0.085***
				(0.011)	(0.027)	(0.017)	(0.019)
N				3059	3057	3057	3057
R2				0.277	0.403	0.446	0.432
Starting 1997							
Employment Growth					0.237***	0.092***	0.094***
					(0.047)	(0.024)	(0.028)
N					3059	3059	3059
R2					0.391	0.396	0.390
Starting 2002							
Employment Growth						-0.015	-0.021
						(0.016)	(0.019)
N						3054	3054
R2						0.279	0.358
Starting 2007							
Employment Growth							0.007
							(0.031)
N							3058
R2							0.151

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 23: Winsorized Change in Number of Hired Laborers, Full and Part Time, County Level

	1997	2002	2007	2012
Starting 1992				
Employment Growth	0.011	0.055*	0.067***	0.058**
	(0.035)	(0.026)	(0.016)	(0.019)
N	3042	3028	2992	3009
R2	0.115	0.205	0.287	0.313
Starting 1997				
Employment Growth		0.174***	0.135***	0.090**
		(0.048)	(0.024)	(0.029)
N		3030	2996	3012
R2		0.148	0.238	0.273
Starting 2002				
Employment Growth			0.121**	0.025
			(0.039)	(0.041)
N			2995	3007
R2			0.208	0.251
Starting 2007				
Employment Growth				-0.004
				(0.045)
N				2983
R2				0.151

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 24: Winsorized Change in Employed Population in Agriculture and Related Industries, County Level

	1990	2000	2010	2015
Starting 1980				
Employment Growth	0.172***	0.038***	0.049***	0.074***
	(0.019)	(0.010)	(0.010)	(0.010)
N	3066	3064	3064	3063
R2	0.473	0.364	0.284	0.317
Starting 1990				
Employment Growth		-0.040*	0.008	0.050***
		(0.016)	(0.017)	(0.015)
N		3064	3063	3062
R2		0.405	0.287	0.307
Starting 2000				
Employment Growth			0.101*	0.226***
			(0.040)	(0.033)
N			3064	3063
R2			0.151	0.297
Starting 2010				
Employment Growth				0.231***
				(0.054)
N				3061
R2				0.140

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 25: Winsorized Change in Acres for Major Crops, County Level

	1982	1987	1992	1997	2002	2007	2012
Corn (Grain)							
Employment Growth	0.013	0.059	-0.010	-0.015	-0.074*	-0.115*	-0.136*
	(0.060)	(0.047)	(0.039)	(0.037)	(0.030)	(0.045)	(0.056)
N	2248	2248	2248	2248	2248	2248	2248
R2	0.276	0.306	0.285	0.295	0.371	0.404	0.418
Soybeans							
Employment Growth	-0.506***	-0.278**	-0.253**	-0.298*	-0.555**	-0.611***	-0.951***
	(0.144)	(0.085)	(0.092)	(0.118)	(0.176)	(0.140)	(0.219)
N	1738	1738	1738	1738	1738	1738	1738
R2	0.444	0.585	0.591	0.575	0.571	0.572	0.546
Wheat							
Employment Growth	-1.551**	-0.384**	-0.292**	-0.252***	-0.136**	-0.222***	-0.209***
	(0.522)	(0.126)	(0.089)	(0.070)	(0.051)	(0.053)	(0.056)
N	2287	2287	2287	2287	2286	2286	2286
R2	0.517	0.417	0.264	0.273	0.291	0.298	0.359
Alfalfa							
Employment Growth	0.138*	-0.023	0.033	-0.006	0.003	0.002	-0.005
	(0.063)	(0.046)	(0.037)	(0.015)	(0.015)	(0.010)	(0.009)
N	2646	2646	2646	2646	2646	2646	2646
R2	0.111	0.259	0.305	0.319	0.323	0.347	0.332
Cotton							
Employment Growth	0.080	0.365	0.436	0.203	-0.185	-0.077	0.089
	(0.214)	(0.258)	(0.371)	(0.510)	(0.423)	(0.238)	(0.417)
N	401	401	401	401	400	400	400
R2	0.230	0.301	0.439	0.522	0.517	0.482	0.429

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 26: Winsorized Change in Acres for Crop Groups, County Level

	1982	1987	1992	1997	2002	2007	2012
Grains							
Employment Growth	-0.277*	-0.084*	-0.104***	-0.053*	-0.053**	-0.080***	-0.050**
	(0.117)	(0.042)	(0.031)	(0.021)	(0.016)	(0.016)	(0.017)
N	2743	2743	2743	2743	2742	2742	2742
R2	0.310	0.216	0.177	0.166	0.222	0.237	0.168
Legumes							
Employment Growth	-0.550***	-0.254**	-0.226*	-0.275*	-0.420*	-0.467***	-0.516*
	(0.147)	(0.093)	(0.100)	(0.135)	(0.178)	(0.129)	(0.219)
N	2123	2123	2123	2123	2122	2122	2122
R2	0.335	0.462	0.460	0.444	0.457	0.475	0.439
Fruit							
Employment Growth	-0.201	-0.184	-0.012	0.046	0.116+	0.026	0.108*
	(0.193)	(0.130)	(0.140)	(0.070)	(0.069)	(0.046)	(0.049)
N	1934	1934	1934	1934	1933	1933	1933
R2	0.103	0.126	0.150	0.103	0.124	0.139	0.158
Vegetables							
Employment Growth	0.137	-0.059	-0.224	0.055	0.064	-0.035	0.051
	(0.199)	(0.148)	(0.179)	(0.121)	(0.107)	(0.083)	(0.094)
N	1413	1413	1413	1413	1412	1412	1412
R2	0.082	0.098	0.144	0.146	0.151	0.160	0.181
Other							
Employment Growth	0.083+	0.032	-0.056	-0.145***	-0.139***	-0.099***	-0.106***
	(0.051)	(0.038)	(0.040)	(0.031)	(0.029)	(0.020)	(0.022)
N	2922	2922	2922	2922	2920	2920	2920
R2	0.138	0.161	0.222	0.192	0.143	0.138	0.176

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 27: Winsorized Change in Acres for Crops by Production Quintile, County Level

	1982	1987	1992	1997	2002	2007	2012
Quintile 2							
Employment Growth	-0.011	-0.612	-1.327+	-0.120	-0.613+	-0.110	0.173
	(0.774)	(0.618)	(0.685)	(0.322)	(0.347)	(0.207)	(0.293)
N	410	410	410	410	409	409	409
R2	0.127	0.173	0.197	0.200	0.203	0.264	0.292
Quintile 3							
Employment Growth	-0.238	-0.371	0.075	0.078	-0.136	-0.000	-0.131
	(0.327)	(0.298)	(0.265)	(0.201)	(0.216)	(0.161)	(0.255)
N	991	991	991	991	990	990	990
R2	0.135	0.169	0.221	0.217	0.250	0.251	0.258
Quintile 4							
Employment Growth	0.611*	0.066	0.063	-0.109	0.160	-0.042	0.074
	(0.278)	(0.207)	(0.239)	(0.125)	(0.114)	(0.073)	(0.063)
N	1837	1837	1837	1837	1836	1836	1836
R2	0.109	0.113	0.118	0.097	0.105	0.121	0.164
Quintile 5							
Employment Growth	0.005	0.003	-0.009	-0.015	-0.014+	-0.023**	-0.017*
	(0.029)	(0.018)	(0.015)	(0.009)	(0.008)	(0.007)	(0.008)
N	3001	3001	3001	3001	2999	2999	2999
R2	0.127	0.202	0.268	0.401	0.432	0.480	0.423

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 28: County-Level Shift-Share, First Stage

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Shift-Share Employment Growth	0.776***	1.022***	0.611***	0.871***	0.985***	1.044***	1.009***
	(0.070)	(0.082)	(0.065)	(0.073)	(0.072)	(0.077)	(0.074)
N	3047	3028	3051	3051	3036	2994	3007
First-stage F-stat	157.793	303.958	160.917	193.541	273.539	273.428	275.076
Starting 1982							
Shift-Share Employment Growth		0.759***	0.175***	0.777***	1.068***	1.116***	1.043***
		(0.077)	(0.032)	(0.073)	(0.079)	(0.081)	(0.076)
N		3031	3057	3057	3040	3000	3012
First-stage F-stat		152.760	35.482	157.281	269.853	292.138	277.416
Starting 1987							
Shift-Share Employment Growth			0.221**	0.528***	0.874***	0.951***	0.855***
			(0.069)	(0.070)	(0.069)	(0.063)	(0.060)
N			3033	3033	3021	2980	2991
First-stage F-stat			13.898	61.004	209.821	245.340	214.366
Starting 1992							
Shift-Share Employment Growth				0.848***	1.045***	1.047***	0.909***
				(0.097)	(0.078)	(0.073)	(0.064)
N				3045	3033	2990	3002
First-stage F-stat				117.845	328.789	341.776	293.620
Starting 1997							
Shift-Share Employment Growth					1.236***	1.152***	0.863***
					(0.084)	(0.068)	(0.060)
N					3038	2997	3010
First-stage F-stat					369.237	395.251	292.849
Starting 2002							
Shift-Share Employment Growth						0.829***	0.492***
						(0.082)	(0.052)
N						2994	3005
First-stage F-stat						126.487	84.664
Starting 2007							
Shift-Share Employment Growth							0.525***
							(0.066)
N							2977
First-stage F-stat							78.241

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 29: Rotemberg Weights for Shift-Shares

NAICS Code	Industry	Rotemberg weight
Panel A: Main Shift-Share		
7225	Restaurants and Other Eating Places	0.22607080
3152	Cut and Sew Apparel Manufacturing	0.08069283
7139	Other Amusement and Recreation	0.07012400
4251	Electronic Markets, Agents, and Brokers	0.06047936
7211	Traveler Accomodation	0.05681864
Panel B: Manufacturing Only		
3152	Cut and Sew Apparel Manufacturing	0.20100387
3132	Fabric Mills	0.10998069
3162	Footwear Manufacturing	0.06647439
3363	Motor Vehicle Parts Manufacturing	0.05553127
3131	Fiber, Yarn, and Thread Mills	0.04715854
Panel C: 2-Digit NAICS Codes		
72	Accommodation and Food Services	0.3188599
31	Food and Textile Manufacturing	0.2233759
56	Administrative Support and Waste Removal	0.1728068
71	Arts, Entertainment, and Recreation	0.1479910
33	Other Manufacturing	0.1112123

Table 30: Winsorized Change in Agricultural Production Value, County Level, Shift-Share with 2-Digit NAICS Codes

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Fitted Employment Growth	-0.252+	-0.289**	-0.336**	-0.370**	-0.402***	-0.632***	-1.149***
	(0.133)	(0.106)	(0.122)	(0.134)	(0.121)	(0.175)	(0.222)
N	3047	3028	3051	3051	3036	2994	3007
First-stage F-stat	148.347	234.632	179.256	185.282	244.581	214.857	205.940
Starting 1982							
Fitted Employment Growth		-0.256	-0.286+	-0.407**	-0.349**	-0.623***	-1.059***
		(0.194)	(0.154)	(0.153)	(0.122)	(0.178)	(0.219)
N		3031	3057	3057	3040	3000	3012
First-stage F-stat		76.242	109.627	127.677	213.731	191.749	200.953
Starting 1987							
Fitted Employment Growth			-0.282	-0.332	-0.363**	-0.750***	-1.566***
			(0.336)	(0.209)	(0.130)	(0.184)	(0.278)
N			3033	3033	3021	2980	2991
First-stage F-stat			22.829	82.399	221.571	197.123	169.624
Starting 1992							
Fitted Employment Growth				-0.184	-0.210+	-0.750***	-1.709***
				(0.204)	(0.112)	(0.167)	(0.252)
N				3045	3033	2990	3002
First-stage F-stat				110.765	305.402	280.954	257.860
Starting 1997							
Fitted Employment Growth					-0.341**	-0.907***	-1.974***
					(0.129)	(0.184)	(0.287)
N					3038	2997	3010
First-stage F-stat					272.105	257.846	223.082
Starting 2002							
Fitted Employment Growth						-1.657**	-2.899**
						(0.541)	(0.883)
N						2994	3005
First-stage F-stat						55.053	38.745
Starting 2007							
Fitted Employment Growth							-0.517
							(0.492)
N							2977
First-stage F-stat							59.279

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 31: Winsorized Change in Agricultural Production Value, County Level, Shift-Share of Manufacturing Only

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Fitted Employment Growth	-0.127	-0.171+	-0.153	-0.185	-0.308**	-0.496**	-0.891***
	(0.097)	(0.089)	(0.107)	(0.131)	(0.118)	(0.179)	(0.217)
N	2998	2979	3001	3001	2987	2946	2959
First-stage F-stat	211.190	236.067	218.821	206.494	259.483	228.841	269.886
Starting 1982							
Fitted Employment Growth		-0.079	-0.086	-0.214	-0.258*	-0.417*	-0.717**
		(0.149)	(0.172)	(0.150)	(0.126)	(0.194)	(0.228)
N		2967	2993	2993	2978	2941	2954
First-stage F-stat		95.496	80.864	123.614	194.059	173.620	225.750
Starting 1987							
Fitted Employment Growth			0.528*	0.037	-0.195	-0.471*	-1.010***
			(0.266)	(0.213)	(0.133)	(0.191)	(0.242)
N			2973	2973	2963	2925	2934
First-stage F-stat			32.151	84.708	227.607	207.783	267.484
Starting 1992							
Fitted Employment Growth				-0.132	-0.202+	-0.570**	-1.087***
				(0.200)	(0.118)	(0.181)	(0.228)
N				3034	3023	2980	2992
First-stage F-stat				120.743	279.840	231.045	312.775
Starting 1997							
Fitted Employment Growth					-0.320*	-0.681***	-1.426***
					(0.142)	(0.177)	(0.231)
N					3031	2990	3004
First-stage F-stat					259.459	245.266	359.829
Starting 2002							
Fitted Employment Growth						-0.869*	-1.478***
						(0.384)	(0.407)
N						2906	2918
First-stage F-stat						98.837	222.935
Starting 2007							
Fitted Employment Growth							-1.221***
							(0.349)
N							2890
First-stage F-stat							184.756

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 32: Winsorized Change in Total Cropland, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.081+	-0.052+	-0.071+	-0.120***	-0.086***	-0.054**	-0.074***
	(0.048)	(0.028)	(0.037)	(0.029)	(0.024)	(0.018)	(0.020)
N	2939	2935	2973	2973	2967	2965	2965
R2	0.890	0.918	0.912	0.915	0.923	0.918	0.922
Starting 1982							
Employment Growth		-0.053	-0.017	-0.112**	-0.095***	-0.073***	-0.101***
		(0.046)	(0.090)	(0.037)	(0.027)	(0.021)	(0.024)
N		2963	3001	3001	2993	2992	2987
R2		0.911	0.907	0.912	0.922	0.916	0.920
Starting 1987							
Employment Growth			-0.368+	-0.232**	-0.130***	-0.112***	-0.171***
			(0.216)	(0.078)	(0.038)	(0.031)	(0.038)
N			3003	3003	2994	2995	2994
R2			0.896	0.901	0.911	0.906	0.913
Starting 1992							
Employment Growth				-0.205***	-0.119***	-0.134***	-0.192***
				(0.060)	(0.036)	(0.038)	(0.045)
N				2975	2968	2968	2966
R2				0.895	0.904	0.905	0.915
Starting 1997							
Employment Growth					-0.027	-0.147*	-0.252***
					(0.061)	(0.060)	(0.071)
N					2985	2984	2980
R2					0.904	0.909	0.918
Starting 2002							
Employment Growth						-0.085	-0.212
						(0.132)	(0.154)
N						3042	3036
R2						0.901	0.912
Starting 2007							
Employment Growth							-0.235
							(0.224)
N							3038
R2							0.905

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 33: Winsorized Change in Total Farmland, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.056	-0.052*	-0.031	-0.082***	-0.078***	-0.091***	-0.116***
	(0.039)	(0.024)	(0.024)	(0.019)	(0.018)	(0.015)	(0.016)
N	3035	3028	3042	3042	3013	3029	3023
R2	0.890	0.914	0.923	0.917	0.910	0.914	0.911
Starting 1982							
Employment Growth		-0.101**	-0.027	-0.098***	-0.094***	-0.113***	-0.134***
		(0.035)	(0.058)	(0.025)	(0.023)	(0.018)	(0.018)
N		3033	3048	3048	3017	3036	3030
R2		0.916	0.923	0.916	0.908	0.912	0.911
Starting 1987							
Employment Growth			0.050	-0.163**	-0.122***	-0.144***	-0.199***
			(0.122)	(0.051)	(0.033)	(0.027)	(0.027)
N			3042	3042	3012	3028	3022
R2			0.905	0.902	0.900	0.906	0.904
Starting 1992							
Employment Growth				-0.134*	-0.124**	-0.180***	-0.244***
				(0.059)	(0.039)	(0.035)	(0.032)
N				3045	3015	3034	3025
R2				0.896	0.899	0.905	0.902
Starting 1997							
Employment Growth					-0.115*	-0.187***	-0.324***
					(0.052)	(0.046)	(0.048)
N					3013	3033	3023
R2					0.898	0.905	0.902
Starting 2002							
Employment Growth						-0.346***	-0.456***
						(0.100)	(0.102)
N						3018	3010
R2						0.900	0.897
Starting 2007							
Employment Growth							-0.372*
							(0.154)
N							3034
R2							0.895

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 34: Winsorized Change in Number of Full-Time Operators and Hired Laborers, County Level Shift-Share

	1997	2002	2007	2012
Starting 1992				
Employment Growth	1.077***	0.943***	0.638***	0.525***
	(0.300)	(0.150)	(0.101)	(0.109)
N	3061	2975	2802	2835
R2	0.927	0.932	0.941	0.943
Starting 1997				
Employment Growth		0.579***	0.380***	0.182+
		(0.156)	(0.073)	(0.110)
N		2976	2803	2835
R2		0.905	0.911	0.911
Starting 2002				
Employment Growth			0.180	-0.052
			(0.129)	(0.140)
N			2765	2792
R2			0.906	0.909
Starting 2007				
Employment Growth				-0.581*
				(0.260)
N				2679
R2				0.900

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 35: Winsorized Change in Value of Hired and Contracted Labor, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.277	-0.188	-0.111	-0.024	0.017	-0.039	-0.606**
	(0.214)	(0.159)	(0.178)	(0.149)	(0.139)	(0.125)	(0.188)
N	2769	2749	2952	2952	2805	2770	2825
R2	0.892	0.889	0.890	0.891	0.893	0.894	0.893
Starting 1982							
Employment Growth		-0.469+	-0.041	0.025	0.120	-0.031	-0.363*
		(0.263)	(0.323)	(0.136)	(0.151)	(0.123)	(0.167)
N		2651	2829	2829	2696	2667	2719
R2		0.893	0.888	0.890	0.891	0.892	0.892
Starting 1987							
Employment Growth			0.858	0.289	0.320+	0.271*	-0.517*
			(0.757)	(0.314)	(0.172)	(0.137)	(0.215)
N			2811	2811	2677	2644	2702
R2			0.891	0.890	0.891	0.894	0.890
Starting 1992							
Employment Growth				-0.341	0.130	0.075	-0.531*
				(0.283)	(0.207)	(0.149)	(0.224)
N				3032	2859	2819	2879
R2				0.889	0.889	0.892	0.892
Starting 1997							
Employment Growth					0.591+	0.471*	-0.799*
					(0.328)	(0.231)	(0.394)
N					2854	2821	2877
R2					0.889	0.893	0.892
Starting 2002							
Employment Growth						-0.086	-1.393*
						(0.341)	(0.564)
N						2709	2756
R2						0.890	0.892
Starting 2007							
Employment Growth							-0.518
							(0.630)
N							2738
R2							0.894

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 36: Winsorized Change in Employed Population in Agriculture and Related Industries, County Level Shift-Share

	1990	2000	2010	2015
Starting 1980				
Fitted Employment Growth	-0.306+	-0.334***	-0.169***	-0.053+
	(0.179)	(0.048)	(0.036)	(0.032)
N	3066	3064	3064	3063
R2	0.915	0.908	0.906	0.908
Starting 1990				
Fitted Employment Growth		-1.958***	-0.461***	-0.248***
		(0.335)	(0.073)	(0.062)
N		3064	3063	3062
R2		0.913	0.907	0.912
Starting 2000				
Fitted Employment Growth			0.233+	0.515***
			(0.119)	(0.117)
N			3064	3063
R2			0.897	0.904
Starting 2010				
Fitted Employment Growth				0.731+
				(0.388)
N				3061
R2				0.889

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 37: Winsorized Change in Total Production Value, County Level, by Working-Age Population Growth

	1997	2002	2007	2012
Starting 1992				
Population Growth	0.009	0.104	-0.243*	-0.456*
	(0.159)	(0.092)	(0.121)	(0.177)
N	1111	1107	1098	1096
R2	0.148	0.096	0.131	0.248
Starting 1997				
Population Growth		0.320*	-0.174	-0.459*
		(0.153)	(0.128)	(0.200)
N		1107	1098	1096
R2		0.080	0.143	0.254
Starting 2002				
Population Growth			-0.823***	-0.908**
			(0.213)	(0.294)
N			1093	1093
R2			0.251	0.298
Starting 2007				
Population Growth				0.385
				(0.350)
N				1089
R2				0.163

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 38: Winsorized Change in Total Production Value, County Level, by Unemployment Rate Growth

	1997	2002	2007	2012
Starting 1992				
Unemployment Rate Growth	-0.076	-0.030	-0.057	-0.100
	(0.050)	(0.066)	(0.115)	(0.121)
N	1110	1106	1097	1095
R2	0.151	0.095	0.127	0.239
Starting 1997				
Unemployment Rate Growth		0.018	0.002	-0.064
		(0.025)	(0.051)	(0.060)
N		1106	1097	1095
R2		0.075	0.142	0.248
Starting 2002				
Unemployment Rate Growth			0.091	-0.140
			(0.093)	(0.090)
N			1092	1092
R2			0.240	0.288
Starting 2007				
Unemployment Rate Growth				-0.074
				(0.053)
N				1088
R2				0.163

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 39: Winsorized Change in Total Cropland, County Level, by Working-Age Population Growth

	1997	2002	2007	2012
Starting 1992				
Population Growth	-0.129**	-0.081*	-0.184***	-0.123***
	(0.048)	(0.038)	(0.042)	(0.035)
N	1091	1090	1090	1088
R2	0.165	0.256	0.280	0.437
Starting 1997				
Population Growth		-0.076	-0.272***	-0.143*
		(0.082)	(0.073)	(0.057)
N		1085	1085	1083
R2		0.160	0.291	0.455
Starting 2002				
Population Growth			-0.453***	-0.175*
			(0.109)	(0.082)
N			1107	1104
R2			0.215	0.403
Starting 2007				
Population Growth				0.149
				(0.130)
N				1105
R2				0.282

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 40: Winsorized Change in Total Cropland, County Level, by Unemployment Rate Growth

	1997	2002	2007	2012
Starting 1992				
Unemployment Rate Growth	-0.001	-0.045*	-0.076*	-0.044*
	(0.018)	(0.020)	(0.036)	(0.022)
N	1090	1089	1089	1087
R2	0.157	0.255	0.256	0.428
Starting 1997				
Unemployment Rate Growth		-0.018	-0.052*	-0.035*
		(0.014)	(0.023)	(0.014)
N		1084	1084	1082
R2		0.159	0.274	0.453
Starting 2002				
Unemployment Rate Growth			0.000	0.001
			(0.043)	(0.025)
N			1106	1103
R2			0.197	0.399
Starting 2007				
Unemployment Rate Growth				-0.012
				(0.025)
N				1104
R2				0.281

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 41: Winsorized Change in Agricultural Production Value, County Level, Employment/Area Quintiles, Beginning in 1978

	1982	1987	1992	1997	2002	2007	2012
Q1							
Employment Growth	-0.084	0.044	0.079	0.009	-0.003	-0.021	-0.101
	(0.055)	(0.046)	(0.049)	(0.046)	(0.058)	(0.062)	(0.092)
N	604	604	607	607	601	593	597
R2	0.252	0.158	0.150	0.181	0.218	0.319	0.435
Q2							
Employment Growth	-0.001	0.104+	0.072	-0.005	-0.036	-0.054	-0.145+
	(0.054)	(0.060)	(0.064)	(0.060)	(0.057)	(0.077)	(0.086)
N	609	606	609	609	605	600	605
R2	0.232	0.210	0.169	0.186	0.163	0.160	0.265
Q3							
Employment Growth	-0.037	0.054	0.045	0.072	-0.002	-0.153*	-0.301***
	(0.058)	(0.060)	(0.060)	(0.062)	(0.063)	(0.075)	(0.080)
N	606	601	606	606	604	597	599
R2	0.267	0.196	0.184	0.174	0.168	0.181	0.234
Q4							
Employment Growth	0.056	0.053	-0.045	-0.105+	-0.143*	-0.262**	-0.477***
	(0.078)	(0.055)	(0.056)	(0.060)	(0.061)	(0.082)	(0.106)
N	608	604	609	609	609	597	596
R2	0.241	0.178	0.142	0.193	0.171	0.129	0.191
Q5							
Employment Growth	-0.045	-0.083	-0.079	-0.052	0.003	-0.100	-0.281**
	(0.080)	(0.063)	(0.060)	(0.071)	(0.061)	(0.081)	(0.108)
N	602	594	602	602	599	588	592
R2	0.205	0.248	0.191	0.243	0.255	0.231	0.240

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 42: Winsorized Change in Total Cropland, County Level, Employment/Area Quintiles, Beginning in 1978

	1982	1987	1992	1997	2002	2007	2012
Q1							
Employment Growth	-0.047*	0.012	0.078***	0.041*	0.032*	0.007	-0.012
	(0.023)	(0.018)	(0.020)	(0.016)	(0.014)	(0.012)	(0.012)
N	534	534	554	554	553	554	552
R2	0.211	0.380	0.309	0.274	0.297	0.386	0.465
Q2							
Employment Growth	-0.004	0.027	0.053***	0.024*	0.012	-0.014	-0.021*
	(0.026)	(0.017)	(0.015)	(0.012)	(0.011)	(0.009)	(0.010)
N	595	592	598	598	598	598	598
R2	0.172	0.353	0.353	0.367	0.451	0.453	0.549
Q3							
Employment Growth	-0.027	-0.007	0.020	0.014	0.010	-0.014	-0.012
	(0.025)	(0.021)	(0.018)	(0.014)	(0.013)	(0.009)	(0.009)
N	598	598	600	600	600	599	600
R2	0.143	0.378	0.396	0.431	0.466	0.446	0.551
Q4							
Employment Growth	-0.028	-0.041*	-0.037*	-0.038**	-0.038**	-0.052***	-0.048***
	(0.024)	(0.017)	(0.016)	(0.013)	(0.012)	(0.010)	(0.011)
N	604	605	608	608	608	608	606
R2	0.171	0.393	0.374	0.372	0.383	0.462	0.583
Q5							
Employment Growth	-0.059+	-0.059***	-0.052**	-0.063***	-0.056***	-0.056***	-0.042***
	(0.031)	(0.016)	(0.016)	(0.012)	(0.010)	(0.010)	(0.009)
N	594	590	598	598	594	591	593
R2	0.173	0.367	0.415	0.413	0.429	0.494	0.615

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 43: Winsorized Change in Average Value of Acre Farmland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.137***	0.264***	0.280***	0.255***	0.397***	0.358***	0.371***
	(0.032)	(0.030)	(0.026)	(0.023)	(0.033)	(0.033)	(0.038)
N	3032	3018	3041	3041	3035	3036	3039
R2	0.346	0.542	0.539	0.487	0.525	0.587	0.377
Starting 1982							
Employment Growth		0.215***	0.213***	0.195***	0.302***	0.276***	0.265***
		(0.022)	(0.023)	(0.018)	(0.025)	(0.028)	(0.031)
N		3025	3049	3049	3042	3043	3047
R2		0.536	0.547	0.487	0.530	0.564	0.321
Starting 1987							
Employment Growth			0.087***	0.114***	0.208***	0.158***	0.070+
			(0.024)	(0.022)	(0.029)	(0.032)	(0.040)
N			3033	3033	3030	3029	3031
R2			0.287	0.379	0.447	0.476	0.527
Starting 1992							
Employment Growth				0.101***	0.203***	0.126***	0.089+
				(0.027)	(0.037)	(0.035)	(0.048)
N				3036	3033	3032	3034
R2				0.306	0.382	0.479	0.591
Starting 1997							
Employment Growth					0.163***	0.122**	0.081
					(0.041)	(0.038)	(0.056)
N					3045	3044	3047
R2					0.290	0.471	0.540
Starting 2002							
Employment Growth						0.058+	0.115*
						(0.035)	(0.052)
N						3049	3050
R2						0.518	0.651
Starting 2007							
Employment Growth							0.120**
							(0.044)
N							3057
R2							0.597

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 44: Winsorized Change in Average Value of Acre Farmland, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.004	0.150*	0.020	-0.072	-0.107	-0.460***	-0.133
	(0.135)	(0.075)	(0.098)	(0.084)	(0.104)	(0.117)	(0.113)
N	3032	3018	3041	3041	3035	3036	3039
R2	0.914	0.938	0.939	0.932	0.935	0.938	0.912
Starting 1982							
Employment Growth		0.163+	-0.076	-0.077	-0.110	-0.343**	-0.127
		(0.087)	(0.154)	(0.082)	(0.089)	(0.106)	(0.101)
N		3025	3049	3049	3042	3043	3047
R2		0.940	0.943	0.934	0.939	0.938	0.909
Starting 1987							
Employment Growth			-0.100	-0.135	-0.272*	-0.952***	-0.649***
			(0.311)	(0.160)	(0.123)	(0.139)	(0.166)
N			3033	3033	3030	3029	3031
R2			0.913	0.924	0.933	0.928	0.937
Starting 1992							
Employment Growth				-0.537***	-0.479***	-1.152***	-0.887***
				(0.159)	(0.129)	(0.141)	(0.154)
N				3036	3033	3032	3034
R2				0.913	0.925	0.919	0.942
Starting 1997							
Employment Growth					-0.172	-0.791***	-0.423*
					(0.120)	(0.135)	(0.168)
N					3045	3044	3047
R2					0.918	0.922	0.940
Starting 2002							
Employment Growth						-1.467***	-0.255
						(0.258)	(0.289)
N						3049	3050
R2						0.925	0.952
Starting 2007							
Employment Growth							0.095
							(0.294)
N							3057
R2							0.952

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 45: Winsorized Change in Variables, County Level, Starting 1978

	1982	1987	1992	1997	2002	2007	2012
Agricultural Production Value							
Employment Growth	-0.017	0.049	0.038	0.010	-0.033	-0.126**	-0.290***
	(0.033)	(0.035)	(0.040)	(0.050)	(0.038)	(0.046)	(0.055)
Employment Growth x Palmer Drought Index	-0.011	-0.041	-0.085	-0.050	-0.000	0.067	0.140
	(0.043)	(0.039)	(0.071)	(0.061)	(0.059)	(0.083)	(0.091)
N	3046	3027	3050	3050	3036	2994	3007
R2	0.192	0.166	0.134	0.147	0.132	0.142	0.238
Total Cropland							
Employment Growth	-0.032*	-0.016	0.014	-0.003	-0.013+	-0.028***	-0.029***
	(0.014)	(0.011)	(0.012)	(0.012)	(0.007)	(0.006)	(0.006)
Employment Growth x Palmer Drought Index	-0.011	0.000	-0.010	-0.003	0.001	-0.006	0.002
	(0.020)	(0.014)	(0.024)	(0.016)	(0.012)	(0.011)	(0.010)
N	2938	2934	2972	2972	2967	2965	2965
R2	0.117	0.352	0.344	0.353	0.402	0.424	0.531

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Drought is represented using the Palmer Drought Severity Index, averaged from the starting year to the given year. The drought variable is also included as a control by itself. Lower values of the Palmer index indicate more drought. The drought measure is winsorized also.

Table 46: Winsorized Change in Variables, County Level, Starting 1978

	1982	1987	1992	1997	2002	2007	2012
Agricultural Production Value							
Employment Growth	-0.113	-0.083	-0.062	-0.090	-0.070	-0.059	0.016
	(0.083)	(0.078)	(0.074)	(0.072)	(0.077)	(0.089)	(0.131)
Precipitation	-0.163*	0.012	0.387+	0.770*	1.369**	1.772**	2.073*
	(0.073)	(0.154)	(0.234)	(0.336)	(0.448)	(0.652)	(0.855)
Employment Growth x Precipitation	0.002	0.003	0.002	0.002	0.001	-0.002	-0.008*
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.003)	(0.004)
N	3045	3026	3049	3049	3035	2993	3006
R2	0.194	0.162	0.132	0.152	0.142	0.148	0.241
Total Cropland							
Employment Growth	-0.013	0.009	0.067*	0.032	0.023	0.008	-0.011
	(0.031)	(0.028)	(0.029)	(0.022)	(0.019)	(0.016)	(0.014)
Precipitation	-0.033	-0.058	0.148*	0.183*	-0.033	-0.339**	-0.308**
	(0.026)	(0.040)	(0.060)	(0.071)	(0.077)	(0.107)	(0.103)
Employment Growth x Precipitation	-0.001	-0.001	-0.001*	-0.001+	-0.001+	-0.001*	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)	(0.000)	(0.000)
N	2938	2934	2972	2972	2967	2965	2965
R2	0.117	0.353	0.343	0.355	0.403	0.436	0.537

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Precipitation is the yearly amount of precipitation in a county from 1948 to 1977, averaged over those years. The precipitation variable is also included as a control by itself and is winsorized.

Table 47: Winsorized Change in Variables, County Level, Starting 1978

	1982	1987	1992	1997	2002	2007	2012
Agricultural Production Value							
Employment Growth	-0.013	0.024	-0.003	-0.044	-0.033	-0.120**	-0.282***
	(0.032)	(0.030)	(0.032)	(0.033)	(0.030)	(0.041)	(0.062)
Emp. Growth x Agricultural Employment Share	-0.613	0.026	0.912	1.351	-0.432	-0.233	-0.009
	(0.990)	(0.940)	(1.035)	(1.299)	(1.238)	(1.667)	(2.136)
N	2839	2821	2842	2842	2830	2790	2803
R2	0.191	0.164	0.130	0.146	0.131	0.137	0.219
Total Cropland							
Employment Growth	-0.036*	-0.030**	-0.005	-0.021*	-0.026***	-0.039***	-0.036***
	(0.014)	(0.012)	(0.011)	(0.009)	(0.007)	(0.006)	(0.006)
Emp. Growth x Agricultural Employment Share	-0.085	0.615+	0.861**	0.953**	0.825**	0.611**	0.545**
	(0.540)	(0.318)	(0.304)	(0.299)	(0.259)	(0.205)	(0.202)
N	2760	2758	2791	2791	2785	2783	2784
R2	0.116	0.359	0.346	0.361	0.411	0.437	0.542

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
I also control for the agricultural employment share by itself, which is winsorized. Employment values are always measured as of the starting year to avoid exogeneity concerns.

Table 48: Winsorized Change in Agricultural Production Value, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth, Less Polluting Industries	-0.026	0.073*	0.073*	0.035	0.044	-0.009	-0.122**
	(0.029)	(0.032)	(0.029)	(0.030)	(0.032)	(0.032)	(0.040)
Employment Growth, More Polluting Industries	-0.003	-0.019	-0.041*	-0.035	-0.051+	-0.104**	-0.145*
	(0.012)	(0.013)	(0.017)	(0.023)	(0.028)	(0.038)	(0.057)
N	3030	3012	3033	3036	3015	2961	2975
R2	0.192	0.164	0.133	0.148	0.134	0.140	0.232
Starting 1982							
Employment Growth, Less Polluting Industries		0.067*	0.049*	0.033	0.012	-0.028	-0.148***
		(0.032)	(0.024)	(0.026)	(0.027)	(0.029)	(0.036)
Employment Growth, More Polluting Industries		-0.004	-0.026+	-0.040*	-0.025	-0.079*	-0.058
		(0.013)	(0.014)	(0.017)	(0.022)	(0.032)	(0.046)
N		3011	3036	3038	3015	2965	2979
R2		0.205	0.152	0.167	0.137	0.139	0.227
Starting 1987							
Employment Growth, Less Polluting Industries			-0.005	-0.037	-0.056+	-0.095**	-0.229***
			(0.026)	(0.029)	(0.030)	(0.035)	(0.054)
Employment Growth, More Polluting Industries			-0.002	-0.010	0.003	-0.044	-0.066
			(0.013)	(0.016)	(0.020)	(0.034)	(0.052)
N			3014	3016	2999	2945	2958
R2			0.134	0.175	0.119	0.153	0.253
Starting 1992							
Employment Growth, Less Polluting Industries				0.017	0.006	-0.073*	-0.172**
				(0.032)	(0.033)	(0.035)	(0.055)
Employment Growth, More Polluting Industries				-0.005	0.022	-0.018	-0.082
				(0.014)	(0.021)	(0.030)	(0.052)
N				3033	3015	2958	2970
R2				0.152	0.107	0.138	0.259
Starting 1997							
Employment Growth, Less Polluting Industries					0.106**	-0.019	-0.098
					(0.036)	(0.033)	(0.063)
Employment Growth, More Polluting Industries					0.003	-0.018	-0.059
					(0.019)	(0.029)	(0.041)
N					3020	2967	2981
R2					0.099	0.164	0.282
Starting 2002							
Employment Growth, Less Polluting Industries						-0.076+	-0.169*
						(0.043)	(0.080)
Employment Growth, More Polluting Industries						-0.016	-0.069+
						(0.023)	(0.040)
N						2954	2967
R2						0.232	0.303
Starting 2007							
Employment Growth, Less Polluting Industries							0.139*
							(0.066)
Employment Growth, More Polluting Industries							-0.006
							(0.024)
N							2901
R2							0.189

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 49: Winsorized Change in Total Cropland, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth, Less Polluting Industries	-0.030*	-0.014	0.008	-0.002	-0.012*	-0.028***	-0.029***
	(0.012)	(0.011)	(0.009)	(0.007)	(0.006)	(0.005)	(0.004)
Employment Growth, More Polluting Industries	-0.003	-0.007	0.005	0.001	-0.001	0.002	0.003
	(0.005)	(0.005)	(0.006)	(0.005)	(0.006)	(0.006)	(0.006)
N	2927	2925	2961	2963	2951	2934	2938
R2	0.115	0.352	0.340	0.351	0.404	0.428	0.535
Starting 1982							
Employment Growth, Less Polluting Industries		-0.004	0.011	-0.002	-0.015*	-0.031***	-0.030***
		(0.011)	(0.011)	(0.008)	(0.006)	(0.005)	(0.005)
Employment Growth, More Polluting Industries		-0.013*	-0.011	0.003	-0.005	-0.004	-0.008
		(0.005)	(0.007)	(0.006)	(0.006)	(0.007)	(0.007)
N		2947	2981	2983	2969	2958	2956
R2		0.297	0.301	0.317	0.396	0.412	0.514
Starting 1987							
Employment Growth, Less Polluting Industries			0.009	0.004	-0.016*	-0.033***	-0.038***
			(0.012)	(0.010)	(0.008)	(0.007)	(0.008)
Employment Growth, More Polluting Industries			0.001	0.007	0.001	-0.007	-0.008
			(0.006)	(0.006)	(0.006)	(0.007)	(0.008)
N			2989	2990	2974	2965	2965
R2			0.168	0.199	0.286	0.304	0.427
Starting 1992							
Employment Growth, Less Polluting Industries				0.005	-0.014	-0.055***	-0.053***
				(0.012)	(0.012)	(0.011)	(0.012)
Employment Growth, More Polluting Industries				-0.005	-0.003	-0.011	-0.014
				(0.007)	(0.007)	(0.009)	(0.011)
N				2969	2955	2940	2940
R2				0.112	0.206	0.288	0.436
Starting 1997							
Employment Growth, Less Polluting Industries					-0.018	-0.064***	-0.063***
					(0.017)	(0.016)	(0.018)
Employment Growth, More Polluting Industries					-0.001	-0.007	-0.009
					(0.009)	(0.010)	(0.012)
N					2969	2955	2952
R2					0.191	0.306	0.442
Starting 2002							
Employment Growth, Less Polluting Industries						-0.073***	-0.043+
						(0.020)	(0.024)
Employment Growth, More Polluting Industries						-0.021*	-0.019+
						(0.009)	(0.011)
N						3000	2997
R2						0.233	0.383
Starting 2007							
Employment Growth, Less Polluting Industries							0.021
							(0.028)
Employment Growth, More Polluting Industries							-0.000
							(0.010)
N							2957
R2							0.250

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 50: Winsorized Change in Property Tax Paid, County Level

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	0.027	0.003	0.125***	0.080*	0.081+
	(0.029)	(0.025)	(0.034)	(0.037)	(0.048)
N	3010	3010	2998	2993	2992
R2	0.192	0.315	0.321	0.462	0.379
Starting 1992					
Employment Growth		0.003	0.163***	0.058+	0.066
		(0.031)	(0.045)	(0.034)	(0.048)
N		3041	3032	3026	3024
R2		0.238	0.266	0.384	0.328
Starting 1997					
Employment Growth			0.201***	0.061	0.035
			(0.052)	(0.041)	(0.050)
N			3044	3039	3038
R2			0.178	0.285	0.251
Starting 2002					
Employment Growth				-0.008	-0.011
				(0.041)	(0.048)
N				3028	3027
R2				0.228	0.226
Starting 2007					
Employment Growth					0.023
					(0.051)
N					3028
R2					0.229

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 51: Winsorized Change in Property Tax Paid, County Level Shift-Share

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	0.126	-0.405*	-0.086	-0.299**	-0.677***
	(0.387)	(0.163)	(0.139)	(0.108)	(0.140)
N	3010	3010	2998	2993	2992
R2	0.900	0.912	0.913	0.927	0.920
Starting 1992					
Employment Growth		-0.517***	-0.198	-0.298**	-0.643***
		(0.152)	(0.125)	(0.113)	(0.138)
N		3041	3032	3026	3024
R2		0.905	0.907	0.919	0.916
Starting 1997					
Employment Growth			0.304*	-0.128	-0.639***
			(0.144)	(0.121)	(0.150)
N			3044	3039	3038
R2			0.902	0.912	0.910
Starting 2002					
Employment Growth				-0.592*	-0.821**
				(0.236)	(0.274)
N				3028	3027
R2				0.905	0.907
Starting 2007					
Employment Growth					-0.743*
					(0.295)
N					3028
R2					0.909

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 52: Winsorized Change in Value of Direct-to-Consumer Sales, County Level

	1997	2002	2007	2012
Starting 1992				
Employment Growth	0.213	0.786***	0.769**	1.245***
	(0.182)	(0.228)	(0.243)	(0.274)
N	2740	2593	2607	2630
R2	0.095	0.195	0.188	0.208
Starting 1997				
Employment Growth		0.157	0.337	0.768**
		(0.306)	(0.323)	(0.291)
N		2595	2597	2628
R2		0.181	0.168	0.180
Starting 2002				
Employment Growth			0.050	0.185
			(0.258)	(0.239)
N			2641	2658
R2			0.182	0.169
Starting 2007				
Employment Growth				-0.266
				(0.227)
N				2674
R2				0.146

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 53: Winsorized Change in Value of Direct-to-Consumer Sales, County Level Shift-Share

	1997	2002	2007	2012
Starting 1992				
Employment Growth	-0.322	-1.155	-1.875+	-0.997
	(0.940)	(0.785)	(0.997)	(0.975)
N	2740	2593	2607	2630
R2	0.884	0.889	0.888	0.891
Starting 1997				
Employment Growth		-1.745+	-1.663+	-0.014
		(0.902)	(1.004)	(0.807)
N		2595	2597	2628
R2		0.892	0.889	0.890
Starting 2002				
Employment Growth			-2.214+	0.616
			(1.280)	(0.933)
N			2641	2658
R2			0.892	0.891
Starting 2007				
Employment Growth				0.510
				(1.148)
N				2674
R2				0.892

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 54: Winsorized Change in Crop to Total Production Ratio, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.037	0.018	0.047+	0.066*	0.109***	0.072***	0.068**
	(0.030)	(0.028)	(0.027)	(0.027)	(0.022)	(0.020)	(0.025)
N	2836	2892	2929	2930	2857	2799	2807
R2	0.187	0.148	0.152	0.272	0.308	0.328	0.417
Starting 1982							
Employment Growth		0.073*	0.086**	0.108***	0.141***	0.088***	0.081**
		(0.029)	(0.027)	(0.029)	(0.025)	(0.022)	(0.026)
N		2895	2917	2924	2854	2796	2807
R2		0.206	0.185	0.288	0.354	0.351	0.445
Starting 1987							
Employment Growth			0.042	0.090*	0.168***	0.097***	0.081*
			(0.031)	(0.036)	(0.034)	(0.027)	(0.034)
N			2990	2990	2925	2872	2872
R2			0.154	0.256	0.296	0.320	0.429
Starting 1992							
Employment Growth				0.039	0.142***	0.080**	0.049
				(0.039)	(0.038)	(0.028)	(0.031)
N				2997	2929	2876	2878
R2				0.244	0.265	0.319	0.427
Starting 1997							
Employment Growth					0.097*	0.060+	0.005
					(0.040)	(0.032)	(0.033)
N					2933	2877	2883
R2					0.164	0.210	0.348
Starting 2002							
Employment Growth						-0.049	-0.077
						(0.042)	(0.048)
N						2853	2865
R2						0.241	0.359
Starting 2007							
Employment Growth							-0.034
							(0.059)
N							2822
R2							0.380

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 55: Winsorized Change in Crop to Total Production Ratio, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.142	0.374***	0.105	0.339**	0.298***	0.272**	0.065
	(0.149)	(0.097)	(0.095)	(0.104)	(0.079)	(0.086)	(0.099)
N	2836	2892	2929	2930	2857	2799	2807
R2	0.895	0.895	0.892	0.900	0.902	0.904	0.901
Starting 1982							
Employment Growth		0.213	-0.062	0.275*	0.195+	0.220*	-0.017
		(0.132)	(0.170)	(0.124)	(0.104)	(0.094)	(0.113)
N		2895	2917	2924	2854	2796	2807
R2		0.900	0.897	0.903	0.906	0.906	0.903
Starting 1987							
Employment Growth			-0.647	0.178	0.234+	0.210+	-0.255+
			(0.532)	(0.246)	(0.138)	(0.115)	(0.145)
N			2990	2990	2925	2872	2872
R2			0.889	0.893	0.896	0.898	0.896
Starting 1992							
Employment Growth				-0.253	0.318*	0.273*	-0.191
				(0.237)	(0.127)	(0.115)	(0.128)
N				2997	2929	2876	2878
R2				0.897	0.898	0.902	0.899
Starting 1997							
Employment Growth					0.552***	0.392***	-0.102
					(0.134)	(0.104)	(0.134)
N					2933	2877	2883
R2					0.896	0.900	0.895
Starting 2002							
Employment Growth						-0.095	-1.348***
						(0.210)	(0.348)
N						2853	2865
R2						0.901	0.894
Starting 2007							
Employment Growth							-1.741**
							(0.589)
N							2822
R2							0.899

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 56: Winsorized Change in Cropland to Farmland Ratio, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.032**	-0.005	0.012+	0.009+	0.002	-0.008+	-0.012+
	(0.011)	(0.007)	(0.007)	(0.006)	(0.005)	(0.005)	(0.006)
N	2933	2925	2963	2957	2948	2958	2953
R2	0.111	0.188	0.111	0.196	0.393	0.384	0.464
Starting 1982							
Employment Growth		0.006	0.012	0.012+	-0.000	-0.014*	-0.017**
		(0.008)	(0.007)	(0.006)	(0.005)	(0.005)	(0.006)
N		2950	2986	2977	2968	2979	2973
R2		0.104	0.084	0.180	0.412	0.391	0.459
Starting 1987							
Employment Growth			0.015	0.030**	0.008	-0.009	-0.014+
			(0.009)	(0.009)	(0.006)	(0.007)	(0.008)
N			2980	2980	2963	2977	2973
R2			0.079	0.139	0.386	0.374	0.463
Starting 1992							
Employment Growth				0.003	-0.007	-0.030**	-0.032**
				(0.010)	(0.009)	(0.010)	(0.012)
N				2961	2945	2961	2952
R2				0.147	0.429	0.429	0.500
Starting 1997							
Employment Growth					-0.034*	-0.039*	-0.031+
					(0.014)	(0.015)	(0.017)
N					2954	2967	2958
R2					0.333	0.393	0.480
Starting 2002							
Employment Growth						-0.055***	-0.017
						(0.016)	(0.019)
N						3012	3003
R2						0.312	0.457
Starting 2007							
Employment Growth							0.009
							(0.022)
N							3021
R2							0.313

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 57: Winsorized Change in Cropland to Farmland Ratio, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.066	0.010	-0.023	-0.026	0.009	0.040*	0.027
	(0.046)	(0.025)	(0.029)	(0.024)	(0.019)	(0.018)	(0.021)
N	2933	2925	2963	2957	2948	2958	2953
R2	0.888	0.896	0.886	0.894	0.915	0.917	0.926
Starting 1982							
Employment Growth		0.044	0.068	-0.024	0.010	0.029	0.020
		(0.034)	(0.076)	(0.027)	(0.021)	(0.020)	(0.024)
N		2950	2986	2977	2968	2979	2973
R2		0.888	0.882	0.891	0.914	0.914	0.923
Starting 1987							
Employment Growth			-0.337+	-0.056	0.023	0.034	0.003
			(0.195)	(0.069)	(0.031)	(0.024)	(0.032)
N			2980	2980	2963	2977	2973
R2			0.885	0.890	0.916	0.915	0.923
Starting 1992							
Employment Growth				-0.090	0.010	0.020	-0.004
				(0.055)	(0.033)	(0.029)	(0.040)
N				2961	2945	2961	2952
R2				0.894	0.920	0.920	0.927
Starting 1997							
Employment Growth					0.083	0.039	0.004
					(0.053)	(0.046)	(0.063)
N					2954	2967	2958
R2					0.913	0.919	0.926
Starting 2002							
Employment Growth						0.211*	0.161
						(0.096)	(0.130)
N						3012	3003
R2						0.910	0.922
Starting 2007							
Employment Growth							0.151
							(0.170)
N							3021
R2							0.910

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 58: Winsorized Change in Machinery to Production Value Ratio, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.105***	0.127***	0.084***	0.097***	0.129***	0.178***	0.156***
	(0.032)	(0.022)	(0.016)	(0.016)	(0.021)	(0.025)	(0.021)
N	3046	3018	3035	3042	3030	2991	3005
R2	0.207	0.169	0.166	0.217	0.187	0.302	0.297
Starting 1982							
Employment Growth		0.068**	0.049**	0.071***	0.103***	0.160***	0.132***
		(0.024)	(0.017)	(0.015)	(0.021)	(0.024)	(0.020)
N		3024	3041	3049	3035	2998	3011
R2		0.123	0.110	0.129	0.151	0.256	0.244
Starting 1987							
Employment Growth			0.049*	0.062**	0.093***	0.184***	0.139***
			(0.021)	(0.019)	(0.028)	(0.028)	(0.025)
N			3020	3021	3011	2973	2983
R2			0.168	0.154	0.135	0.241	0.227
Starting 1992							
Employment Growth				0.043	0.066	0.213***	0.173***
				(0.033)	(0.043)	(0.039)	(0.033)
N				3036	3029	2987	2999
R2				0.174	0.163	0.302	0.256
Starting 1997							
Employment Growth					0.041	0.233***	0.136**
					(0.057)	(0.048)	(0.042)
N					3034	2994	3008
R2					0.146	0.221	0.185
Starting 2002							
Employment Growth						0.110*	0.067
						(0.053)	(0.042)
N						2987	2999
R2						0.242	0.231
Starting 2007							
Employment Growth							0.001
							(0.042)
N							2973
R2							0.213

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 59: Winsorized Change in Machinery to Production Value Ratio, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.061	0.243***	0.146*	0.129+	0.137+	0.267**	0.343***
	(0.160)	(0.066)	(0.069)	(0.071)	(0.073)	(0.083)	(0.067)
N	3046	3018	3035	3042	3030	2991	3005
R2	0.899	0.895	0.895	0.901	0.899	0.910	0.910
Starting 1982							
Employment Growth		0.050	-0.060	0.091	0.039	0.208**	0.283***
		(0.093)	(0.143)	(0.066)	(0.071)	(0.069)	(0.062)
N		3024	3041	3049	3035	2998	3011
R2		0.892	0.890	0.893	0.895	0.905	0.905
Starting 1987							
Employment Growth			0.288	-0.091	-0.158	0.285**	0.398***
			(0.360)	(0.158)	(0.124)	(0.100)	(0.091)
N			3020	3021	3011	2973	2983
R2			0.895	0.897	0.894	0.906	0.905
Starting 1992							
Employment Growth				-0.205	-0.088	0.525***	0.765***
				(0.177)	(0.149)	(0.121)	(0.123)
N				3036	3029	2987	2999
R2				0.901	0.899	0.913	0.911
Starting 1997							
Employment Growth					0.132	0.895***	0.995***
					(0.211)	(0.134)	(0.151)
N					3034	2994	3008
R2					0.899	0.910	0.907
Starting 2002							
Employment Growth						0.917**	1.099***
						(0.304)	(0.301)
N						2987	2999
R2						0.912	0.909
Starting 2007							
Employment Growth							0.508+
							(0.288)
N							2973
R2							0.902

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 60: Winsorized Change in Fertilizer to Production Value Ratio, County Level

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	-0.092*	-0.005	0.004	-0.010	0.001	0.005	0.026
	(0.038)	(0.022)	(0.017)	(0.015)	(0.016)	(0.017)	(0.019)
N	3006	2988	3019	3023	3002	2963	2962
R2	0.158	0.152	0.163	0.254	0.219	0.221	0.255
Starting 1982							
Employment Growth		-0.032	-0.005	-0.025	-0.016	-0.021	0.026
		(0.033)	(0.024)	(0.020)	(0.018)	(0.022)	(0.024)
N		2986	3010	3018	2996	2959	2954
R2		0.257	0.251	0.354	0.307	0.304	0.329
Starting 1987							
Employment Growth			0.038	-0.012	0.002	0.008	0.062*
			(0.032)	(0.026)	(0.026)	(0.026)	(0.028)
N			2995	2996	2978	2942	2937
R2			0.146	0.267	0.216	0.208	0.253
Starting 1992							
Employment Growth				-0.026	-0.024	-0.017	0.081*
				(0.038)	(0.030)	(0.035)	(0.039)
N				3019	2999	2961	2955
R2				0.237	0.186	0.181	0.248
Starting 1997							
Employment Growth					-0.078+	-0.007	0.052
					(0.047)	(0.044)	(0.047)
N					3000	2968	2964
R2					0.092	0.091	0.173
Starting 2002							
Employment Growth						-0.091+	0.068
						(0.054)	(0.054)
N						2960	2959
R2						0.101	0.190
Starting 2007							
Employment Growth							0.146*
							(0.063)
N							2933
R2							0.230

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 61: Winsorized Change in Fertilizer to Production Value Ratio, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Starting 1978							
Employment Growth	0.320+	0.191*	0.143+	0.061	0.109*	0.121+	0.260***
	(0.164)	(0.078)	(0.083)	(0.059)	(0.048)	(0.063)	(0.057)
N	3006	2988	3019	3023	3002	2963	2962
R2	0.892	0.890	0.892	0.901	0.896	0.896	0.899
Starting 1982							
Employment Growth		0.176	0.284	0.023	0.024	0.075	0.217**
		(0.158)	(0.224)	(0.089)	(0.061)	(0.073)	(0.081)
N		2986	3010	3018	2996	2959	2954
R2		0.899	0.898	0.910	0.903	0.903	0.906
Starting 1987							
Employment Growth			0.931+	0.017	-0.015	0.099	0.417***
			(0.556)	(0.170)	(0.097)	(0.096)	(0.092)
N			2995	2996	2978	2942	2937
R2			0.893	0.906	0.900	0.899	0.903
Starting 1992							
Employment Growth				-0.218	0.123	0.280**	0.667***
				(0.177)	(0.102)	(0.100)	(0.115)
N				3019	2999	2961	2955
R2				0.904	0.897	0.896	0.903
Starting 1997							
Employment Growth					0.211	0.502***	0.767***
					(0.185)	(0.109)	(0.160)
N					3000	2968	2964
R2					0.891	0.892	0.898
Starting 2002							
Employment Growth						0.594+	1.182***
						(0.314)	(0.359)
N						2960	2959
R2						0.892	0.898
Starting 2007							
Employment Growth							0.438
							(0.373)
N							2933
R2							0.904

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

Table 62: Winsorized Change in Total Net Profit, County Level

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	-0.226*	-0.253**	-0.481***	-0.381***	-1.115***
	(0.090)	(0.083)	(0.110)	(0.098)	(0.150)
N	3048	3048	3021	3014	3024
R2	0.105	0.126	0.156	0.169	0.268
Starting 1992					
Employment Growth		-0.023	-0.438**	-0.153	-0.874***
		(0.137)	(0.149)	(0.128)	(0.210)
N		3038	3014	3008	3017
R2		0.093	0.109	0.108	0.234
Starting 1997					
Employment Growth			-0.373*	-0.509**	-0.954***
			(0.186)	(0.175)	(0.282)
N			3025	3021	3036
R2			0.085	0.131	0.204
Starting 2002					
Employment Growth				-0.800*	-0.644+
				(0.336)	(0.338)
N				3002	3011
R2				0.158	0.148
Starting 2007					
Employment Growth					-0.348
					(0.279)
N					3008
R2					0.169

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 63: Winsorized Change in Total Net Profit, County Level Shift-Share

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	-0.653	0.387	-1.167**	-0.621	-3.917***
	(1.283)	(0.904)	(0.421)	(0.409)	(0.635)
N	3048	3048	3021	3014	3024
R2	0.890	0.895	0.899	0.896	0.910
Starting 1992					
Employment Growth		0.461	-1.579**	-0.935*	-2.738***
		(0.787)	(0.513)	(0.407)	(0.753)
N		3038	3014	3008	3017
R2		0.894	0.896	0.894	0.907
Starting 1997					
Employment Growth			-2.658***	-1.703***	-4.530***
			(0.639)	(0.508)	(0.865)
N			3025	3021	3036
R2			0.894	0.897	0.906
Starting 2002					
Employment Growth				-3.050+	-1.639
				(1.721)	(1.972)
N				3002	3011
R2				0.900	0.900
Starting 2007					
Employment Growth					-2.061
					(1.938)
N					3008
R2					0.904

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 64: Winsorized Change in Profit Margin, County Level

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	-0.187***	-0.128**	-0.299***	-0.138***	-0.341***
	(0.056)	(0.044)	(0.065)	(0.035)	(0.055)
N	3020	3019	2999	2954	2972
R2	0.339	0.315	0.217	0.430	0.285
Starting 1992					
Employment Growth		-0.093	-0.407***	-0.125*	-0.413***
		(0.092)	(0.106)	(0.062)	(0.088)
N		3028	3008	2959	2979
R2		0.110	0.102	0.159	0.188
Starting 1997					
Employment Growth			-0.544**	-0.326***	-0.510***
			(0.166)	(0.097)	(0.124)
N			3017	2969	2991
R2			0.119	0.193	0.178
Starting 2002					
Employment Growth				-0.456*	-0.144
				(0.199)	(0.156)
N				2956	2974
R2				0.332	0.128
Starting 2007					
Employment Growth					-0.132
					(0.196)
N					2939
R2					0.174

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 65: Winsorized Change in Profit Margin, County Level Shift-Share

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	-1.219	0.503	-0.542*	0.111	-1.221***
	(1.017)	(0.481)	(0.231)	(0.152)	(0.209)
N	3020	3019	2999	2954	2972
R2	0.889	0.892	0.897	0.894	0.909
Starting 1992					
Employment Growth		0.691	-0.919*	-0.045	-0.894**
		(0.517)	(0.368)	(0.173)	(0.307)
N		3028	3008	2959	2979
R2		0.891	0.893	0.891	0.903
Starting 1997					
Employment Growth			-2.196***	-0.444	-1.812***
			(0.561)	(0.278)	(0.397)
N			3017	2969	2991
R2			0.894	0.894	0.901
Starting 2002					
Employment Growth				-1.454	-0.345
				(1.058)	(0.977)
N				2956	2974
R2				0.897	0.895
Starting 2007					
Employment Growth					-1.882
					(1.448)
N					2939
R2					0.905

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 66: Winsorized Change in Percent of Farms with Net Losses, County Level

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	0.022	0.028*	0.082***	0.097***	0.090***
	(0.026)	(0.013)	(0.013)	(0.011)	(0.010)
N	3057	3057	3055	3054	3054
R2	0.189	0.308	0.445	0.433	0.474
Starting 1992					
Employment Growth		0.017	0.104***	0.125***	0.118***
		(0.026)	(0.020)	(0.016)	(0.013)
N		3055	3053	3052	3052
R2		0.354	0.515	0.477	0.518
Starting 1997					
Employment Growth			0.176***	0.184***	0.146***
			(0.029)	(0.022)	(0.018)
N			3059	3058	3058
R2			0.431	0.418	0.432
Starting 2002					
Employment Growth				0.109***	0.088***
				(0.025)	(0.021)
N				3055	3056
R2				0.316	0.331
Starting 2007					
Employment Growth					0.027
					(0.028)
N					3058
R2					0.367

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 67: Winsorized Change in Percent of Farms with Net Losses, County Level Shift-Share

	1992	1997	2002	2007	2012
Starting 1987					
Employment Growth	-0.295	-0.397***	-0.242***	-0.018	-0.009
	(0.364)	(0.114)	(0.059)	(0.041)	(0.045)
N	3057	3057	3055	3054	3054
R2	0.885	0.892	0.895	0.913	0.911
Starting 1992					
Employment Growth		-0.150	-0.132+	0.058	0.071
		(0.115)	(0.074)	(0.050)	(0.049)
N		3055	3053	3052	3052
R2		0.895	0.898	0.916	0.916
Starting 1997					
Employment Growth			0.042	0.210**	0.169*
			(0.111)	(0.070)	(0.070)
N			3059	3058	3058
R2			0.898	0.917	0.914
Starting 2002					
Employment Growth				0.099	0.141
				(0.120)	(0.122)
N				3055	3056
R2				0.913	0.907
Starting 2007					
Employment Growth					0.159
					(0.174)
N					3058
R2					0.905

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 68: Winsorized Change in Acres by Organizational Structure, County Level

	1982	1987	1992	1997	2002	2007	2012
Sole Proprietorship							
Employment Growth	-0.015	-0.018*	-0.018*	-0.018**	-0.018**	-0.032***	-0.028***
	(0.012)	(0.009)	(0.007)	(0.007)	(0.007)	(0.005)	(0.005)
N	2993	2996	3020	3020	2299	2418	2531
R2	0.085	0.258	0.343	0.301	0.303	0.356	0.351
Partnership							
Employment Growth	-0.045	-0.004	0.030	-0.008	-0.047***	-0.050***	-0.030*
	(0.041)	(0.028)	(0.026)	(0.017)	(0.014)	(0.012)	(0.014)
N	2452	2312	2654	2654	2454	2553	2561
R2	0.080	0.094	0.109	0.112	0.135	0.163	0.170
Family Corp.							
Employment Growth	-0.348*	-0.148	-0.061	0.011	-0.122*	-0.123*	-0.085
	(0.154)	(0.131)	(0.091)	(0.078)	(0.059)	(0.049)	(0.057)
N	868	806	1410	1410	1227	1315	1341
R2	0.210	0.256	0.179	0.231	0.319	0.329	0.314
Non-Family Corp.							
Employment Growth	0.757	-0.237	0.096	0.217*	0.158	0.232	0.253
	(0.625)	(0.298)	(0.119)	(0.097)	(0.139)	(0.160)	(0.205)
N	281	245	454	454	250	297	297
R2	0.365	0.329	0.215	0.199	0.303	0.350	0.386
Other							
Employment Growth	-0.201	-0.294	0.133*	0.142**	-0.130	-0.295*	-0.318*
	(0.296)	(0.202)	(0.063)	(0.055)	(0.114)	(0.127)	(0.158)
N	783	760	1023	1023	432	523	590
R2	0.198	0.165	0.124	0.139	0.199	0.310	0.311

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 69: Winsorized Change in Acres by Organizational Structure, County Level Shift-Share

	1982	1987	1992	1997	2002	2007	2012
Sole Proprietorship							
Employment Growth	-0.102+	-0.108***	-0.088**	-0.131***	-0.101***	-0.118***	-0.148***
	(0.056)	(0.026)	(0.032)	(0.024)	(0.023)	(0.017)	(0.018)
N	2993	2996	3020	3020	2299	2418	2531
R2	0.887	0.907	0.916	0.912	0.913	0.918	0.917
Partnership							
Employment Growth	0.029	0.014	-0.238*	-0.108	-0.062	-0.075+	0.001
	(0.179)	(0.120)	(0.116)	(0.074)	(0.048)	(0.041)	(0.048)
N	2452	2312	2654	2654	2454	2553	2561
R2	0.889	0.891	0.887	0.890	0.893	0.896	0.897
Family Corp.							
Employment Growth	-0.280	-0.170	0.309	0.422+	-0.275	-0.215	-0.326
	(0.503)	(0.394)	(0.371)	(0.256)	(0.249)	(0.219)	(0.270)
N	868	806	1410	1410	1227	1315	1341
R2	0.903	0.906	0.899	0.904	0.908	0.907	0.902
Non-Family Corp.							
Employment Growth	1.980	-0.631	-0.237	1.128*	0.174	0.191	0.291
	(1.539)	(1.100)	(0.656)	(0.444)	(0.622)	(0.659)	(0.776)
N	281	245	454	454	250	297	297
R2	0.913	0.909	0.912	0.908	0.910	0.904	0.911
Other							
Employment Growth	1.991	-1.253+	0.077	0.557**	-0.009	-0.349	-0.404
	(1.689)	(0.702)	(0.329)	(0.203)	(0.417)	(0.595)	(0.644)
N	783	760	1023	1023	432	523	590
R2	0.899	0.895	0.899	0.900	0.899	0.907	0.904

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

10 Appendix B: Data and Code Details

10.1 Coding Processes

I use R for all data cleaning and analyses (R Core Team, 2025). For data cleaning, I make use of the tidyverse packages, as well as readr, Matrix, easystats, and here (Wickham et al., 2019; Müller, 2025; Bates et al., 2025; Lüdecke et al., 2022). I use BLSLoadR to access the Bureau of Labor Statistics data (Bureau of Labor Statistics, 2026; Schmidt and Rembert, 2026). I use JJ Chen's implementation of Rotemberg weights in line with Goldsmith-Pinkham et al. (2020) (Chen, 2026). I use the sf package for choropleths and other spatial data, alongside the maps package and tigris (Pebesma, 2018; Becker et al., 2025; Walker, 2025). I also use reshape2, scales, and ggrepel for data visualization alongside the tidyverse packages (Wickham, 2007; Wickham et al., 2025; Slowikowski, 2024). When it comes to the regressions, I use the packages fixest, splines, broom, purrr, modelsummary, marginaleffects, and tinytable to generate and format the outputs (Bergé, 2018; Robinson et al., 2025; Wickham and Henry, 2025; Arel-Bundock, 2022, 2025; Arel-Bundock et al., 2024).

Because of the licensing and distribution constraints for both IPUMS and ICPSR, which I get data from, I am not able to post all the data online. However, the code is available on Github, minus the accompanying data, in this repository.

Table 70: NAICS Codes Removed from CBP Data

Omitted NAICS code stem	**Category**
1	Agriculture etc.
21	Mining
311	Food Manufacturing
312	Beverage Manufacturing
3253	Agricultural Chemical Manufacturing
3331	Agriculture, Construction, and Mining Machinery Manufacturing
4244	Grocery Merchant Wholesalers
4245	Farm Product Raw Material Merchant Wholesalers
4442	Lawn and Garden Equipment and Supplies Stores
924	Administration of Environmental Quality Programs